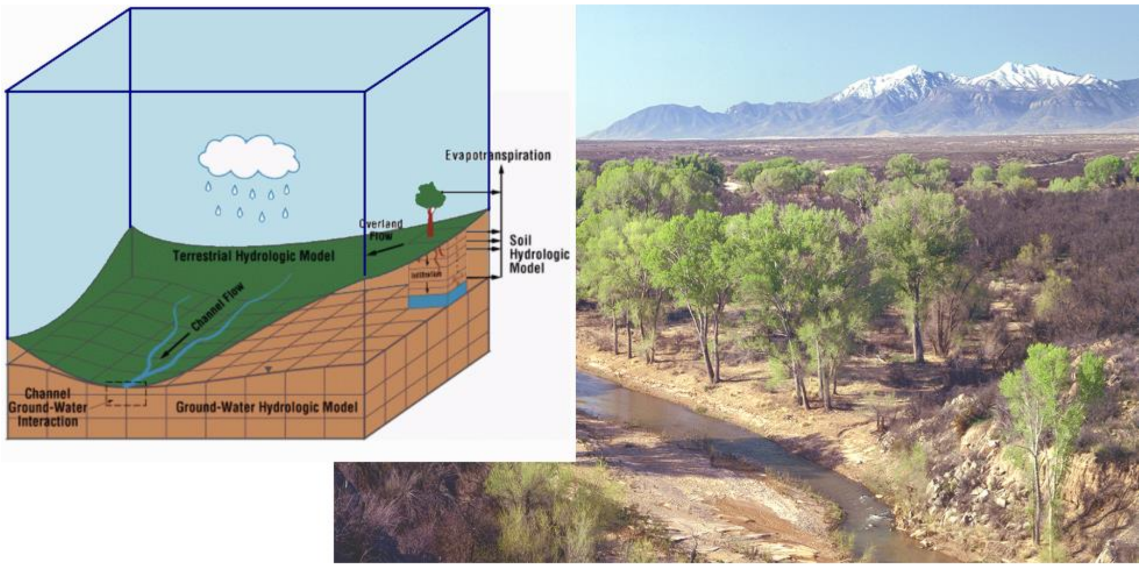
Topic	Surface Water Modelling	
Module	1.	Hydrological Modelling



Source: George H. Leavesley, USGS

Learning objectives: After completion of this module the learner will be able to

1. explain the principles and objectives of hydrological modelling (Level 2: Understand)
2. explain the structure and variables of a hydrological model (Level 2: Understand)
3. explain the main types of hydrological models (Level 2: Understand)
4. explain the steps to develop a hydrological model (Level 2: Understand)
5. develop a simple hydrological model (Level 6: Create)
6. select model parameters (Level 5: Evaluate)
7. explain the challenges of model calibration and validation (Level 2: Understand) and use this knowledge to calibrate and validate a model (Level 4: Analyse)
8. discuss the importance of considering uncertainties and judge the level of relevance of various uncertainty sources for a given application (Level 5: Evaluate)
9. implement a hydrological model (Level 3: Apply) to compare and contrast the effect of land use change and implementation of various infrastructure measures on catchment hydrology (Level 4: Analyse)

Note: sections that are not examinable are indicated by (beginning of 'not examinable') and (end of 'not examinable')

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Principles of Hydrological Modelling

KEY POINTS

- The development of a hydrological model requires: 1) selection and development of a structure (and associated computer code or suitable elements from existing computer code) to represent the physical medium and processes of the modelled system, 2) estimation of the model parameters and 3) model validation and characterisation of the associated uncertainties.
- The different approaches that can be used to represent the physical medium and processes lead to different model types. The process description can be empirical (black box), conceptual (grey box) or physically-based (white box). Furthermore, processes can be described in a stochastic or deterministic manner. The spatial representation can be lumped, semi-distributed or distributed. Models can be distinguished in particular by the nature of the spatial unit within which the equations describing the processes are solved.
- A hydrological model is made up of different modules that can be both dependent and independent. Examples of such modules are modules used to convert input data into the required format, distribute the various water fluxes at the soil-vegetation-atmosphere interface, generate discharges at the basin outlet and route streamflows along the river and through lakes and reservoirs.
- The level of complexity used for the different representations or components of a model must be well-balanced. Processes must be represented in a manner that is compatible with the spatial and temporal resolutions of the model.
- Model parameters can be predicted on the basis of measurements, field observations or thematic maps, for instance using regional models. They can also be estimated by calibration.
- A model can be calibrated manually, automatically or semi-automatically. The objective of calibration is to determine the set of parameters that minimises the differences between simulated values and observations. Calibration must always be subject to careful reasoning.
- Performance criteria can be qualitative or quantitative and based on an objective function.
- Ideally, calibration should be multi-criteria and multi-objective. To calibrate each of the different components of a model in a differentiated and optimal manner, the calibration must consider different variables, time scales, temporal windows, etc.
- A number of parameter sets can generally be found to produce equivalent model performance, leading to an equifinality problem.
- Model validation is intended to check that the model provides a reasonable representation of reality, adapted to the targeted objective. Validation can be of the internal and/or external type.
- Validation schemes can be based on split sample tests, proxy basin tests, differential split sample tests, proxy basin differential split sample tests or blind validation tests.
- Model validation does not indicate that the model represents the hydrosystem in a pertinent manner. This is especially true for its behaviour in extreme situations that have not yet been observed.

- The main sources of uncertainty in variables simulated by a model are related to its structure, its parameters and the data used.
- Simulation uncertainties are often characterised by combining the results of simulations produced for different sets of parameters sampled in the parameter space. Bayesian algorithms such as the Metropolis algorithm are well-suited for this sampling.

BACKGROUND AND OUTLINE

Hydrological modelling is one of the main techniques used to meet the challenges of predicting or forecasting hydrological variables. The aim of the hydrological model is essentially to estimate the various hydrological variables required for a given application, study or design project. Hydrological modelling is also used in addition to observations to reconstruct and/or verify observed data or as a complement to experiments in scientific research in hydrology. Hydrological modelling consists in representing a given hydrosystem (and the corresponding hydrological phenomena) mathematically in order to simulate a part or all of its hydrological behaviour. Hydrological modelling is generally carried out on the scale of a drainage basin (urban or rural), i.e. the catchment scale, which is the scale that will be dealt with in particular in this chapter. Figure 1 shows a catchment with various elements that might be the goal in the design, evaluation or management of the catchment, using a hydrological model.

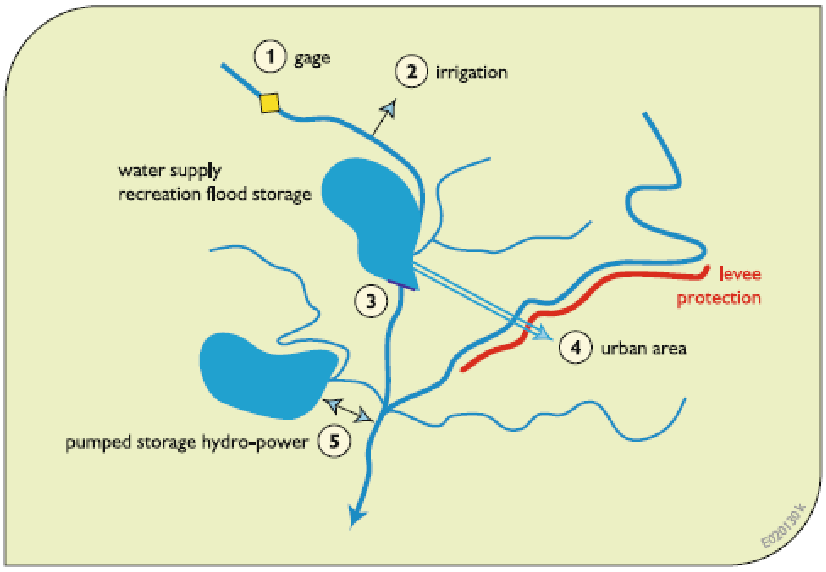

Figure 1. Example for elements in a catchment.

The objective of modelling is often to simulate, at the basin outlet, the runoff or discharges resulting from various meteorological forcing variables (e.g., precipitation, temperature). Precipitation is generally the main forcing variable, which is the reason that hydrological models are often referred to as rainfall-runoff models. Factors affecting the hydrological system response, e.g. the hydrograph at various locations within the catchment:

- Rainfall intensity & duration
- Catchment size
- Catchment slope

- Catchment shape
- Catchment storage
- Channel type and morphology
- Channel storage
- Land use/land cover
- Soil type & infiltration
- Percent impervious

Yet the actual shape and timing of the response hydrograph for a given catchment have been shown to be a function of many physiographic, land use, and meteorological variables. Rainfall intensity and duration are the major driving forces of the rainfall–runoff process, followed by catchment characteristics that translate the rainfall input into an output hydrograph at the outlet of the basin. Size, slope, shape, soils, and storage capacity are all important parameters in catchment geomorphology. Land use and land cover parameters can significantly alter the natural hydrologic response through increases in impervious cover, altered slopes, and improved drainage channel networks.

Examples for engineering applications of hydrological models are: Change in land use (e.g. impacts or restoration to pre-development conditions to manage the impact of urbanisation); Infrastructure (e.g. assessing and mitigation of the impact of a road and railway line); Structural flood mitigation measures (such as dams and stopbanks); Future development scenarios; Change in dam operations; Altered catchment conditions assessment; Climate change impact; Parameter sensitivity tests; amongst others.

Catchment modelling has become the dominant flow estimation technique. This is because it:

- Allows different options to be simulated.
- Can be used with no data, limited data and in data rich situations
- Can be used to transfer estimations from one location to another
- Default parameters are available
- Well designed catchment modelling systems will reproduce flow behaviour over a range of flows
- A full hydrograph produced and can be used to assess storage
- Can be reused by others for similar problems
- Helps to document the process of how the flow estimation was carried out

The scope is to provide practical guidance on the application of these models. Because of the wide variety of flood estimation problems, no modelling framework is suitable for all problems.

Section 1 of this module first describes the principles and objectives of hydrological modelling along with the main types of hydrological models and their structure and variables. Section 2 deals with methods and problems related to the representation of a given hydrosystem and the corresponding hydrological phenomena that occur within it. Section 3 presents the problems encountered when estimating the parameters of these models. Section 4 is devoted to the methods used to validate models and characterise the associated uncertainties. Sections 5 and 6 provide some guidelines for choosing a suitable modelling approach. For further information on any of the different aspects of hydrological modelling, the reader can refer to a wide range of publications devoted to this topic (ASCE, 1996; Ambroise, 1999; Beven, 2000; Singh and Woolhiser, 2002; Duan *et al.*, 2003).

1 INTRODUCTION TO HYDROLOGICAL MODELLING

1.1 Principles and Objectives of Hydrological Modelling

Principles of hydrological modelling

A model is a simplified representation of a given physical system and the different processes that govern its operation. The system can be represented physically by a physical model.[1] It is however more often represented mathematically. Such a representation is formalized by a set of equations expressing the laws or concepts considered to appropriately describe the behaviour of the system. The hydrosystem most often modelled in hydrology is the drainage basin. Smaller hydrological units can however be modelled (e.g., a given hillslope or parcel of land). Whatever the hydrosystem, the objective of hydrological modelling is to simulate part or all of its hydrological behaviour. The hydrological behaviour of a given hydrosystem is extremely complex (Musy and Higy, 2011). This is mainly related to the complexity and heterogeneity of the physical medium itself (e.g., geomorphology of the basin, land use, soil properties, geology and characteristics of the hydrographic network). This explains the multitude of hydrological processes involved which, in addition, function on different spatio-temporal scales (Figure 2).

All hydrological models are consequently an extremely simplified representation of the real hydrosystem and its operation. For convenience, these models focus on the processes and spatio-temporal scales that are critical with respect to the hydrological phenomena to be studied. For example, a model developed to simulate the rainfall-runoff relationship during flood periods often ignores processes related to evaporation and groundwater recharge. On the other hand, these processes must be taken into account if the purpose of the model is to estimate the seasonality and interannual variability of water resources. Note however that for interannual variability, processes related to surface runoff can generally be neglected. A model is therefore highly dependent on the final objective of the simulation. Whatever the objective, there are many ways to simplify the representation of the physical medium and the associated hydrological processes.

For this reason, there is a multitude of modelling possibilities for any given hydrosystem. Note that the term model is often used incorrectly to designate a “modelling system”. A modelling system corresponds to the computer code of generic or custom software developed to model the hydrological behaviour of a drainage basin. The actual model corresponds to the application of this code to a particular site.

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[1] In hydrology, physical models are scale models of hydraulic works, river reaches or drainage basins. They are used to quantify the influence of works such as discharge channels, spillways and retention basins in a simple manner for cases where solving mathematical equations would be excessively complex or impossible. The main difficulties encountered in the application and interpretation of these models are related to the scale and the transposition of results in the field. It is impossible to avoid distortion of the phenomena, which often leads to different scale ratios for the different runoff parameters. Analog physical models also exist. They are based on the similarity between the studied phenomenon (e.g., hydraulic head field and the flow of water in the ground) and another physical phenomenon and (e.g., electrical field and the flow of electrical current in an aqueous solution).

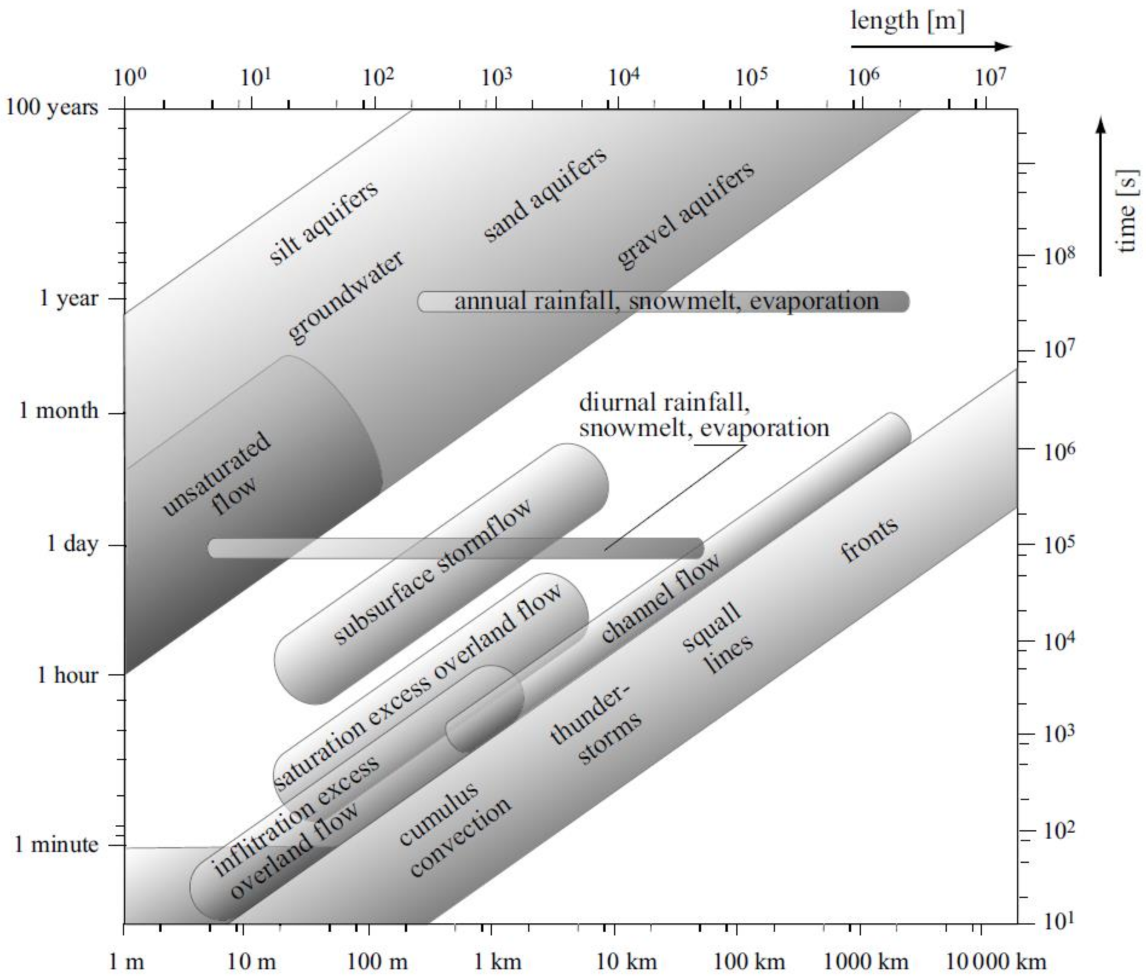


Figure 2. Temporal and spatial scales of various hydrological processes (adapted from Blöschl and Sivapalan, 1995).

Objectives of hydrological modelling

Hydrological modelling can have many different objectives. From a scientific standpoint, the objective is to improve understanding of the behaviour of hydrological systems by modelling the dynamics of the processes involved. As opposed to observations or experiments, modelling can be used to describe and represent, in their full complexity and variability, some or all of the many processes governing the hydrological behaviour of one or all of the compartments of the considered hydrosystem. For a given hydrosystem and under certain conditions, such studies can be used to: 1) identify the most important processes and any non-linearities in the hydrological behaviour of the hydrosystem, 2) test different hypotheses and conceptual frameworks and 3) identify any lack of understanding of the behaviour of certain components. Modelling can also be used to simulate the hydrological behaviour of the considered hydrosystem under changed conditions for which no observations are available.

Rainfall-runoff modelling also has operational objectives, for instance to simulate the hydrological response of the hydrosystem to a given forcing. How such simulations will be used depends on the analysis context.

Simulations can be used to reconstruct historical hydrological events that were observed but not measured. They can also be used for hydrological forecasting, the objective being to estimate possible future changes of different hydrological variables. In another application, simulations can be used to produce the hydrological scenarios required for medium- or long-term management of water resources and associated hydrological risks or to manage development projects. In this context, modelling can provide the hydrological information required for example to design hydraulic works (e.g., dams, drainage networks) or flood protection schemes (e.g., dikes). Simulations can also be used to estimate changes in the response of the considered hydrosystem resulting from modification of the characteristics of the drainage basin (e.g., future projects, changes in land use), modification of meteorological forcing (e.g., climate change) or other nonstationary conditions.

In hydrology, modelling can be implemented on other scales than that of the drainage basin. When coupled with general circulation models, hydrological models can be used for example to simulate water fluxes on the continental scale. On a given hill slope within a drainage basin, they can be used to simulate surface runoff and the associated erosion. For a given parcel of land, the can be used to estimate the water balance of the soil-vegetation atmosphere system and the water requirements of the vegetation.

1.2 Structure and Variables of a Hydrological Model

The typical structure and variables of a hydrological model are summarised in Figure 3. The model, designated *G*, can be formally defined by the following expression:

$$\mathbf{Z}_{sim}(t) = G\,[\mathbf{X}(t), \mathbf{Y}(t), \mathbf{BC}(t), \boldsymbol{\theta}] + \boldsymbol{\varepsilon}_Z(t, \boldsymbol{\theta}) \qquad (1)$$

where $\mathbf{Z}_{sim}(t)$ are the values simulated at time t of the various output variables of the hydrological model and $\mathbf{X}(t)$, $\mathbf{Y}(t)$ and $\mathbf{BC}(t)$ are respectively the values of the various input variables, state variables and boundary conditions at time t. Furthermore, $\boldsymbol{\theta}$ is the vector of model parameters and $\boldsymbol{\varepsilon}_Z(t, \boldsymbol{\theta})$ the vector of model errors that are a function of time and parameter set $\boldsymbol{\theta}$. All these variables will be dealt with below.

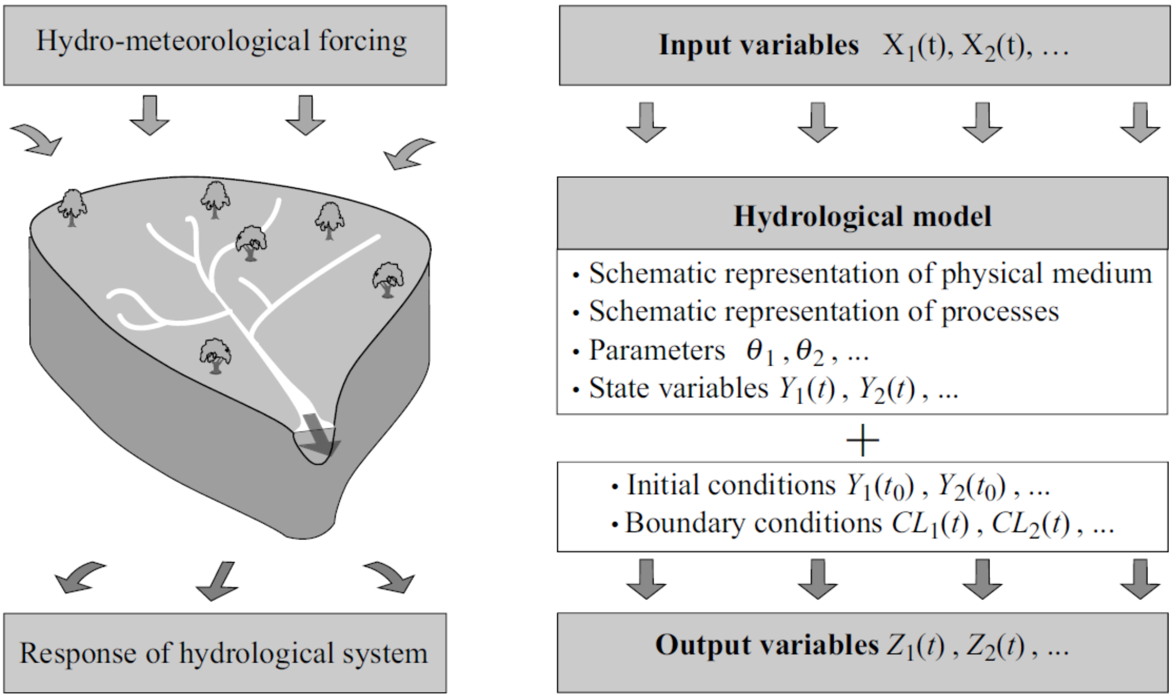


Figure 3. Typical structure and variables of a hydrological model.

Input variables, output variables and model error

The variables used by a model to describe the hydrometeorological forcing phenomena represent the model inputs, X(t). These variables, often referred to as forcing variables, are generally independent and the corresponding data are available in the form of time series. The variables used depend on the model but deal essentially with meteorological phenomena such as precipitation, reference evapotranspiration and temperature. The corresponding data are often available only at certain points corresponding to the different measurement stations located within the drainage basin. In some cases, the data may reflect a spatial distribution, as is the case with precipitation fields derived from radar observations. Other model input variables include inflows such as those from upstream drainage basins and outflows such as pumping for irrigation or diversion to neighbouring drainage basins.

The model outputs, for which the values at time t are given by the formal expression $\mathbf{Z}_{sim}(t, \boldsymbol{\theta}) = G\,[\mathbf{X}(t), \mathbf{Y}(t), \mathbf{BC}(t), \boldsymbol{\theta}\,]$, are the variables that the model produces by simulation. The main output variable of any hydrological model is the discharge, or sometimes the water level, at one or more points in the hydrographic network. In addition, other variables of the hydrological cycle are often simulated (e.g., actual evaporation rate, water table height, snow cover and snowpack water equivalent). The model outputs can be compared to the values observed for the corresponding hydrological variables, if available. The differences between the simulated outputs and the observed values of a given hydrological variable represent the model error, also referred to as residuals. The error varies with time (see Figure 4). For output variable Z, it can be expressed by:

$$\varepsilon_Z(t,\boldsymbol{\theta}) = Z_{obs}(t) - Z_{sim}(t,\boldsymbol{\theta}) \qquad (2)$$

where $\varepsilon_Z(t, \boldsymbol{\theta})$ is the error at time t for hydrological variable Z, $Z_{obs}(t)$ the value observed for this variable and $Z_{sim}(t)$ the value simulated for this variable by model G at time t. The error is a function of the parameter set $\boldsymbol{\theta}$.

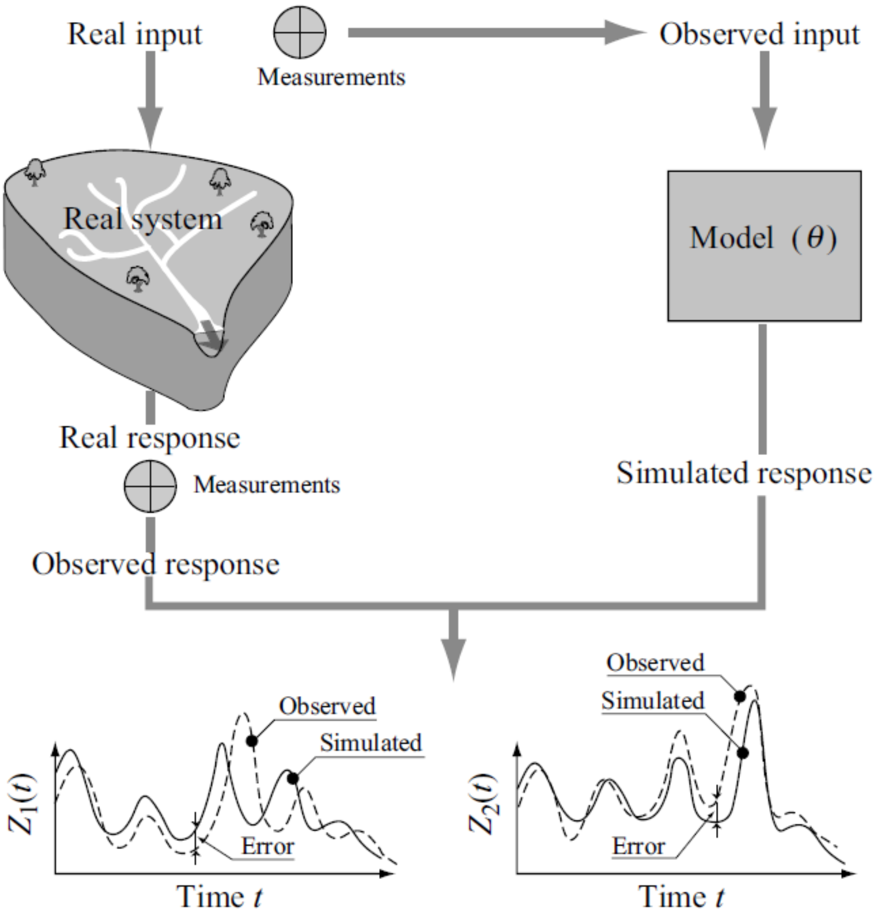


Figure 4. Illustration of simulation error produced by a model for two output variables $Z_1(t)$ *and* $Z_2(t)$*.*

Model structure

The simplified representation of the hydrosystem used to transform the model input variables into output variables defines the structure of model *G*. It concerns:

- the representation of the physical medium defining the hydrosystem.
- the representation of the hydrological behaviour of the hydrosystem with respect to the targeted hydrological phenomena, requiring the development of equations for certain hydrological processes.

The schematic representation of the physical medium concerns first of all the description of its geometry, including for example the elevation contours of the basin, the relief, the structure of the hydrographic network, the geometry of surface reservoirs and the spatial organisation of the terrain. It also concerns the description of its physical properties (geology, soil types, land use). Representation of the physical medium requires its spatial discretisation, i.e., division of the medium into distinct hydrological compartments, each governed by its own laws of behaviour. These compartments may or may not correspond to real hydrologic entities of the system. Different types of spatial discretisations are presented in Section 2.3.

To represent the hydrological phenomena of interest for the considered hydrosystem, equations must be provided for certain processes. First of all, a choice must be made concerning the main hydrological processes to be represented to describe the behaviour of the hydrosystem with respect to the phenomenon of interest. Then mathematical expressions that provide a “reasonable” representation of these processes must be chosen.

The development of the equations also requires details on the nature of the interactions between the different spatial discretisation elements selected to represent the medium. The equations typically used for this purpose are the laws of conservation of mass, energy and momentum, the state relationships (i.e., relationships between the different variables defining the state of the considered spatial unit) and the flux relationships expressing the various fluxes as a function of the pressure or pressure gradient. Different types of models are possible depending on the nature of the equations selected to represent the processes. These types of models are described in Section 2.2.

State variables and model parameters

Each of the equations used by a model expresses a relationship between different quantities, some of which vary with time. They define the model state variables, previously designated $\mathbf{Y}(t)$ (e.g., water volume or level, wetted perimeter). Some of these variables can represent model outputs (e.g., water table height at different points in the drainage basin).

Given the many simplifications in the representation of the medium and the phenomena, the state variables used to model the hydrosystem are rarely of the same nature as the variables used to describe the state of the true physical system. For example, the state of soil moisture, defined by its humidity profile, is often described in a simplified manner by a unique variable describing the average degree of saturation of the soil. Similarly, the state of the snowpack, defined by different variables such as snow depth, density and degree of maturity, is often described simply by its water equivalent. The other quantities appearing in the equations of the model are constants.[2] Some of these are related to measurable physical characteristics of the medium (e.g., hydraulic conductivity of soils). Others are abstract quantities (e.g., storage capacity of the conceptual reservoirs). These various constants defined the model parameters, previously described by the vector $\boldsymbol{\theta}$. Depending on the nature of the discretisation used to represent physical medium, both the state variables and the parameters of the model can vary in space.

Initial conditions and boundary conditions of the model

For a given hydrologic modelling application, it is always necessary to specify the initial conditions and boundary conditions of the simulation.

The initial conditions are the initial values assigned to the state variables $\mathbf{Y}(t)$, i.e., the values $\mathbf{Y}(t_0)$ that these variables take on the first simulation time step (e.g., values of water level at every point of the hydrographic network for $t = t_0$). The temporal evolution of the state variables subsequently depends on the boundary conditions specified for the simulation.

The boundary conditions $\mathbf{BC}(t)$ define the interactions between the boundaries of the considered hydrosystem and its neighbourhood. They are used to estimate, for each time step of the period covered by the simulation, the values of the model state variables at the

[2] In certain hydrological models, referred to as dynamic, the constants can vary with time. This is the case for example for parameters used to describe the growth stage of a given vegetal cover or the degree of impermeabilisation of a drainage basin subject to urbanisation. Temporal variations of these constants are slow compared to those of hydrological state variables.

boundaries of the hydrosystem (e.g., lateral or vertical boundaries). The boundary conditions can specify the value of certain extensive variables such as water table height or intensive variables such as mass and energy flux exchanges across the boundaries of the hydrosystem. They can be defined in the form of a zero-flux condition (e.g., no exchange of discharge between the considered drainage basin and neighbouring drainage basins) or a functional relationship (e.g., equilibrium relationship between the outflow and the water level at the corresponding channel cross-section).

1.3 Main Types of Hydrological Models

A large number of hydrological models exists, which vary in terms of their nature and the complexity. They are hard to classify into unambiguous types given the variety of possible classification criteria. Various classifications have been proposed in the literature (Singh, 1995; ASCE, 1996; Chocat, 1997; Refsgaard, 1997; Ambroise, 1999). A hydrological model classification is shown in Figure 5.

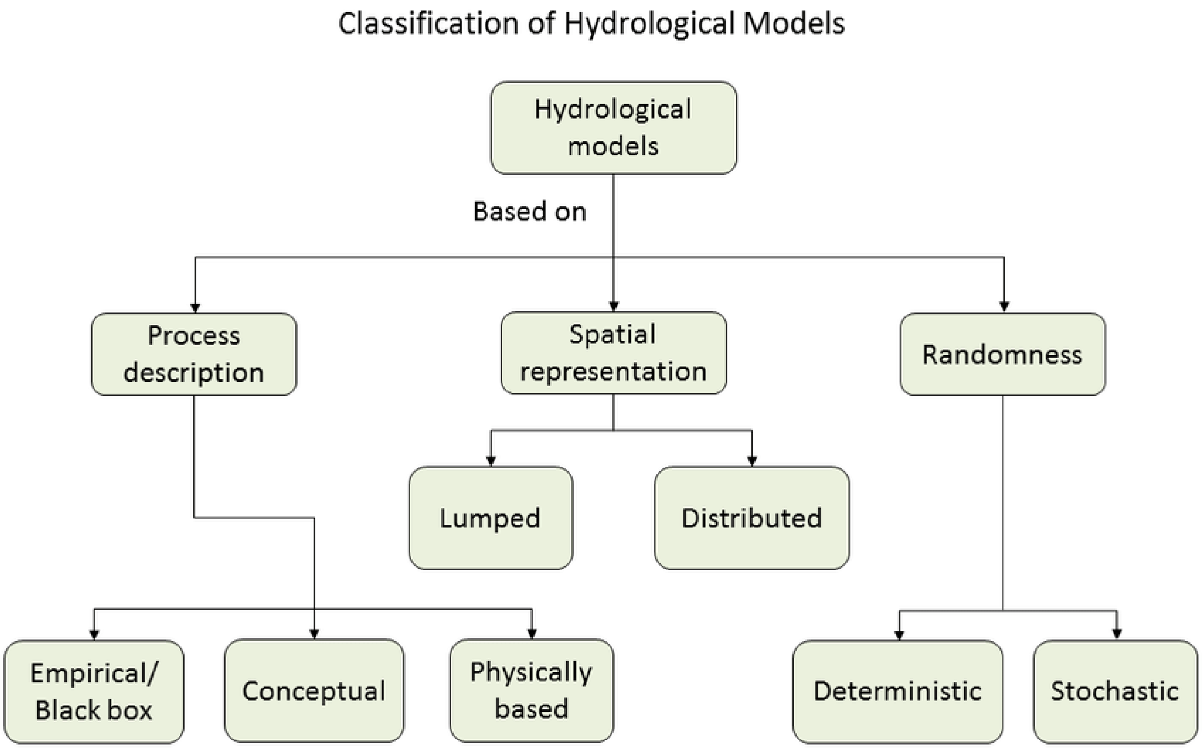


Figure 5 Classification of hydrological models. Note that that the spatial representation can be further refined and classified as 'lumped', semi-distributed' and 'distributed'.

These hydrological models vary depending on the viewpoint adopted:

- Depending on the nature of the relationships used to represent the processes, models can be classed as empirical, conceptual or physically-based. These models are presented in Section 2.3.
- Depending on the manner in which the physical medium is represented and in particular depending on the nature of the spatial unit in which the equations used to describe the processes are solved, models can be classed as lumped, semi-distributed and distributed. These models are presented in Section 2.2.

- Depending on the way the hydrological variables of the relationships between these variables are considered, models can be classed as deterministic or stochastic. The principle of these models is presented in Addendum 1.
- Depending on the nature of the time periods considered for the simulation, the models can be classified as event-based or continuous-simulation models. The principle of these models is also presented in Addendum 1.

Figure 6 presents an example of how the physical system has been represented in a model, the fluxes in the physical system and the partitioning of these fluxes shown as time series.

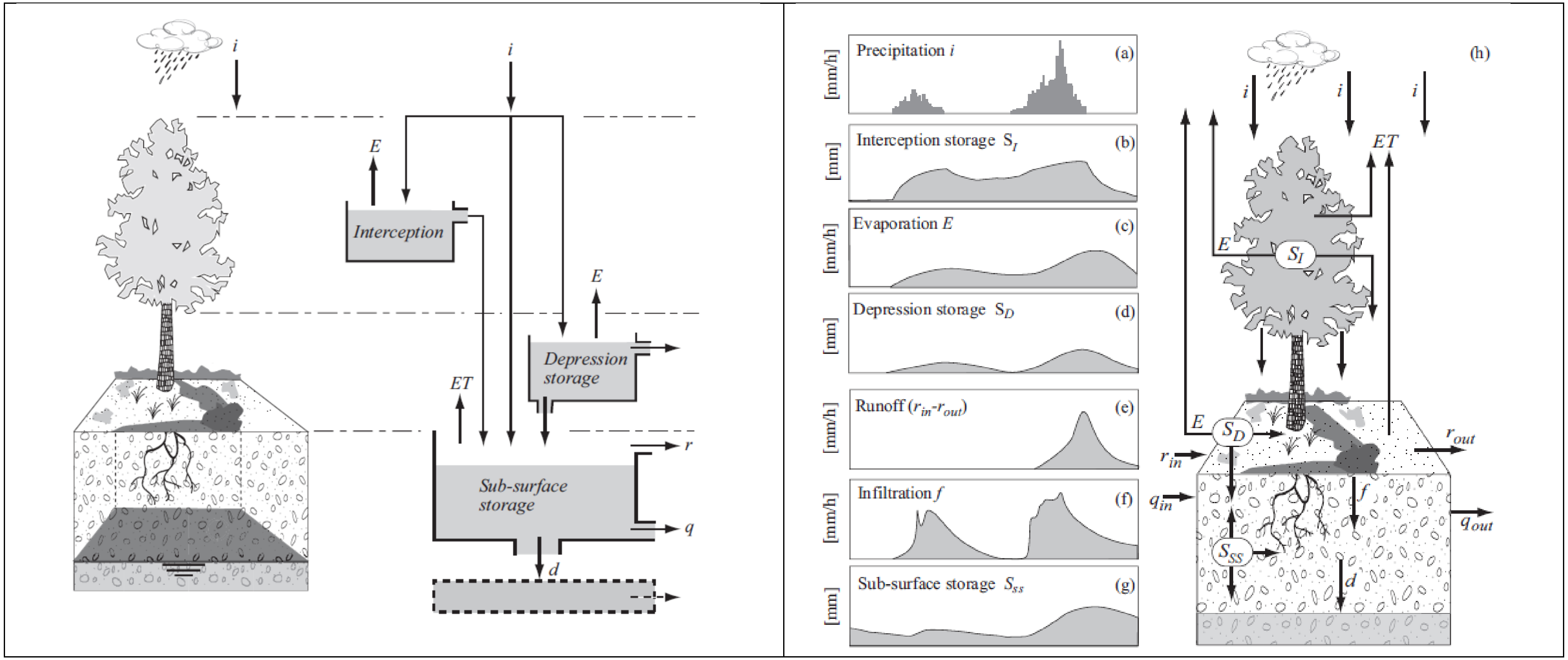


Figure 6. left: The physical system and the model representation; right: The fluxes in the physical system and the partitioning of these fluxes shown as time series.

Another distinction between models can be made on the basis of the time step selected for the simulation. For example, for flood forecasting models developed in urban hydrology, the time step can be infra-hourly. For models used to manage water resources on the scale of very large drainage basins, the time step can be weekly, monthly or even annual. The principles of various hydrological models (among many others) currently used in different regions of the world are illustrated in the Appendix.

Addendum 1 Deterministic, stochastic, event-based and continuous-simulation models

Deterministic and stochastic models

Deterministic models stipulate that, for the given initial conditions and boundary conditions, the relationship between the inputs and the outputs of the system considered are unequivocal. In other words, for a given model input, there exists one and only one output. Stochastic models simulate processes that are partially or totally random. For a given set of initial conditions and boundary conditions, different applications of the stochastic model give, for a given input, different outputs. The underlying assumption is that the considered phenomenon (e.g., precipitation, flood discharge, low-flow discharge) is the result of so many causes that are so complex that the relationship between the model inputs and outputs is not unequivocal. Stochastic models therefore generate a possible realization of the output variable, conditioned by the values of the model input variables. This realisation results from the change, generally with time, of one or more appropriate random variables for the studied phenomenon.

Stochastic or data-driven models are also used in hydrological modelling. Stochastic streamflow models are designed to mimic our historical hydrologic experience, while simultaneously enabling us to recognise the range of statistically possible future hydrologic conditions and the risk of failure associated with water infrastructure. Examples are autoregressive statistical models with exogenous variables (ARX) or autoregressive moving-average models with exogenous variables (ARMAX) (Box and Jenkins, 1970), that can be used stochastically. More recently, machine learning is used to model the hydrological system. At the core, machine learning is a set of tools that allow us to build and train models that extract and reproduce the spatial and temporal patterns in the datasets they encounter (Figure 7). Improved prediction quality in hydrologic machine learning (ML) models has been achieved not by infusing process-based assumptions into the models, but by conducting extensive training of the models with large quantities of a priori data.

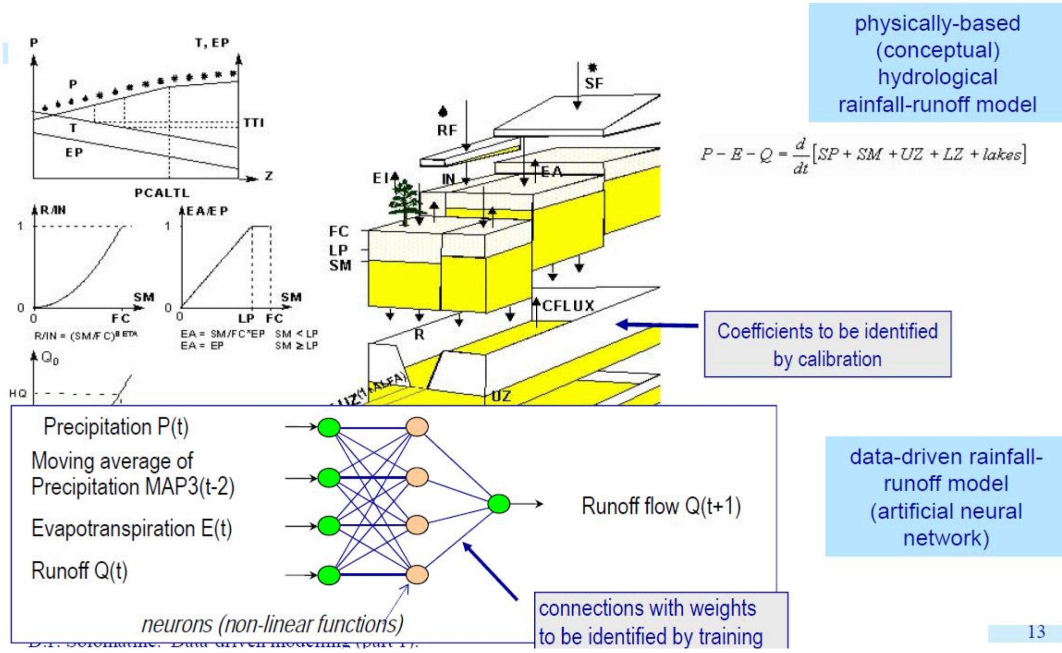


Figure 7. Data-driven vs physically based and conceptual models.

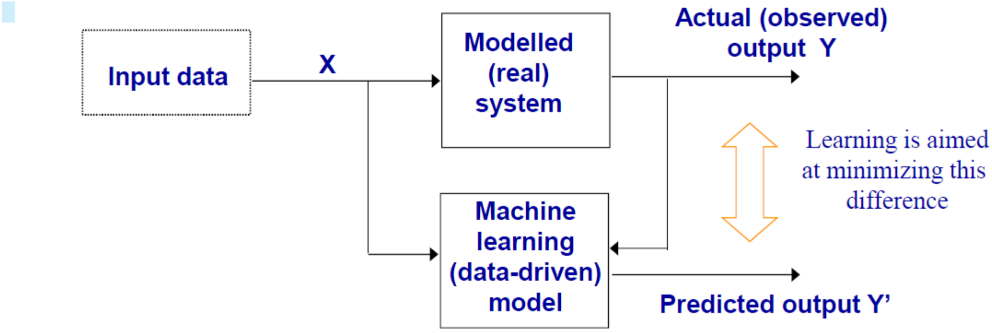


Figure 8. Machine learning (data-driven) model: aim is to minimise difference between actual and predicted output.

How does a data-drive model (DDM) function? (see Figure 8)

- DDM learns the target function Y = f(x) describing how the real system behaves
- Learning = process of minimising the difference between observed data and model output. X and Y may be non-numeric.
- After learning, being fed with the new inputs, the DDM can generate output close to what the real system would generate

An example is a linear regression model, with the independent variable X (input) and dependent variable Y (output), shown in Figure 9. Linear regression describes the observed relationship with a certain coefficient of determination R^2. The parameters a_1 and a_2 are unknown and are found by feeding the model with data and solving an optimisation problem (training). The model then predicts output for the new input without actual knowledge of what drives Y.

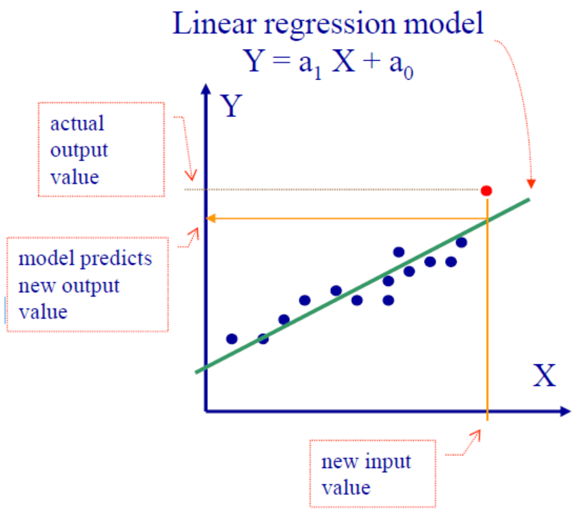


Figure 9. A simple data driven model: linear regression model.

Another example is a non-linear regression model, where for example rainfall, runoff and the autocorrelation between the runoff at different time steps can be utilised (Figure 10).

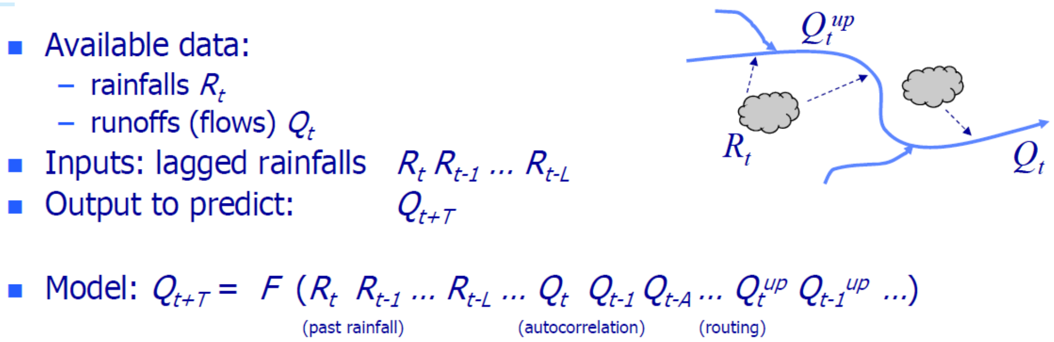


Figure 10. A more comprehensive regression model.

The challenge for the modeller is to identify the appropriate autocorrelator lags. These lags embody the physical properties of the catchment. The information must then be used to develop the non-linear regression function F.

Various stochastic approaches can be applied to estimate model parameters, or the uncertainty related to a given model (Section 4.2). Similarly, numerous stochastic models have been developed to generate meteorological scenarios that are subsequently used as input to deterministic hydrological models. Deterministic and stochastic models are therefore increasingly often used together. In this case, the deterministic model makes it possible to simulate identifiable physical phenomena and the stochastic model considers phenomena of a predominantly random character (e.g., precipitation) as well as the uncertainty inherent to parameter estimation.

Event-based and continuous-simulation models

Event-based models are used to simulate selected hydrological events without considering periods between events. They are mainly developed to simulate the transformation of rainfall into runoff for flood discharge forecasting or prediction. Use of an event-based model requires estimation of the initial conditions the simulation for each event considered. This represents one of the main difficulties of event-based approaches. When simulating the design flood, average initial conditions can be used. An example flood hydrograph with its different components is shown in Figure 11.

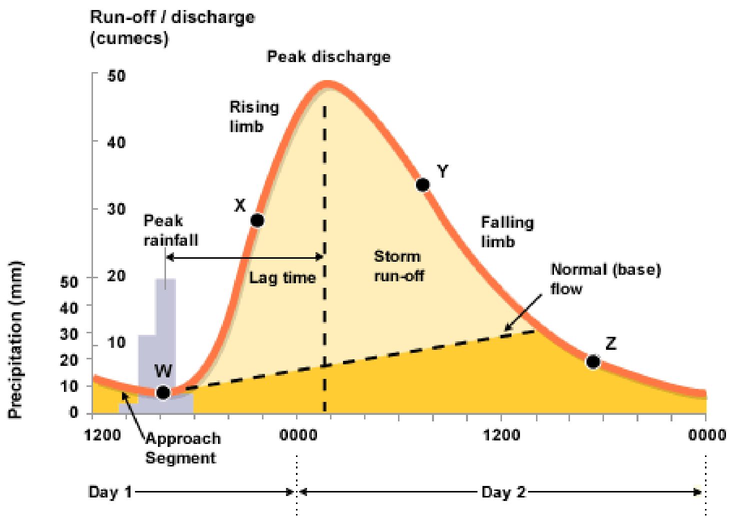


Figure 11. Event-based modelling.

Continuous-simulation models are used to simulate on a continuous basis the hydrological behaviour of the basin. The simulation can be run over long time periods covering a variety of hydrometeorological situations extending from low flows to flood discharges (see Figure 12). These models must consider all the processes significantly influencing the response of the drainage basin and not only the processes implied by particular hydrological phenomenon (e.g., the generation of floods). They also require estimation of the initial conditions. However, the influence of initial conditions on the expected results becomes negligible after a certain time (Section 5).

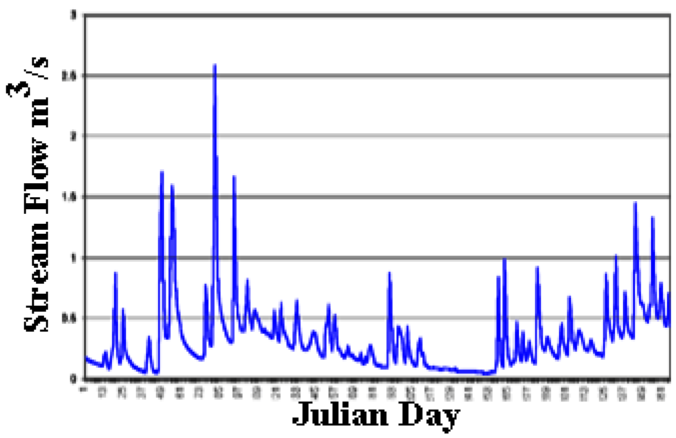


Figure 12. Continuous hydrological modelling.

Engineers use hydrological models to tackle a wide range of challenges. Models such as the original HEC-1 Flood Hydrograph Package (HEC, 1981), the revised HEC-HMS (HEC, 1998, 2006, 2010, 2016) are widely used for event-based floodplain analysis, and also for continuous simulations. To address urban hydrology challenges, the Storm Water

Management Model (SWMM) was developed in 1971 for the EPA to address in detail the quantity and quality variations in urban runoff (Metcalf et al., 1971). The SWMM model can be used for storm event or continuous simulation and has been through a number of updates and improvements over the years (Huber and Dickinson, 1988; SWMM5, 2005; Rossman, 2010). This model is considered one of the most comprehensive for simulating urban runoff dynamics, drainage design, hydraulic analysis, and nonpoint source water quality. With more demands from end users, the SWMM model was enhanced into a commercial modelling package—XP-SWMM (eXPert Stormwater and Wastewater Management Model) by XP Software (2010). Besides all the capabilities of SWMM, the XP-SWMM model has a convenient graphical user interface and the ability of linking the 1-D flow simulation to a 2-D overland flow model to generate floodplain maps. As an important addition to the water resources modelling field, the Danish Hydraulic Institute (DHI) Water and Environment, Inc., provides a whole suite of hydrologic and hydraulic modelling tools (MIKE URBAN, MIKE FLOOD, etc., now combined in MIKE+) to address engineering problems of rainfall–runoff, pipe flows, overland flows, water quality, and sediment transport. For instance, the MIKE URBAN model enables users to simulate water, wastewater, and storm water in one integrated GIS-based framework that is powered by ESRI's ArcObjects components (DHI, 2011). The basic module of MIKE URBAN - Model Manager incorporates the U.S. EPA's fully dynamic storm water and wastewater network modelling package (SWMM 5) and the standard distribution network model (EPANET) to simulate hydraulics and water quality. In addition, the 2-D flow simulation module provides an easy and flexible tool for computing and visualizing the flows above and below ground in one fully integrated dynamic simulation.

1.4 Approach for the Development of a Hydrologic Model

In any modelling approach, it is essential to first identify the modelling objective and framework, i.e., the hydrosystem and the nature of the hydrological phenomena to be modelled, the data available to describe the physical medium and the hydrometeorological variables of interest. This conditions the main hydrological processes to be considered, the possible spatial and temporal resolutions and the choice of realistic concepts and reasonable simplifications.

When the modelling framework has been defined, the general approach to be followed when developing a model follows three steps:

- Model construction
- Model calibration
- Model evaluation, implying in particular its validation

The principles of these three steps are presented below. They will be described in more detail in Section 2, 3 and 4.

Data preparation is also an important step in the development of the model. It is often carried out in parallel with model construction. It includes the mapping and preliminary processing of hydrometeorological data. Mapping aspects concern all the data required to describe the physical medium. These data are often used for the discretisation of the drainage basin or for the estimation of certain model parameters (Section 3). Data preparation also involves compiling valid hydrometeorological time series. These time series are then used either as model input or for the model calibration and validation steps.

Once all these steps have been carried out and the initial and boundary conditions specified, the model can finally be used for the targeted hydrological application.

Model construction principles

To build the structure of the model, the hydrosystem must be schematized and the mathematical and numerical problems related to its spatial discretisation and to the concepts chosen to represent the physical medium and processes must be formulated. Model construction is necessarily accompanied by the development of the computer program that will be used to solve the associated mathematical and numerical problems. The construction of the model and development of the corresponding computer program is often based on existing modelling system. In this case, the modelling concepts along with the numerical tools required to solve the mathematical problem are reused. On the other hand, the schematization of the physical medium and its spatial discretisation in particular must in principle be adapted to the hydrosystem targeted by the study.

Model calibration principles

The objective of model calibration is to: 1) optimise the structure of the model and in particular 2) estimate the parameters appearing in each of the model equations, in each of the spatial discretisation elements. Some of these parameters can sometimes be derived from measured data or thematic maps available for the drainage basin. Most of the parameters must however be adjusted by calibration. Calibration is carried out by comparing, on the basis of selected criteria and for different hydrological variables, observed data and model output data obtained by simulation.

Model evaluation principles

Evaluation concerns the different quality criteria that must be satisfied by the model. The model must essentially be valid, parsimonious, robust and sensitive.

- Validity signifies that the representation of the physical medium and processes that occur within it is possible or acceptable. Validation is generally carried out by comparing, on the basis of selected criteria and for different hydrological variables, observed data and model output data obtained by simulation. This validation is carried out on a dataset that is different from the one used for model calibration. Validation concerns the structure selected to describe the physical medium and processes as well as the values assigned to the different model parameters.
- Parsimony refers to the complexity of the model. A model is parsimonious if it has few parameters and requires a limited quantity of input data. Parsimony is directly related to the choices made concerning the structure of the model.
- Robustness and sensitivity imply that the results obtained by simulation with the model: 1) do not diverge when the values of input variables are slightly modified, e.g., due to errors and uncertainties that are associated with them, but 2) are sensitive to variations of the factors for which the effects are to be estimated.

The approach for developing a hydrological model is summarised in Figure 13.

2 REPRESENTING THE PHYSICAL MEDIUM AND PROCESSES

Whatever the modelling objective, construction of a hydrological model is always based on the choice or design of its structure, i.e., the representation of the processes to be modelled and the spatial representation of the drainage basin.

2.1 Constructing a Model Two main approaches can be followed when constructing a hydrological model. The first approach is often referred to as the top-down approach and the modelling objectives in this case are limited to a few or often only a single objective. The model is developed to reproduce the measured hydrological response of the studied system in the simplest manner possible. The approach is inductive and systemic. Only the most important processes for the given objective are considered. The representation of the processes or the geometry of the drainage basin can be gradually refined as new observations become available.

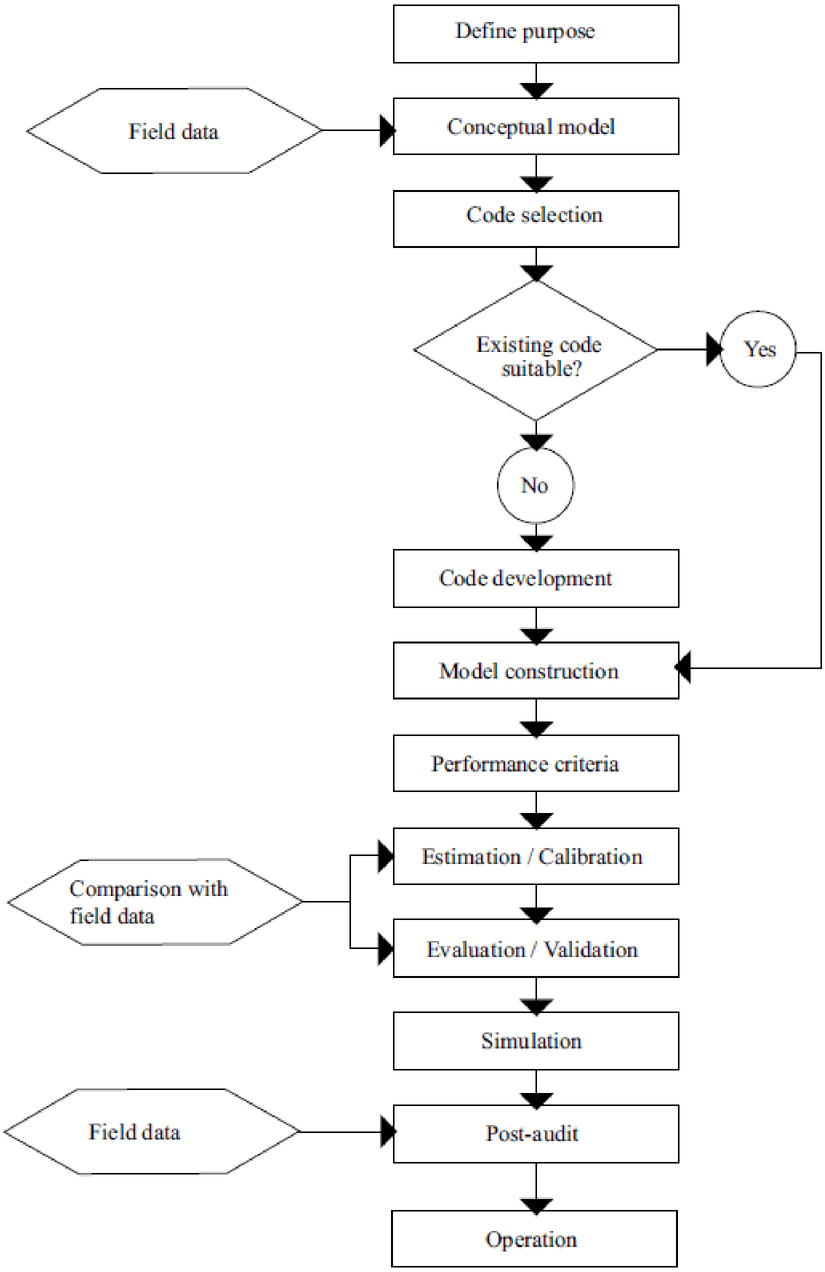
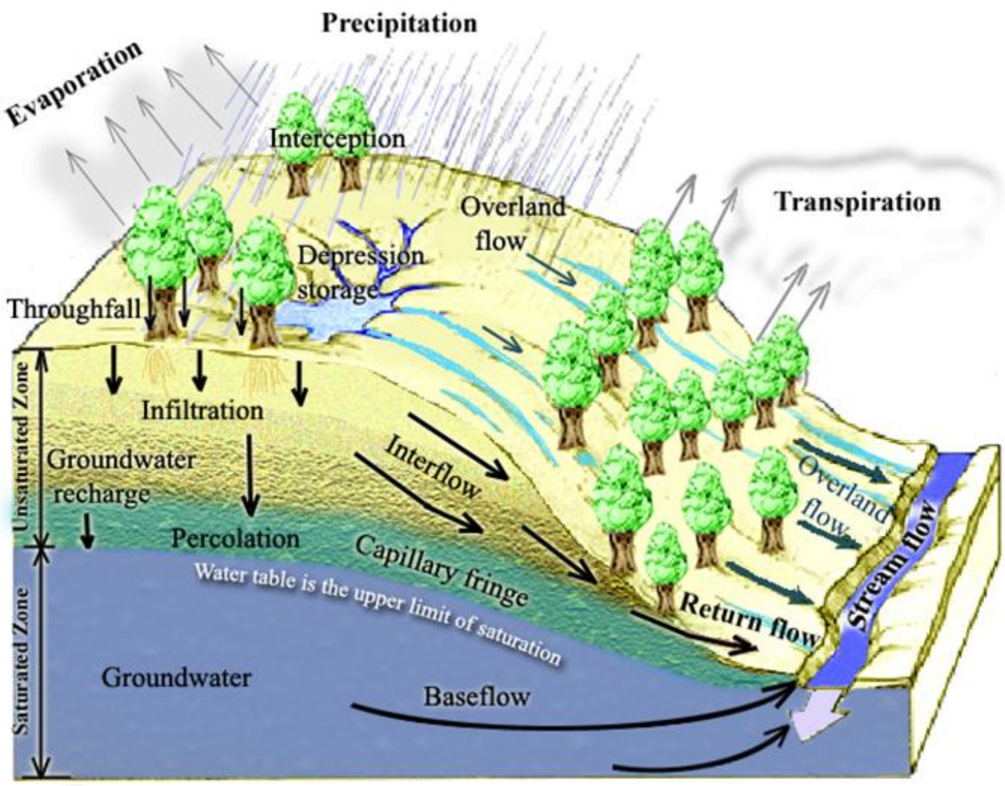


Figure 13. Illustration of the approach used to develop a hydrological model (modified from Refsgaard, 1997).

This approach is often used in operational hydrology. It generally leads to the development of models that are relatively parsimonious and efficient for the targeted objective.
The second approach is referred to as the bottom-up approach and the modelling objectives in this case are not pre-defined. It is assumed that the behaviour of the overall hydrosystem can be simulated properly as long as the processes governing the behaviour of the hydrosystem are sufficiently known and these processes and their interdependence are correctly represented. The bottom-up approach is of a more scientific nature and mainly deductive. It leads to models with complex structures and that are difficult to assess given that the required experimental data often do not exist. This perhaps explains why these models seldom leave the laboratories in which they are developed. To be used in operational hydrology, they often require a number of simplifications. Such models can be easily simplified *a posteriori* if one of the processes initially taken into account proves to have no significant influence on the targeted hydrological behaviour.
Whatever the approach used, the construction of a hydrological model is ideally based on: 1) observation and experimentation of the targeted hydrological phenomenon and associated hydrological processes for the considered hydrosystem, 2) formulation of the assumptions required to represent the physical medium and these processes and 3) testing of these assumptions, i.e., their validation or invalidation. These three successive steps often make up an iterative and a priori infinite cycle aimed at improving the mathematical description and knowledge of some or all of the components of the hydrological cycle.

2.2 Representing the Processes

The different approaches used to represent the processes of the hydrological system (see Figure 14) leads to empirical, conceptual or physically-based models (see Figure 5).



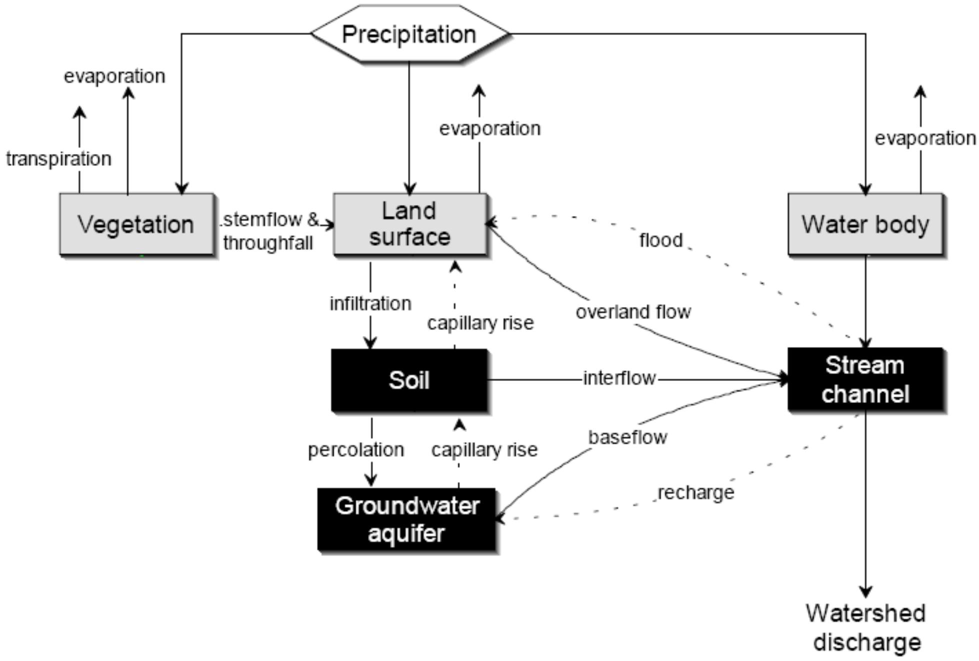


Figure 14.Main processes in the hydrological system (top) and as represented in HEC-HMS (bottom).

The underlying principles of these models will now be reviewed. Whatever the type of representation, the models are generally made up of different modules, each designed to describe, with a varying degree of independence, the main processes explaining the behaviour of the drainage basin. The modules most commonly encountered are presented further on in this section under "Typical components of a hydrological model".

Exercise 1

Construct a hydrological model concept by selecting an adequate process representation for the processes shown in Figure 14. Make sure to define the purpose of the hydrological model first. How would your model concept change if the purpose changes? List all model parameters and discuss potential sources for these model parameters. Which data are needed to apply the model that you have constructed?

Empirical models

Empirical models are based on the relationships observed between the inputs and outputs of the considered hydrosystem. They express the relationship between the input and output variables of the system (e.g., rainfall-runoff relationship) using a set of equations developed and adjusted using data obtained for the system. An empirical model is not designed to describe the causes of the considered hydrological phenomenon nor to explain the operation of the hydrosystem. The hydrosystem is considered to be a black box. Examples of empirical models include the SCS-CN model used to estimate the net rainfall resulting from a given total rainfall and the unit hydrograph model used to estimate the hydrograph of the runoff resulting from this net rainfall. ARX type models can also be classed in this category.

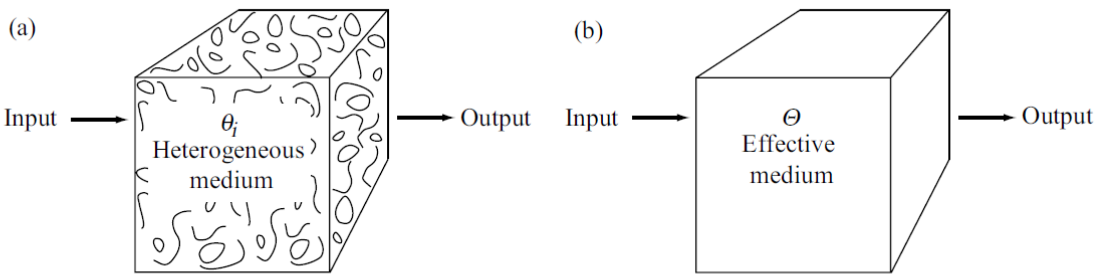
Physically-based models

Physically-based models represent the hydrological operation of the hydrosystem by coupling different sub-models, each dedicated to certain hydrological processes. They are generally based on a fi ne spatial discretisation of the physical medium (Section 2.3). Whatever the nature and spatial resolution of this discretisation, note that the spatial variability of the physical medium and processes can only be explicitly described for spatial scales that are larger than those of the discretisation. For smaller scales, it must be described in a conceptual manner -which is not frequently the case (Addendum 2).

The development of physically-based models on the drainage basin scale goes back to the 1980s. One of the first physically-based models is the well-known European Hydrological System, often referred to as SHE for "Système Hydrologique Européen" (Abbott *et al*., 1986a,b). The underlying principles of this model are presented in the Appendix. Several other models of this type have since been developed and are the subject of numerous publications (Noilhan and Planton, 1989; Habets *et al*., 2008; Liang *et al*., 1994; Wood *et al*., 1992).

Addendum 2 Representation of the internal variability within the discretisation elements of physically-based models

The choice of the spatial resolution of a model determines the respective parts of the spatial variability that can be represented explicitly and implicitly. In practice, three main approaches can be used to account for the variability within the discretisation elements of a given model. The first consists in using so-called effective parameters in the equations describing the processes. These parameters are supposed to make it possible to simulate the overall behaviour of the considered discretisation element. An "effective" conductivity is for example often used to describe saturated flow in heterogeneous porous media (Figure 15). The difficulty of this approach lies in the estimation of the values of these effective parameters. For certain types of heterogeneity, an explicit relationship can be found between the value of the effective parameter and the spatial variability of the parameter within the discretisation element. The method used most often to estimate the effective parameters is empirical (Figure 15), yet often implying the use of the calibration procedure. The use of effective parameters does not often provide full satisfaction, in particular if the processes are non-linear and/or conditioned by one or more operating thresholds (e.g., infiltration processes).



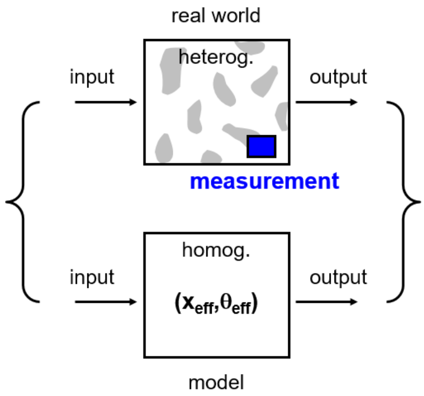


Figure 15. Spatial variability of the physical properties of the medium (e.g., hydraulic conductivity) within a given discretisation element (e.g., heterogeneous porous medium) and the effective parameter used in the model representation (e.g., equivalent hydraulic conductivity).

The second approach consists in using a statistical representation of the physical property considered. The statistical distribution of the property is used to define a set number of classes of behaviour and the model is applied to each of them. This is the same principle as that of the distributed models presented below in Section 2.3. In certain cases, the distribution can be chosen to obtain an analytical solution of the targeted behaviour, which simplifies the application of the model.

The third approach consists in proposing a parameterization for the targeted process, i.e., a conceptual model that can be used to represent the average behaviour at the scale of the considered element. This parameterization can be derived from the corresponding fundamental physical laws. For common hydrological applications, free-surface flow can be modelled for example using the Barré de Saint-Venant model. This model, derived from the Navier-Stokes equations, considers free-surface flows in an integrated manner over a flow section (the Navier-Stokes equations define the basic physical equations of hydrodynamics).

The equations that define the parameterization are not necessarily related to the equations that describe the processes at the scale of these processes. A classical parameterisation is the Darcy model used to describe the mean flow velocity within saturated porous media. The associated parameter is a hydraulic conductivity of the spatial element considered. One difficulty of this approach lies in the fact that adequate parameterisation depends in principle on the spatial and temporal resolutions used.

Conceptual models

Conceptual hydrological models are designed to represent the main hydrological processes in a reasonable manner without requiring parameterisation of the physical laws that govern them. The representation is conceptual in that it is based on the hydrologist's perception of the hydrological behaviour of the drainage basin. This perception is derived from the hydrologist's expertise and theoretical, empirical and/or intuitive understanding of the operation of the studied hydrosystem. For surface runoff, for example, one modelling concept assimilates the drainage basin with a rectangular inclined plane of constant slope. For such a configuration, it is possible to derive runoff equations in a simple manner. Most conceptual models that consider the drainage basin can be represented by a combination of conceptual reservoirs linked together by various exchange relationships (Addendum 3). This is for example the case for the SWMM model with 8 reservoirs and 27 parameters developed at Stanford University for urban drainage basins (Crawford and Linsley, 1966) and the GR3J model with 3 reservoirs and 3 parameters developed for rural drainage basins in France by Cemagref (Edijatno and Michel, 1989; see also Appendix). These reservoirs are sometimes considered to represent the main hydrologic compartments of the basin.

Addendum 3 Conceptual reservoir models

A wide variety of reservoir models are used in hydrological modelling (see the inventory proposed by Perrin, 2000). A few examples are shown in Figure 16. Reservoir models, also called "capacitive models", can first be differentiated by the way they describe the space. They are most often lumped but can also be distributed or semi-distributed. Reservoir models can also be differentiated by the degree of simplification used to represent the processes involved and also by the number and type of reservoirs and the way in which the reservoirs are organised to represent these processes.

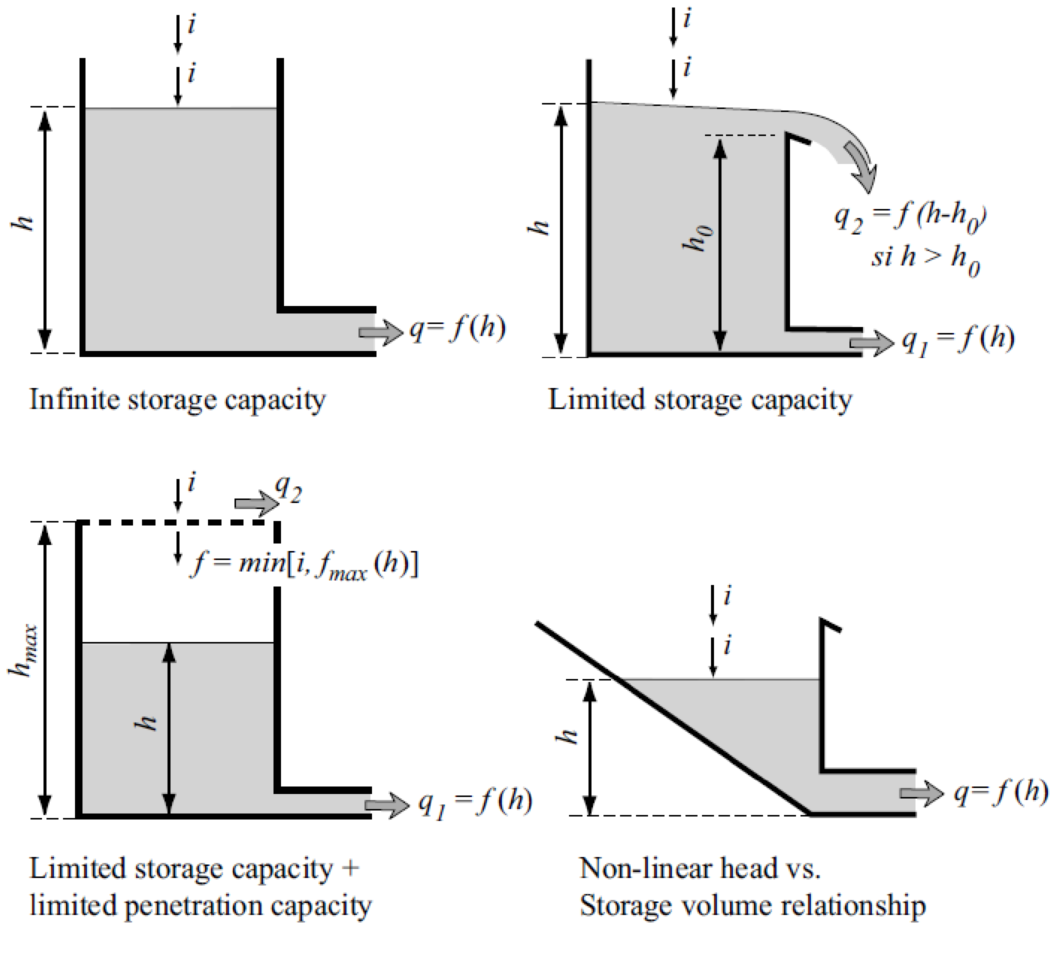


Figure 16. Examples of conceptual reservoirs used in the structures of reservoir-type hydrological models.

Whatever the reservoir implied in the structure of a reservoir model, its behaviour is governed by a system of three types of equations:

- the continuity equation that expresses mass conservation of water in the reservoir,

- a storage equation that defines the volume of water stored in the reservoir for a given water level,
- various discharge equations expressing the inflows and outflows to and from other compartments of the modelled system as a function of the water level in the reservoir.

The variations of the water volume stored in a reservoir depend on the volumes of inflows and outflows. Inflows can be of many types (e.g., direct precipitation or surface runoff). The same is true for outflows (e.g., percolation or evapotranspiration).

A reservoir can have a linear or non-linear behaviour. Non-linearity often results from the non-linear character of one or more of the reservoir discharge equations, in particular for the outflows. For example, the outflow may be expressed as an exponential function of the storage or may depend on the value of certain water-level thresholds.

A reservoir can have different types of outlet works. For example, low-level outlets and bottom sluices provide continuous outflows that vary with time as long as the reservoir is not empty. On the other hand, overflow weirs produce outflows only when the water in the reservoir has exceeded a certain level.

Typical components of a hydrological model

When designed to simulate the hydrological behaviour of hydrosystems, hydrological models, whether empirical, conceptual or physically-based, combine different modules or sub-models, each designed to represent one or more of the hydrological processes considered to be important.

Although sometimes interdependent, these modules are often independent, i.e., the output of one module is the input to another module. Depending on the hydrological model, some of these sub-models may not be considered or may be considered in an extremely simplified manner. Typical modules of a hydrological model are presented in Figure 17. The first module is generally designed to transform available meteorological data into input data with the spatial and temporal resolution required by the simulation model. This module is often used to estimate meteorological forcing data for each element of the selected spatial discretisation. It makes use of data combination and spatialisation techniques.

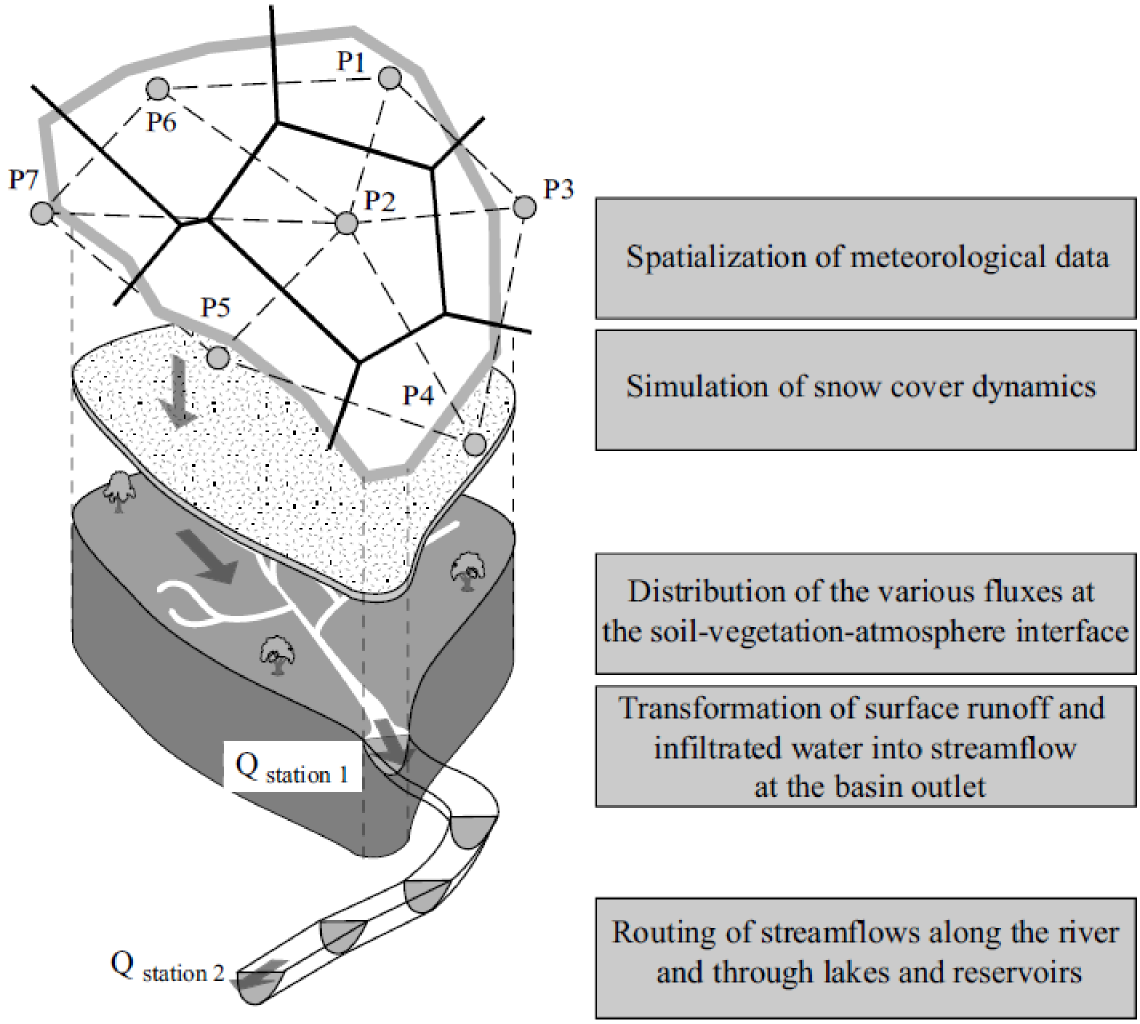


Figure 17. Typical modules of a hydrological model.

The second module generally focuses on the processes that determine the distribution of the various water fluxes at the soil-vegetation-atmosphere interface. In varying degrees of detail, this module represents interception by vegetation, depression storage, evaporation from water bodies, evapotranspiration from vegetal cover and infiltration. In hydrological modelling, this module can be associated with the estimation of rainfall excess, i.e., the net rainfall contributing to surface runoff.

The rainfall-excess module is often accompanied by a snowmelt module that treats phenomena related to the presence of solid precipitation, snow cover and ice surfaces. The way that surface runoff (resulting from the net rainfall calculated by the rainfall excess and, if applicable, the snowmelt modules) generates streamflow at the outlet of a given basin is modelled by another module, often based on what is referred to as the rainfall-runoff transformation function.

Another module can be used to handle precipitation that has infiltrated in the soil and that percolates downward into the water table before entering the hydrographic network via base flow. Different approaches can be followed to represent the related processes. They are either hydrological, similar to those used for rainfall-runoff analysis, or hydrogeological.

The propagation of streamflows along a river and through lakes, artificial reservoirs and hydraulic structures is handled by another complementary module often based on what hydrologists refer to as the "routing" function.
Note that the modules related to rainfall excess and snowmelt deal with predominantly vertical processes while the modules related to the rainfall-runoff relationship and routing deal with predominantly lateral processes.

2.3 Representation of the Physical Medium and Spatial Discretisation

As already mentioned, lumped, distributed, and semi-distributed models can be distinguished by the way they represent the physical medium and, in particular, by the spatial unit in which the equations describing the processes are solved (Figure 18).

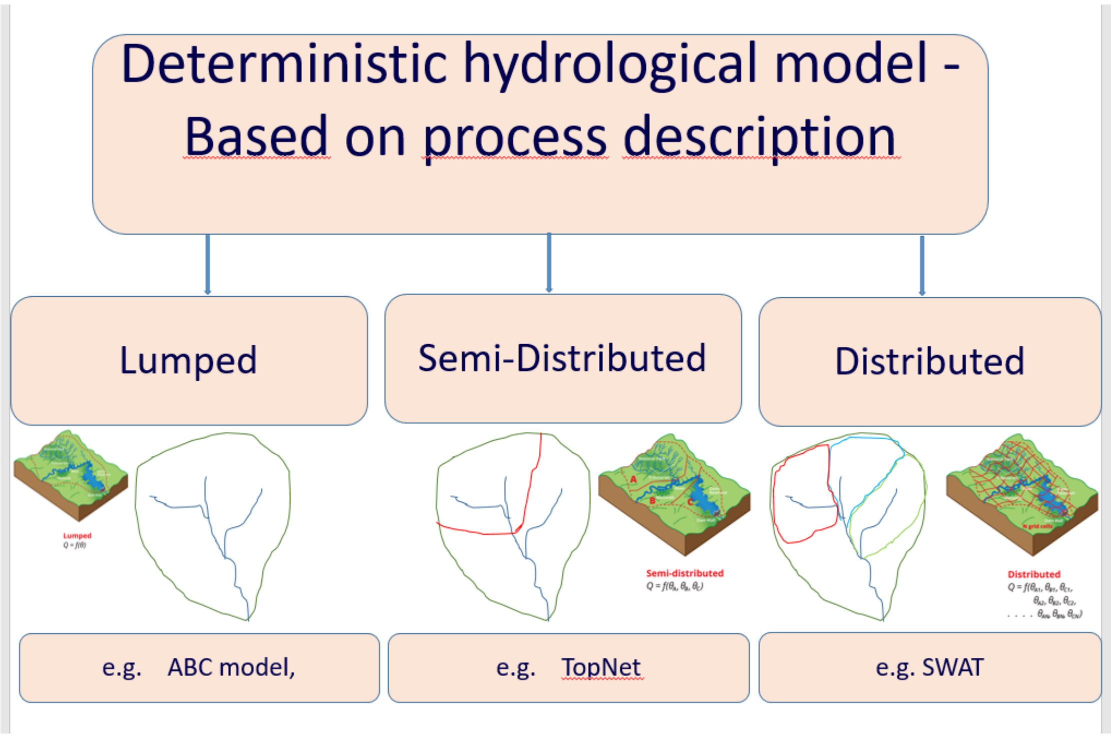


Figure 18. Process description in the spatial unit: Lumped, semi-distributed and distributed.

Lumped hydrological models describe the drainage basin as a single hydrological unit. The spatial variability of the hydrological processes considered to describe its behaviour is therefore not taken into account. The GR3J model developed in France and a number of other models are of this type (Edijatno and Michel, 1989) (see Appendix).
The drainage basin can also be represented in a discretized manner by dividing it up into distinct spatial units. Depending on the spatial discretisation and the nature of the relationships between the discretisation elements, the models are referred to as distributed, semi-distributed or distributed. These types of discretisation are described below. A more detailed presentation can be found in the work of Grayson and Blöschl (2000) or Indarto (2002).

Distributed models

Distributed models attempt to represent the physical medium and in particular the surface of the drainage basin using a relatively fine spatial discretisation that respects the spatial

organisation of the discretisation elements. Most of them have been derived from work aimed at explaining the hydrological response of a considered basin on the basis of its geomorphology (Rodriguez-Iturbe and Valdes, 1979; Bérod *et al.*, 1995; Rodriguez *et al.*, 2003).
Distributed models simulate the evolution of the system at every point of this discretisation along with the exchanges of water with neighbouring elements. Each element of the discretisation is in principle characterised by its own parameters. Given the determining influence of topography on the displacement of water at the surface of a drainage basin, a discretisation is mainly dictated by the manner in which the topography is described.
Different types of discretisations are possible (

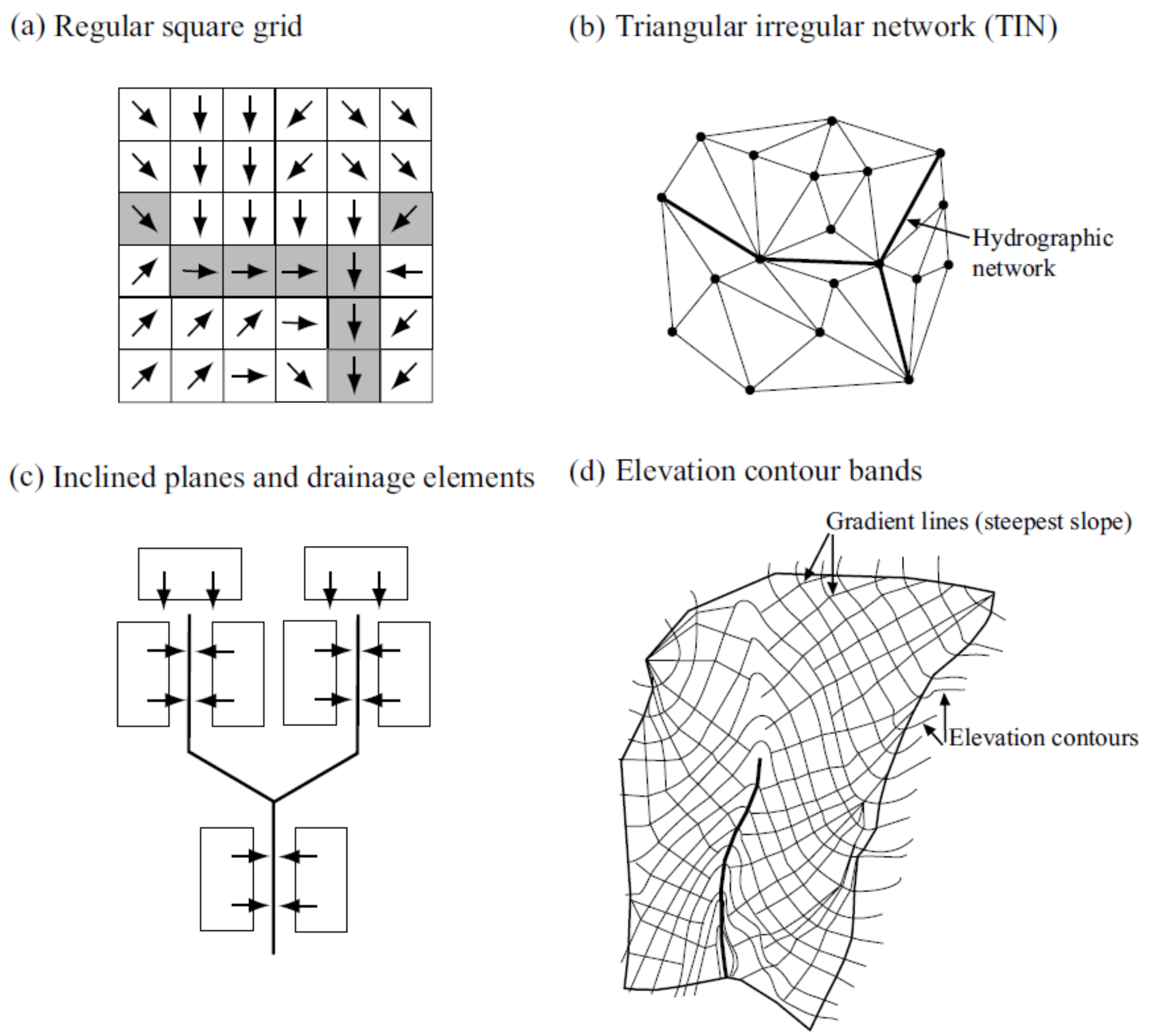


Figure 19). Some are based on elevation contours. Depending on whether the boundaries of the discretisation elements are parallel or orthogonal to the elevation contours, these discretisations delimit bands of equal elevation (e.g., Topog model, Vertessy *et al.*, 1993) or stream tubes (e.g., Thales model, Grayson *et al.*, 1995). Other discretisations describe the drainage basin as a cascade of inclined planes (hill slopes) and drainage elements (channels) organised in a tree structure (e.g., Socont model, Bérod *et al.*, 1995; Kineros model, Smith *et al.*, 1995). The most widely used discretisations are based on spatially referenced digital

elevation data. Given that the raster format is the classical representation mode of Digital Elevation Models (DEM), the physical medium is most often spatially discretized into a regular grid of square cells (e.g., MODCOU, Ledoux *et al*., 1989).

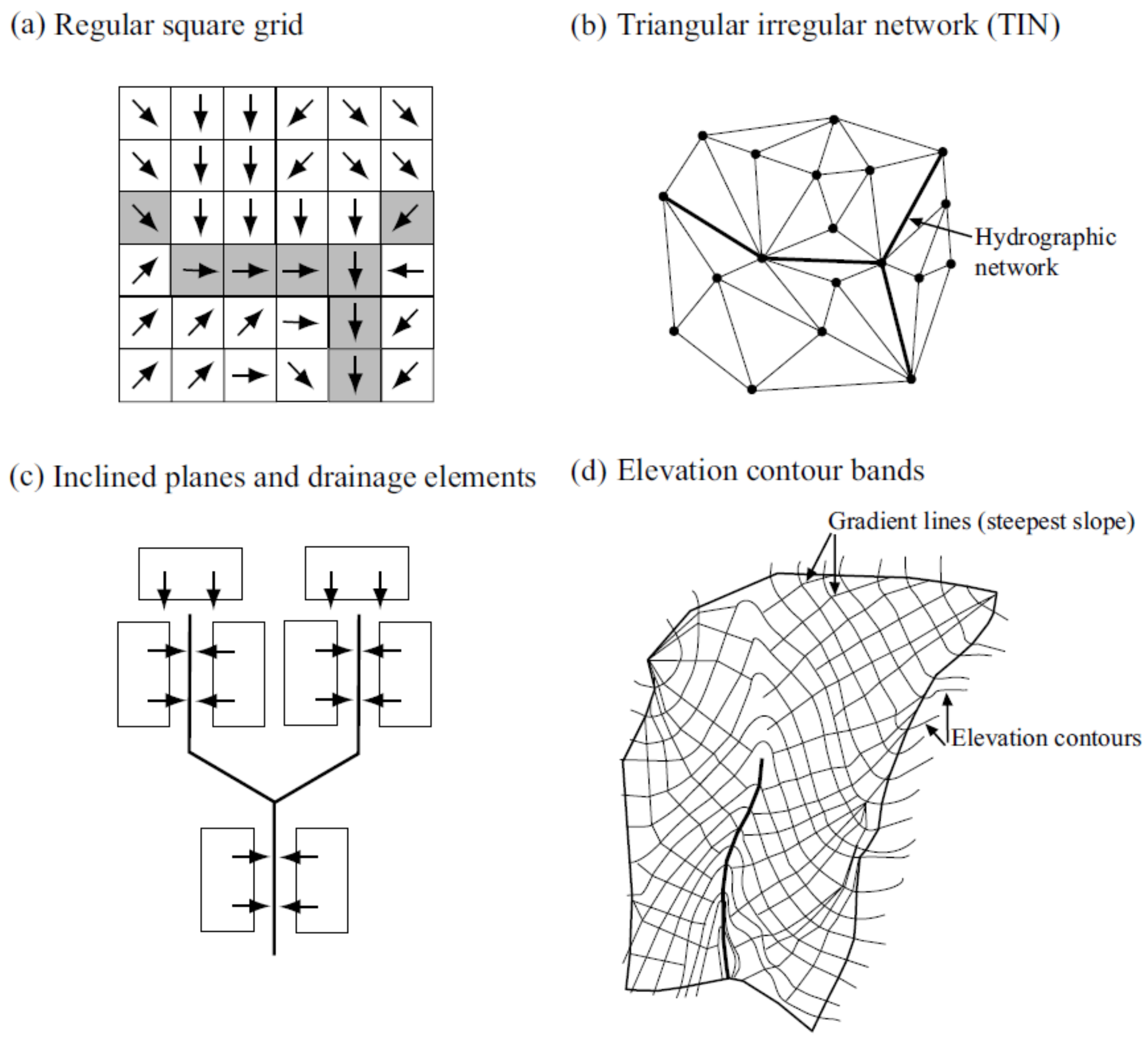


Figure 19. Examples of possible discretisation schemes for hydrological models (modified from Grayson et al.*, 2000).*

One of the models the most representative of this type of discretisation is the SHE model (Abott *et al*., 1986a,b) (see Appendix). The modelled space can also be divided up into irregular triangular grid cells on the basis of a TIN (Triangular Irregular Network) model but this method is less widely used (tRIBS physically-based model, Ivanov *et al*., 2004).
These different discretisation schemes each have their advantages and disadvantages. For example, a raster-type discretisation can be used to identify the drainage network in a simple manner and makes it easier to develop algorithms to solve the associated equations. It however imposes a sub-optimal description of the basin boundaries, the layout of the hydrographic network and the spatial variability of the characteristics of the physical medium. On the other hand, a TIN-type spatial discretisation generally requires fewer points to provide

a very satisfactory representation of the topography and spatial variability of the characteristics of the medium. At the same time, however, it requires more complicated algorithms to solve the associated equations.

Semi-distributed models

For a semi-distributed model, discretisation consists in delimiting hydrological units for which it is assumed that the behaviour can be reasonably described by a lumped or distributed hydrological model. These hydrological units, generally assumed to function independently, are emptied one into another and/or into the hydrographic network that drains them from upstream to downstream. Many discretisation methods and criteria are possible.

One possible type of discretisation consists in delimiting Relatively Homogeneous Hydrological Units (RHHUs), i.e., zones in which the hydrological processes are comparable both in terms of their nature and relative impact. These hydrological units are generally identified on the basis of the most pertinent physiographic characteristics of the basin with respect to the key processes. The characteristics often used are those of the hydrographic network, topography (elevation, orientation, influencing the meteorological conditions, the drainage network), land use (e.g., characteristics of the vegetal cover, type of urbanization) and pedological and geological characteristics of the soils and subsoils (Amerman, 1965). The CEQUEAU model (Girard *et al*., 1972) uses this type of discretisation (see Appendix). In general, RHHUs are also determined in such a way that meteorological forcing is not excessively heterogeneous within each of the RHHUs. In mountainous regions, this often leads for example to RHHUs based on elevation bands.

For very large drainage basins, the hydrological units are generally sub-basins of the complete drainage basin. Most of the approaches used to delimit these sub-basins are empirical, based on the topography, structure of the hydrographic network and locations of the key points in the network. The sub-basins can for example be defined as the smallest spatial units in which it is possible to calibrate the hydrological model. In this case, the divisions depend greatly on the distribution of the stream-gage stations over the drainage basin. In another case, the sub-basins can be defined on the basis of the points of the hydrographic network where the model must be able to simulate discharge data. The divisions can also be conditioned by the locations of hydraulic structures that modify the natural behaviour of the basin. This criterion can lead to relatively fine discretisations as is the case in urban environments where artificial drainage elements, impervious areas connected and not connected to the drainage network and special equipment such as stormwater infiltration/retention works must be considered separately.

The different partitioning criteria, based for certain on the homogeneity of the hydrological processes and meteorological forcing and for others on the presence of key points, can be considered in a combined manner. The GSM-SOCONT model (Hingray *et al*., 2006b) has been used in this way to model the Rhone upstream of Lake Geneva (Figure 20).

Distributed models

In a distributed model, it is assumed that the hydrological behaviour of the drainage basin can be explained on the basis of the behaviour of a limited number of distinct hydrological units. These units are chosen to be representative of the main types of hydrological behaviours observed within the basin. They are generally referred to as hydrologically similar response units (HRUs) (Gurtz *et al*., 1999).

The hydrological response of the drainage basin to a given meteorological forcing is simply obtained by a weighted sum of the hydrological responses simulated for the different HRUs.

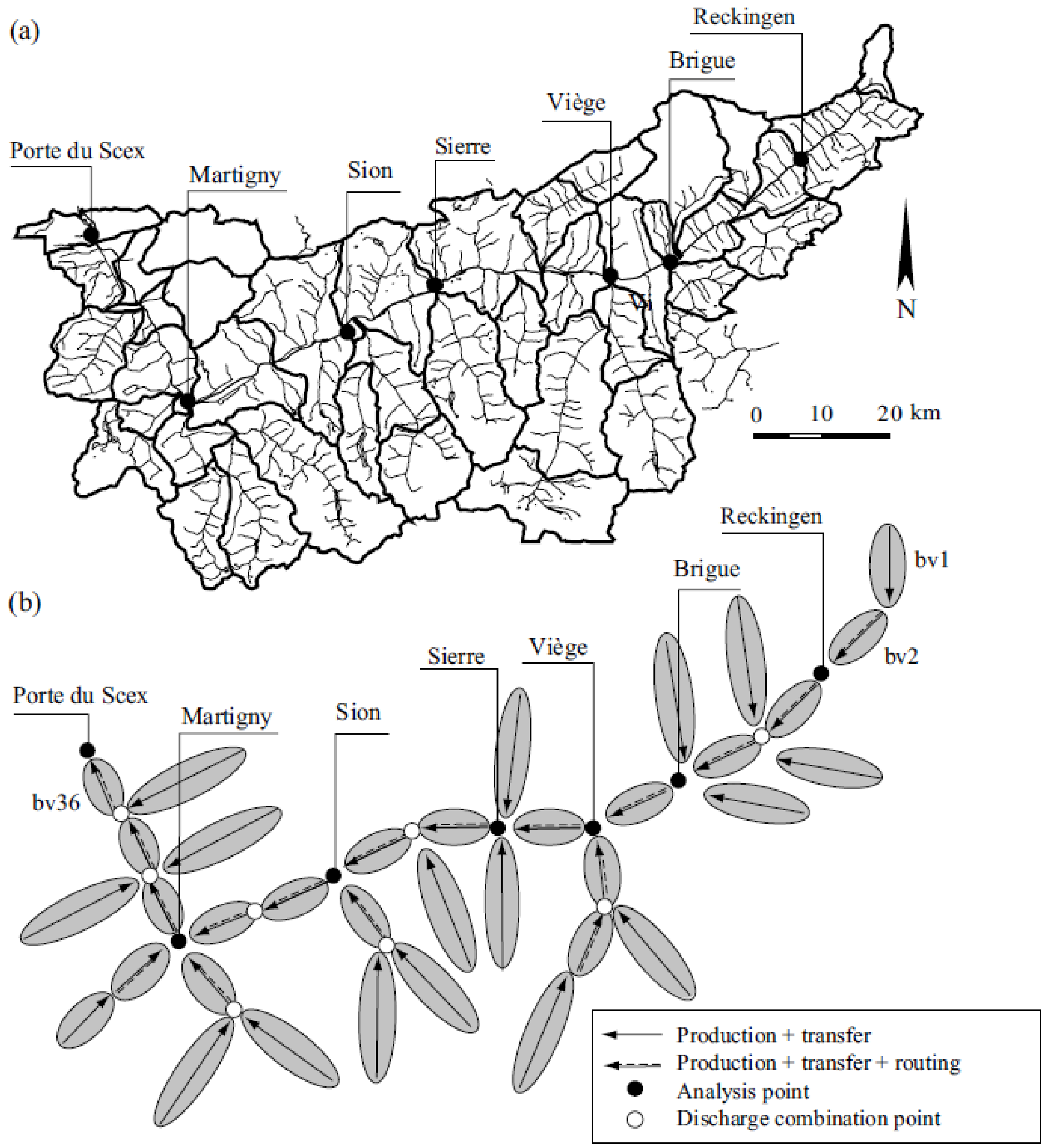


Figure 20. Discretisation of the Rhone drainage basin upstream of Lake Geneva into sub-basins (adapted from Hingray et al.*, 2006b). a) The Rhone drainage basin (area: 5240 km^2 upstream of Porte du Scex), the hydrographic network and the 35 sub-basins selected for discretisation. b) Structure of the hydrological model with: 1) subbasins generating rainfall excess (production) and transforming it into streamflow (transfer) 2) river reaches used to route basin inflows. The discretisation is based on: 1) the natural topographic boundaries of the basins, 2) the points in the network where it is possible to calibrate the model and 3) the points in the network where discharge simulations are required.*

The weight applied to the response of a given HRU is generally proportional to the area occupied by this unit within the drainage basin. This approach reduces the number of calculations required to simulate the behaviour of the basin. Distributed models do not describe predominantly horizontal flows within the drainage basin or do so in an extremely simplified manner (e.g., assuming a permanent equilibrium between the levels of the reservoirs used to represent groundwater storage within each unit). The units are often assumed to be independent of each other and their spatial organisation is frequently ignored. With this approach, each elementary area of the basin must be assigned to one of the HRUs. The physical characteristics used for this are mainly the same as those used to identify relatively homogeneous hydrological units (Section 2.3). They can be used in a combined manner (Figure 21). (beginning of 'not examinable')

In certain models, each elementary area is assigned to an HRU on the basis of a single characteristic. For example, for a drainage basin in a mountainous region, where the hydrological response is highly dependent on snow accumulation and snowmelt processes, assignment is often based on elevation alone. This is for example the principle used for the HBV model (Bergström, 1976; see Appendix). For TOPMODEL (Beven *et al*., 1984), assignment is based on a topographic index calculated at every point in the basin from the local slope and the upstream area drained at the point (see Appendix – note that TOPMODEL was used as basis for the development of TopNet developed and used throughout New Zealand by NIWA - National Institute of Water and Atmospheric Research). When the discretisation is based on a single characteristic, a statistical distribution function is sometimes used to describe the distribution of this characteristic, hence the term "distributed" models.

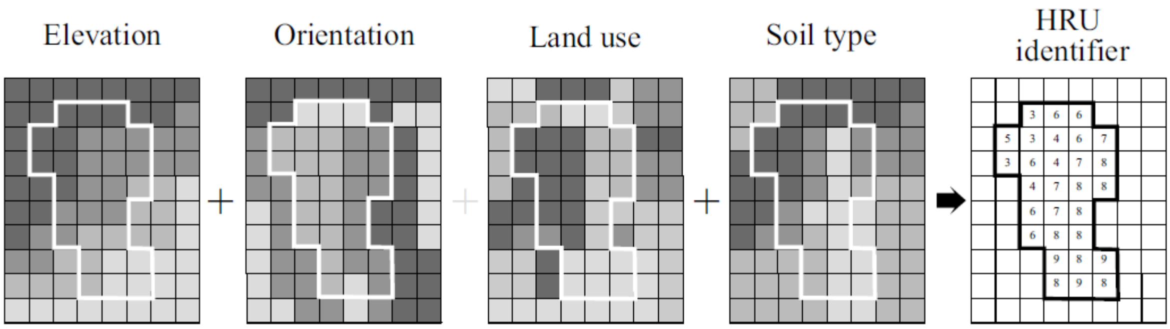


Figure 21. Illustration of HRUs obtained on the basis of different characteristics of the physical medium described in raster format (modified from Zappa, 1999). (end of 'not examinable')

2.4 Compatibility Between Spatial Representations and Processes

A key question when constructing a hydrological model concerns the optimum level of complexity to be targeted. According to Grayson and Blöschl (2000), this depends on modelling objectives, knowledge of the system and the required level of validation. Whatever the case, the level of complexity selected for the different model components must be well-balanced. This is because the pertinence and accuracy that can be achieved by the model is determined by the component with the coarsest representation. In other words, the different levels of complexity used to represent the different processes explaining the behaviour of the hydrosystem must be compatible with each other in the same way as the levels of complexity used to represent the physical medium. They must also be compatible with the data available for model input, calibration and validation. Figure 22 illustrates various degrees of simplification that can be used when representing the geometry of the drainage basin and the transformation of surface runoff into a discharge at the basin outlet.

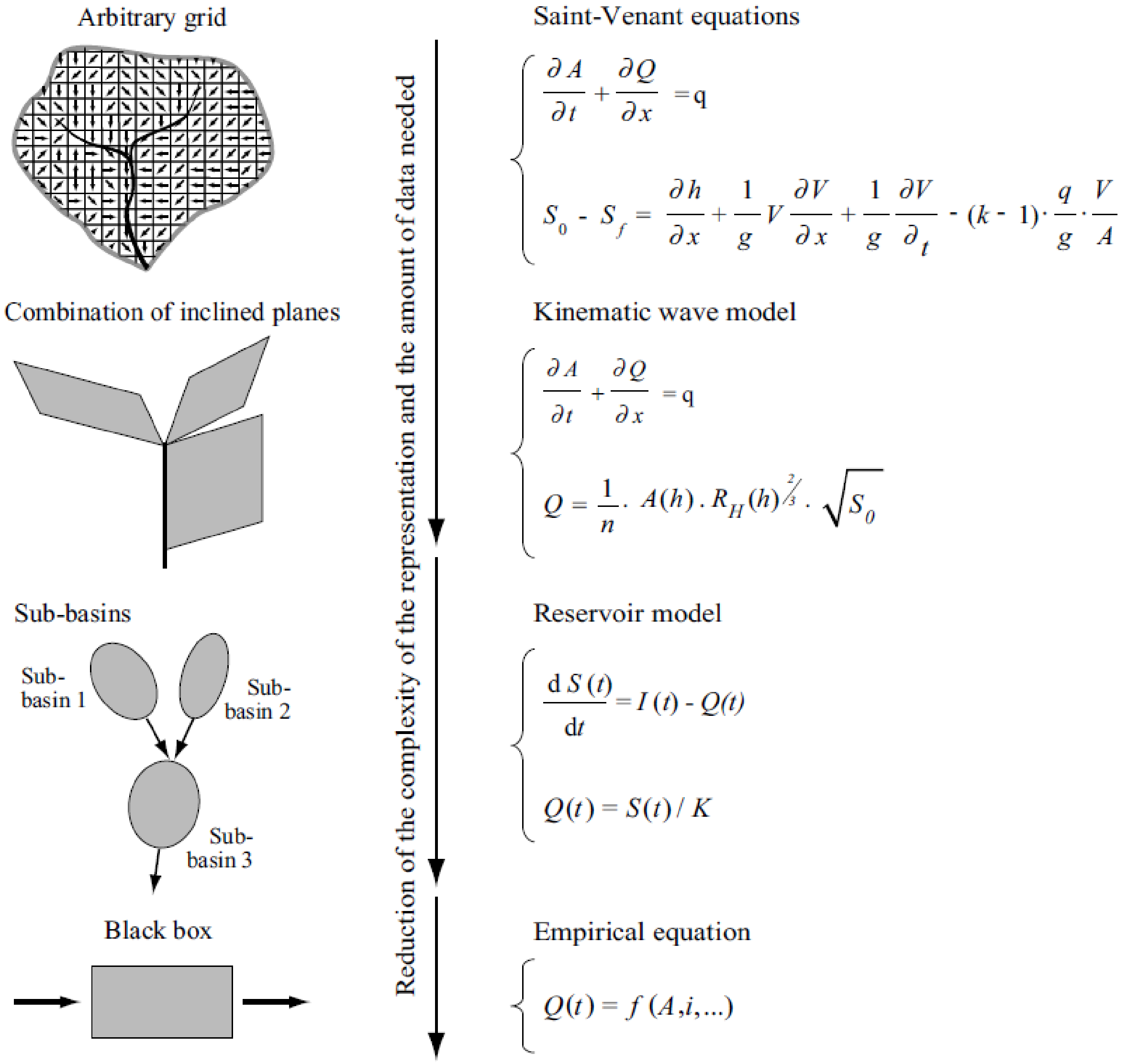


Figure 22. Different degrees of simplification that can be used when representing the geometry of the drainage basin and the corresponding surface runoff.

The targeted level of complexity also depends on the optimal temporal and spatial resolutions for the considered hydrological problem. For a given type of model, the optimal resolutions depend on the available data and the dynamics of the hydrological processes considered. This problem has been relatively well analysed for streamflow routing however this is not the case for modelling of the rainfall-runoff relationship.

The approach proposed by Obled *et al*. (2009), based on analysis of the spatial variability of the precipitation, provides valuable insight into this issue. Their analysis is summarised in Addendum 4 for the interested reader.

Addendum 4 Selecting the temporal and spatial resolution for the modelling of the rainfall-runoff transformation (beginning of 'not examinable')

The objective of the analysis proposed by Obled *et al*. (2009) was to determine the optimum time and space steps for the construction of the parsimonious conceptual model for the simulation of the rainfall-runoff transformation.
In their work, the authors first note that, for a given rain-gage network, the accuracy of the estimation of the precipitation falling on the unit area increases with the duration. On the other hand, the accuracy of the description of the hydrological dynamics of the drainage basin increases as the considered time step decreases. The necessity of limiting uncertainties in the estimation of precipitation therefore leads to selecting the maximum duration that still represents the determining hydrological processes in a reasonable manner.
In process-oriented models, the optimum time step is the smallest time step required to represent all the processes. In certain contexts, this can be a few seconds (e.g., infiltration excess overland flow on steep drainage basins, erosion processes).
In conceptual hydrological models, the required time step depends on the dynamics of the output signal, the discharge. It is longer than for process-oriented models, the drainage basin filtering the high frequencies of the input signal (precipitation) and those of the processes related to the production of rainfall excess in the basin. To accurately represent the shape of the basin discharge response to a given precipitation input, Obled *et al*. (2009) suggest that, in this case, the maximum permissible time step Δt_o is roughly between a fifth and a third of the rise time t_r of the instantaneous unit hydrograph of the basin. Therefore:

$$t_r/5 \leq \Delta t_o \leq t_r/3 \qquad (3)$$

where Δt_o is the time step of the model and consequently the time step used to represent the input signal.
This analysis is based on the spatial correlation of the precipitation fields considered to be the most critical with respect to the generation of floods. Geostatistical analyses of these fields have shown that the decorrelation distance between precipitation depths, i.e., the distance at which the correlation between precipitation depths becomes null, is a function of the time step Δt over which the precipitation has accumulated. In other words, the spatial variability of rainfall increases as the accumulation time decreases. The relationship identified for different regions in France is:

$$a = b * t^c \qquad (4)$$

where a is the decorrelation distance [km], Δt the accumulation time [hours] and b and c constants (b between 18 and 30, c between 1/3 and 1/2).
Using a classical decay model to describe the decreasing correlation with distance (e.g., exponential or spherical), it is possible to evaluate the maximum size for which a point precipitation value reasonably represents the mean precipitation value over a spatial domain centred on the point. For a spherical decay model, 80% of the variance of the mean value on a square with side length $a/2$ is explained by the value at the centre of the square. The explained variance is 72% and 60% for a square with a side length of respectively $2a/3$ and a.
The maximum space step for which it is possible to consider that the rainfall is reasonably uniform over a domain of area $A = (\Delta x)^2$ is therefore roughly between two values:

$$a/2 \leq \Delta x \leq 2a/3 \qquad (5)$$

Given the maximum time step Δt_o that can be used to represent the flood dynamics of the considered basin, equations (4) and (5) can be used to estimate the maximum possible space step Δx or the maximum possible size A of the domain over which the rainfall can be assumed to be reasonably uniform.

$$100\ \Delta t_o \leq A \leq 400\ \Delta t_o \qquad (6)$$

with A in km^2 and Δt_o in hours. This relationship between the maximum possible spatial and temporal resolutions is only valid from a statistical viewpoint for the context targeted by the analysis and under the assumptions of the analysis. Obled *et al*. (2009) discuss the precautions that must be taken when using such an approach.

Expression (6) answers the question concerning the optimal discretisation of the drainage basin. If the area A of the domain is greater than the size of the basin, the basin can be described by a lumped model. If not, it must be divided into sub-basins. The number of sub-basins to be considered can be determined in an iterative manner, the maximum permissible time step decreasing with the number of sub-basins. This is because the rise time of the instantaneous unit hydrograph of the basin increases with the size of the basin. **(end of 'not examinable')**

Exercise 2

Develop a hydrological model using HEC-HMS.

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3 ESTIMATING MODEL PARAMETERS

All hydrological models include parameters that must be estimated. The number of parameters varies greatly with the degree of conceptualization of the model and the spatial discretisation scheme. Lumped conceptual models generally have four to ten parameters while distributed models may have several dozen or more.

Parameter estimation often refers to model calibration, one of the most delicate steps in the development phase. A number of publications and been devoted to this topic (e.g., Duan *et al*., 2003, 2006).

A wide variety of parameter estimation methods exists. This is because there are many types of parameters and no single estimation method has proved satisfactory for all of them. Parameter estimation implies making choices concerning a calibration strategy that of course depends on the modelling objectives. The strategy also depends on the experience, priorities and personal preferences of the modeler. It influences the way the hydrological simulation model represents the hydrosystem as well as the pertinence and performance of the model itself.

Model parameters can be estimated using either a deterministic or probabilistic approach. The fi rst is the most widely used and is aimed at identifying the optimal set of parameters for the considered model and data. The second approach, more recent, has a statistical context and is aimed at identifying not only the best values for the parameters but also their probability distributions. This approach is closely related to the problem of modelling uncertainty and will be dealt with in Section 4.

The objective below is to provide an overview of: 1) certain problems related to parameter estimation using a deterministic approach and 2) certain methods widely used for parameter estimation. This presentation is not intended to be exhaustive and the reader can refer to numerous publications devoted to this problem for more information.

3.1 Nature of Parameters and Estimation Methods

As a first approximation, the estimation of parameters depends on their nature, the type of model and the available data. Two approaches are possible.

The first consists in assigning values on the basis of data describing the drainage basin. These data can be derived from measurements, field observations or thematic maps. This approach is often used to specify parameters that have a physical meaning. It can however be applied to estimate all types of parameters if appropriate regional models are available.
The second approach consists in estimating the parameters by means of a calibration procedure. This is carried out indirectly using hydrometeorological observations.

Specifying parameters that have physical meaning
Depending on the model, certain parameters may have physical meaning and are therefore directly related to the physical characteristics of the medium.
Some of them may be estimated relatively easily from thematic maps, DEMs or other data obtained by remote sensing. The type of land use on a given element of the discretisation can for example be estimated from a land-use map. The slopes and upstream drainage areas at every point of the drainage network can be extracted from DEMs.
Theoretically, some parameters can be estimated from measurements made on the studied drainage basin. This is for example the case for certain properties of the physical medium such as the hydraulic conductivity of the soil. However, two major difficulties are encountered.

- First of all, the physical medium within each discretisation unit is generally very heterogeneous. Measurements, usually made at certain points, cannot be used directly to characterise the properties of the considered discretisation element. Using the measurements requires that it is possible to: 1) find a method to describe the internal variability of the parameter and the associated behaviour of the discretisation element and 2) relate the point values to the associated model parameters (Addendum 2).
- Next, even if the above problem can be solved, it is impossible from a practical viewpoint to measure, at every point, the characteristics required for the estimation. Consequently, the point measurements must be distributed using the classical spatial interpolation methods. This spatial interpolation leads to a very imperfect view of the real variability of the characteristics of interest, in particular because the point measurements are not necessarily representative.

These problems in terms of scale and representativeness explain why it is often necessary to calibrate parameters that have a physical meaning. This is for example the case for the physically-based SHE model (Abbott *et al*., 1986a,b). Note that when an automatic calibration procedure is used for this, the physical meaning of certain parameters can be considerably modified or disappear altogether.

Specifying parameters using regional models
Whether or not they have physical meaning, the parameters can also be estimated on the basis of the characteristics of the physical medium when a regional model capable of predicting them is available.
These predictive models must be constructed by applying a hydrological regionalisation procedure. They are often available to estimate the parameters of hydrological models frequently used for a given region. Such models are of many different types. For example, pedotransfer functions (Soutter *et al*., 2007) are predictive models relating the parameters describing the hydrodynamic properties of soils to various measurable pedological characteristics by means of appropriate mathematical functions (Addendum 6).

A similar predictive approach makes use of tables or nomographs proposing standard values or possible parameter ranges for different types of media. Each discretisation unit chosen to represent the basin is matched to one of the classes figuring in these tables. The value indicated for the class is assigned to the corresponding discretisation unit. Land-use maps are for example often used to estimate the values of certain parameters such as basin roughness coefficients affecting surface runoff.
Note that estimates obtained using these predictive models can give poor simulation results. These prior estimates are consequently very often subsequently adjusted using a calibration/validation procedure as will be discussed below.

Estimating parameters by calibration
Whether they have physical meaning or not, many model parameters are determined using a calibration procedure. The objective of this deterministic approach is to adjust parameter values so as to obtain the best possible agreement between the simulation results obtained with the model and observed data. In other words, model calibration involves finding parameter values that minimize the simulation error $\varepsilon_Z(t, \boldsymbol{\theta})$ for the various output variables considered (Figure 23).
The procedure used to calibrate hydrological models generally consists in choosing:

- the dataset to be used for the calibration,
- one or more criteria for the evaluation of the performance of the hydrological model,
- a method to identify optimum values of the parameters with respect to the selected performance criteria.

The calibration method always depends on the modelling objectives, type of model and available data. The parameter estimates obtained in this step along with the associated uncertainties must be evaluated in a subsequent validation step (Section 4).

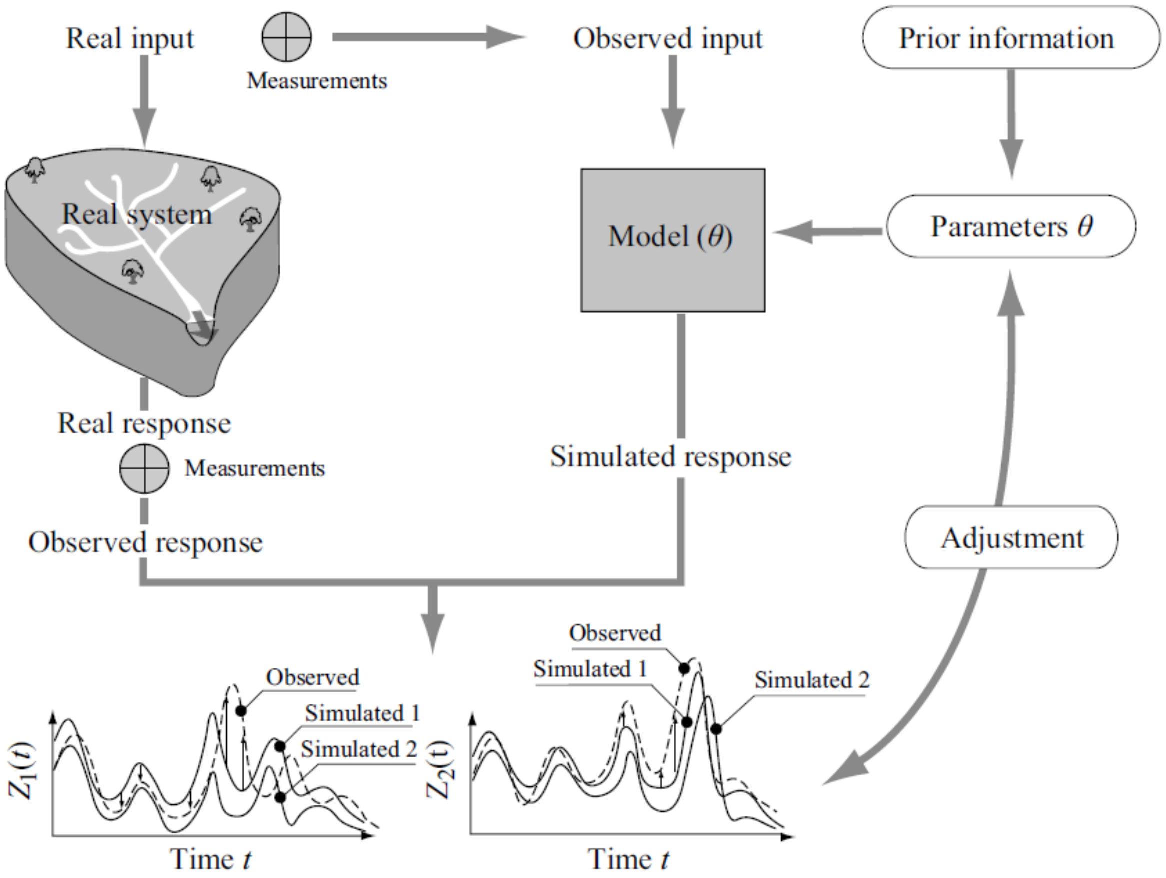


Figure 23. Schematic procedure used to calibrate parameters θ *of a hydrological model. Calibration involves minimising the error between the observed and simulated values of various model output variables (e.g.,* $Z_1(t)$ *and* $Z_2(t)$*).*

3.2 Performance and Equifinality Criteria

Different possible criteria that can be used to assess model performance along with the main difficulties related to this assessment will now be discussed.

Model performance criteria

Before calibrating a hydrological model, one or more performance criteria must first be defined. Many criteria are possible. They can also be used in the model evaluation step. Model performance can first be evaluated qualitatively by visually comparing observed and simulated values of a given variable on a graph. Several types of graphs can be considered, expressing for example:

- the temporal evolution of the variable of interest for different time steps and over a given temporal window,
- the spatial structure of the variable of interest at a given time if the model is distributed or semi-distributed and if the corresponding observations are available,
- the statistical distribution of the variable of interest (e.g., statistical distribution of the daily discharges or maximum flood discharges at an hourly time step).

Model performance can also be evaluated quantitatively using a given numerical performance criterion that is a mathematical function of the error obtained. It is generally based on a metric that can be used to estimate, for any hydrological variable of interest, the norm of the error vector $\boldsymbol{\varepsilon}_{\mathbf{Z}}(\boldsymbol{\theta}) = \mathbf{Z}_{obs} - \mathbf{Z}_{sim}(\boldsymbol{\theta})$ (i.e., vector $[\varepsilon_Z(t_1, \boldsymbol{\theta}), \varepsilon_Z(t_2, \boldsymbol{\theta}), \ldots \varepsilon_Z(t_n, \boldsymbol{\theta})]$) or, in other words, that can be used to measure the distance between the observed and simulated values of the considered variable. This function is often referred to as the objective function, criterion function, error function or likelihood function. This function F can be formally expressed as:

$$F(\mathbf{Z}_{obs}, \mathbf{Z}_{sim}(\boldsymbol{\theta})) = F(\mathbf{Z}_{obs}, G(\mathbf{X}, \boldsymbol{\theta})) \quad (7)$$

where $\mathbf{Z}_{obs}$ and $\mathbf{Z}_{sim}(\boldsymbol{\theta})$ are the vectors containing respectively the n observed and n simulated values of the variable $Z(t)$ for the considered simulation period T, $\boldsymbol{\theta}$ is the vector containing the p parameters of the model G and $\mathbf{X}$ is the vector containing the n values of the m model input variables.

Several of the most widely used objective functions are presented in Addendum 5. Other objective functions can be found in the literature (e.g., Singh, 1995, Moriasi et al., 2007; 2015). They reflect various optimisation objectives that are sometimes conflicting. The most commonly encountered objectives are designed to obtain a good match between: 1) measured and simulated flow volumes over the simulation period, 2) observed and simulated discharges over time - whether for the entire simulation period or for certain selected events, 3) flood characteristics (e.g., peak discharges, rise time, flood volumes) and 4) low-flow characteristics (e.g., decay factors, available reserve).

Addendum 5 Some objective functions commonly used in hydrology

The main objective functions used to calibrate and evaluate hydrological models are described below. The corresponding mathematical expressions are indicated in Table 1. For all the expressions, $Q_o(i)$ and $Q_s(i)$ are respectively the observed and simulated discharges for time step i, n is the number of observations, $\overline{Q}_o$ and $\overline{Q}_s$ are the mean discharges for respectively the n observations and n simulated values, η is a fraction of the mean discharge to be defined (e.g., $\varepsilon = \overline{Q}_o/20$), used to avoid problems related to null values and $Q_{po}(k)$ and $Q_{ps}(k)$ are respectively the observed and simulated peak flood discharges for event k. These formulations can of course be used for variables other than discharge. Sums appearing in certain expressions are weighted (e.g., higher weights for no-rain days when estimating parameters influencing the reconstruction of low flows).

Table 1. Mathematical expressions for a number of objective functions with discharge Q *at the outlet of the considered basin as the variable of interest.*

Criterion	Expression
Bias	$F = \frac{1}{n}\sum_{i=1}^{n}\left(Q_o(i) - Q_s(i)\right)$
Squared correlation coefficient	$F = \frac{\left[\sum_{i=1}^{n}\left(Q_o(i)-\bar{Q}_o\right)\left(Q_s(i)-\bar{Q}_s\right)\right]^2}{\sum_{i=1}^{n}\left(Q_o(i)-\bar{Q}_o\right)^2 \cdot \sum_{i=1}^{n}\left(Q_s(i)-\bar{Q}_s\right)^2}$
Least squares	$F = \sum_{i=1}^{n}\left(Q_o(i) - Q_s(i)\right)^2$
Nash-Sutcliffe	$F = 1 - \frac{\sum_{i=1}^{n}\left(Q_o(i)-Q_s(i)\right)^2}{\sum_{i=1}^{n}\left(Q_o(i)-\bar{Q}_o\right)^2}$
Nash-Sutcliffe with logarithms	$F = 1 - \frac{\sum_{i=1}^{n}\left(\ln\left(Q_o(i)+\eta\right)-\ln\left(Q_s(i)+\eta\right)\right)^2}{\sum_{i=1}^{n}\left(\ln\left(Q_o(i)+\eta\right)-\ln\left(\bar{Q}_o+\eta\right)\right)^2}$
Clabour-Moore	$F = \frac{\sum_{i=1}^{n}\left\lvert Q_o(i)-Q_s(i)\right\rvert}{\sum_{i=1}^{n}Q_o(i)}$
Peak flood discharges	$F = Q_{po}(k) - Q_{ps}(k)$

Moriasi et al. (2015) suggest a rating of performance for some key performance criteria (Table 2), which might be used to communicate the quality of model calibration and validation. However, a high rating alone does not guarantee that the model is in fact fit for purpose. Nor should such ratings be used as the only quality assurance.

Table 2. Performance evaluation criteria rating suggested by Moriasi et al. (2015).

Measure	Output Response	Temporal Scale[a]	Performance Evaluation Criteria: Very Good	Good	Satisfactory	Not Satisfactory
Watershed scale						
R^2	Flow[b]	D-M-A	$R^2 > 0.85$	$0.75 < R^2 \leq 0.85$	$0.60 < R^2 \leq 0.75$	$R^2 \leq 0.60$
	Sediment/P[c]	M	$R^2 > 0.80$	$0.65 < R^2 \leq 0.80$	$0.40 < R^2 \leq 0.65$	$R^2 \leq 0.40$
	N	M	$R^2 > 0.70$	$0.60 < R^2 \leq 0.70$	$0.30 < R^2 \leq 0.60$	$R^2 \leq 0.30$
NSE	Flow	D-M-A	NSE > 0.80	0.70 < NSE ≤ 0.80	0.50 < NSE ≤ 0.70	NSE ≤ 0.50
	Sediment	M	NSE > 0.80	0.70 < NSE ≤ 0.80	0.45< NSE ≤ 0.70	NSE ≤ 0.45
	N/P[c]	M	NSE > 0.65	0.50 < NSE ≤ 0.65	0.35< NSE ≤ 0.50	NSE ≤ 0.35
PBIAS (%)	Flow	D-M-A	PBIAS < ±5	±5 ≤ PBIAS < ±10	±10 ≤ PBIAS < ±15	PBIAS ≥ ±15
	Sediment	D-M-A	PBIAS < ±10	±10 ≤ PBIAS < ±15	±15 ≤ PBIAS < ±20	PBIAS ≥ ±20
	N/P[c]	D-M-A	PBIAS < ±15	±15 ≤ PBIAS < ±20	±20 ≤ PBIAS < ±30	PBIAS ≥ ±30
Field scale						
R^2	Flow	M	$R^2 > 0.85$	$0.75 < R^2 \leq 0.85$	$0.70 < R^2 < 0.75$	$R^2 \leq 0.70$
d	Flow	M	$d > 0.90$	$0.85 < d \leq 0.90$	$0.75 < d < 0.85$	$d \leq 0.75$

[a] D, M, and A denote daily, monthly, and annual temporal scales, respectively.
[b] Includes stream flow, surface runoff, base flow, and tile flow, as appropriate, for watershed- and field-scale models.
[c] Where there were no differences, PEC were grouped for the output responses.

Bias criterion B

This criterion measures the tendency of a model to underestimate or overestimate the observed discharges. It must be minimised (the optimum value being zero). The bias criterion is often normalised by dividing it by the mean observed discharge. It then gives the relative error for the simulated and observed fl ow volumes at the basin outlet over the simulation period. This criterion is generally used to calibrate model parameters that condition the simulated hydrological balance in such a way that the balance is close to the one estimated on the basis of observations.

Least-squares criterion
This criterion is widely used in the field of statistics. It provides an unbiased estimate of the parameters as long as the errors are not correlated between themselves and are identically distributed according to a normal law with constant variance and a mean of zero. These conditions are not generally satisfied in hydrology. To approach these conditions, the data should ideally first be transformed (e.g., logarithmic transformation of discharges, modelling of residuals using an autoregressive (AR) process).

Nash-Sutcliffe criterion (Nash and Sutcliffe, 1970)
This criterion expresses the fraction of the variance of the discharges explained by the hydrological model. In other words, it expresses the relative difference between the error of the tested hydrological model and the error of the reference model with respect to the mean of the discharges.
The closer the value of this criterion is to 1, the better the agreement between the simulation and observations. NSE = 1 indicates perfect correspondence between simulations and observations; NSE = 0 indicates that the model simulations have the same explanatory power as the mean of the observations; and NSE <0 indicates that the model is a worse predictor than the mean of the observations: This is one of the criteria most widely used by hydrologists. It is strictly equivalent to the least squares criterion (it therefore assumes the same conditions concerning the statistical structure of the errors). The great advantage of the Nash Sutcliffe criterion is that it is normalised. It can therefore be used to compare the performance of a model in different simulation contexts and for different drainage basins.
The disadvantage of this criterion is that high values are heavily weighted. To increase the weight of low values, the same formulation is sometimes used on the logarithm (Table 1) or square root of the variable. Such formulations should be preferred in particular when focusing on low flows.
It must be noted that the Kling–Gupta efficiency (KGE) has gained increasing popularity in recent years as and it is used either in addition or instead of NSE more frequently.

Kling-Gupta efficiency (Gupta et al., 2009)
The Kling–Gupta efficiency (KGE) is based on a decomposition of NSE into its constitutive components (correlation, variability bias and mean bias), addresses several perceived shortcomings in NSE (although there are still opportunities to improve the KGE metric and to explore alternative ways to quantify model performance) and is increasingly used for model calibration and evaluation:

$$\mathrm{KGE} = 1 - \sqrt{(r-1)^2 + (\alpha-1)^2 + (\beta-1)^2},$$

where r is the linear correlation between observations and simulations, α a measure of the flow variability error, and β a bias term

$$\mathrm{KGE} = 1 - \sqrt{(r-1)^2 + \left(\frac{\sigma_{\mathrm{sim}}}{\sigma_{\mathrm{obs}}} - 1\right)^2 + \left(\frac{\mu_{\mathrm{sim}}}{\mu_{\mathrm{obs}}} - 1\right)^2},$$

where σ_{obs} is the standard deviation in observations, σ_{sim} the standard deviation in simulations, μ_{sim} the simulation mean, and μ_{obs} the observation mean (i.e. equivalent to $\bar{Q}_{obs}$). Like NSE, KGE = 1 indicates perfect agreement between simulations and observations.

Clabour and Moore criterion (Clabour and Moore, 1970)
This criterion weights all the residuals equally. It is therefore well suited to cases requiring good simulation of both high and low discharge values.

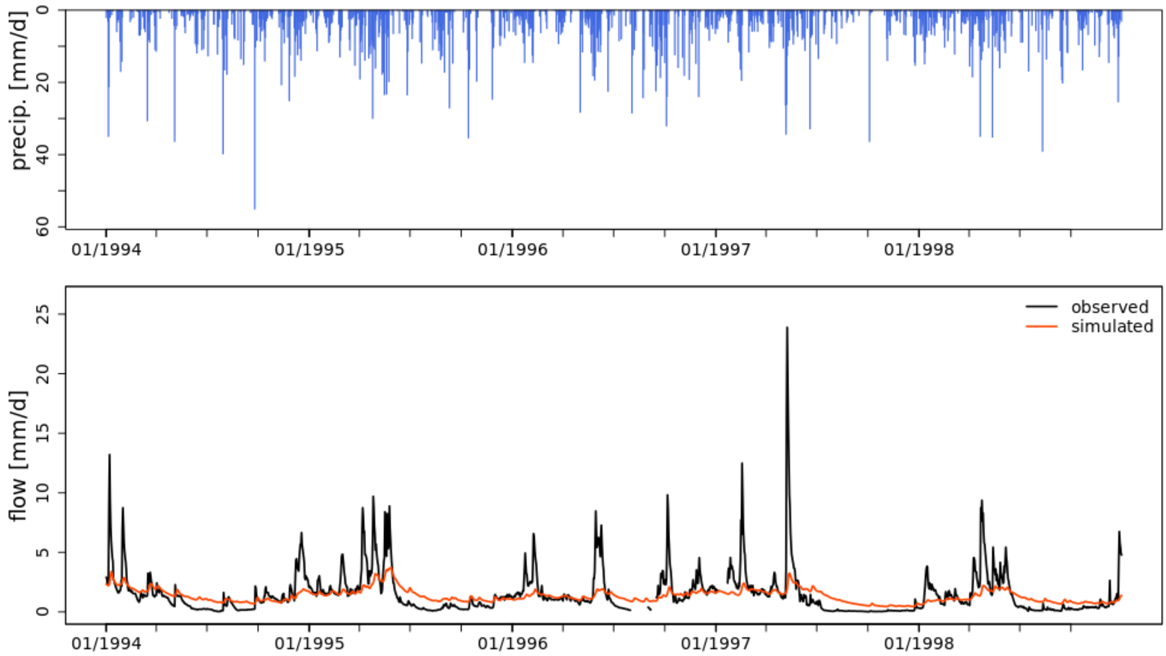
Peak flood discharge difference
The difference between observed and simulated values of peak flood discharges can be adapted by adding an error on the rising time of the flood hydrograph. This is generally not recommended when calibrating a hydrological model.
The surface defined by the values of the objective function at every point in the parameter space is referred to as the response surface of this function. For a model with two parameters, this surface is 3-dimensional. For a model with *N* parameters, it has *N*+1 dimensions. For a model with two parameters, the surface is often represented graphically by iso-criterion curves, i.e., curves along which the value of the criterion is the same at every point (Figure 25). This representation is not possible for the general case, unless *N*-2 values of the *N* parameters are fixed.

Graphical performance measures should always be combined with the assessment of objective functions. Useful measures are scatter plots, time-series plots, cumulative plots, flow and load duration curves and maps. Table x provides a summary of useful graphical performance measures for hydrological model calibration and validation.

Table 4. Summary of graphical performance measures for H/WQ model calibration and validation.

	Purpose	Advantages/Disadvantages
Direct comparison		
Scatter plots	Compare observed and simulated data with no dependent variable. A least square regression line can be fitted to observe deviation from the 1:1 line.	**Advantages:** Divergence from the 1:1 line provides a visual understanding of the underlying behavior of the model, including any bias or systematic variance. **Disadvantages:** Data points clumped in the low intensity, high frequency range and few in the high intensity, low frequency range can artificially make a model's performance look good.
Time-series plots	Compare observed and simulated data with time as a dependent variable.	**Advantages:** Helps inspect and support troubleshooting event-specific prediction issues, including mismatches in magnitude of peaks and shape of recession curve, and outliers. Time series plots can also guide selection of parameters to be used for calibration. **Disadvantages:** Time series plots become cluttered with too many data points.
Derived comparison		
Cumulative plots	Compare cumulative observed and simulated values with time as dependent variable.	**Advantages:** Allows identification of any systematic temporal divergence between observed and simulated values. **Disadvantages:** Cumulative plots may still converge, with major temporal mismatches. They should be used as a preliminary model performance-screening tool.
Flow and load duration curves	Compare observed and simulated values with probability as a dependent variable.	**Advantages:** Provides insight into model performance over different flow/load regimes (i.e., low, medium, high; Pfannerstill et al., 2014). **Disadvantages:** Needs a larger number of data points to derive meaningful conclusions. Duration curves are most useful for long-term monthly, daily, or subdaily calibrations.
Maps	Map showing the output of interest at the desired spatial scale. Examples include showing annual sediment loss for each subwatershed.	**Advantages:** Useful for presenting field-scale to watershed-scale model results for understanding the spatial performance of the model. Pollutant hotspots within a watershed can be quickly identified using color-codes. **Disadvantages:** Choices of color-coding and grouping within a map can sometimes be misleading. For example, red colored areas may or may not represent critical areas depending on actual values plotted.

Figure 24 shows examples of time-series plots, flow duration curve and a scatter plot.



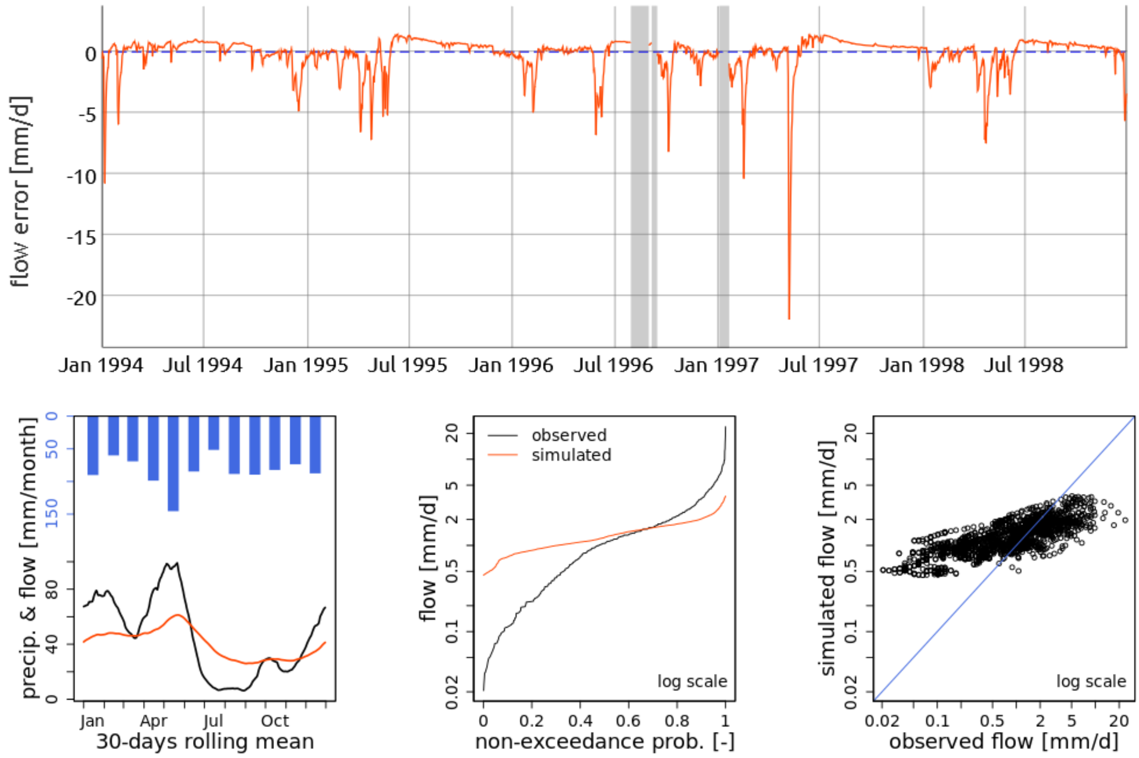


Figure 24. Plots of the time series, flow error, rolling mean, flow duration curve and scatter plot before model calibration.

Exercise 3

Use airGRteaching (https://sunshine.irstea.fr/app/airGRteaching) to explore manual and automatic calibration of a hydrological model. Also explore using different types of plots to assess the model calibration exercise.

Model calibration difficulties

The search for a set of optimal model parameters for a given performance criterion involves finding the set of parameters that gives the best value of this criterion. If the criterion is numerical, this means finding the global optimum of the response surface for the corresponding objective function. Ideally, this set of parameters is located at the centre of more or less circular and concentric iso-criterion curves (Figure 25a).

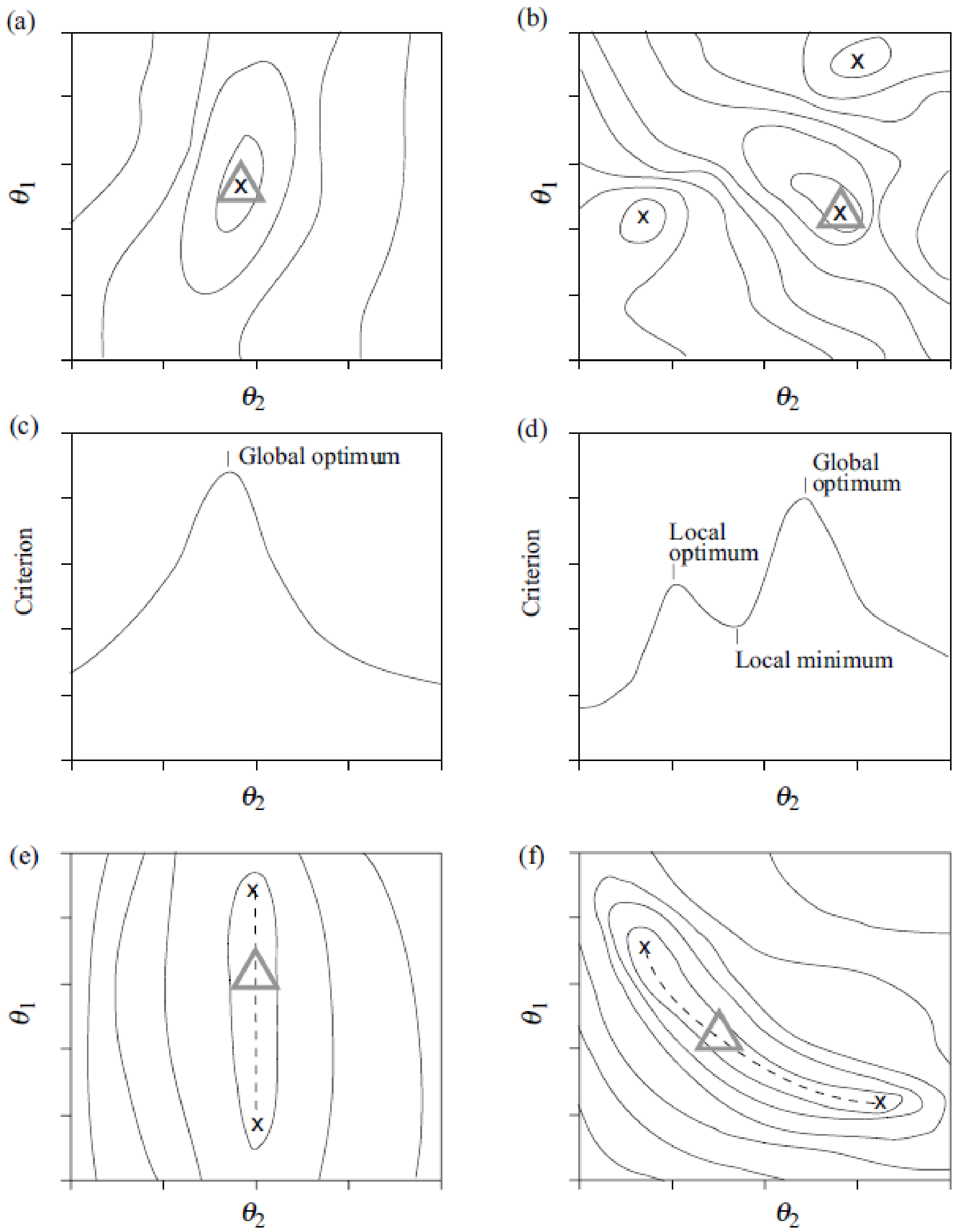


Figure 25. Typical iso-criterion performance curves for two parameters θ_i *and* θ_j *of a model. a) Curves for the ideal case where the response surface is convex. b) Case where different local optima exist. c,d) Cross-sectional views of the two response surfaces shown respectively in graphs a and b. e) Case where the selected performance criterion is insensitive to one of the two parameters (here the parameter* θ_1*). f) Case where the parameters are interdependent with respect to the performance criterion. The global optimum for each configuration is indicated by a triangle and any local or secondary optima by x's.*

It is generally relatively easy to identify the corresponding point. In practice, various problems illustrated by the other response surfaces presented in Figure 25 can complicate this task.

- Possible existence of local optima (Figure 25b,d): Such optima correspond to parameter sets that provide better model performance than all their nearby neighbours, although not as good as that of the global optimum.

- Possible interdependence between certain model parameters (Figure 25f): When this is the case, model performance remains unchanged if the modification of one parameter can be compensated by modification of the other parameters.
- Low sensitivity of model performance to the modification of certain parameters (Figure 25e): If this is the case, a major change in the value of one parameter causes only weak changes in model performance.

In this context, a number of parameter sets can generally be found to give equivalent model performance for a selected performance criterion. This questions the concept of an optimal parameter set and introduces what is commonly referred to as the equifinality problem (Beven and Binley, 1992; Beven, 1996). This problem increases with the number of parameters.

3.3 Manual or Automatic Calibration

The model can be calibrated manually, automatically or by a combination of the two. Manual calibration consists in manually adjusting parameter values in an iterative manner. Model performance is evaluated after each adjustment. The operator stops the calibration when the performance reached is considered to be reasonable. This judgment is generally based on the combination of numerical and graphical performance criteria. Automatic calibration methods are designed to identify the parameter set that provides the optimal value of a selected numerical performance criterion. Optimisation is carried out by applying iterative and systematic changes to parameter values so as to minimise simulation errors for the corresponding objective function. Automatic calibration requires a mathematical optimisation algorithm and the corresponding computer code. The optimisation algorithms mainly follow two approaches referred to as local or global (Addendum 6). For more information on the topic, see the different publications by Duan *et al*. (2003, 2006).

Addendum 6 Principles and limitations of optimisation algorithms (beginning of 'not examinable')

Local approaches

Local approaches have been developed for the ideal optimisation case where the response surface is convex. These approaches explore the parameter space in a progressive and evolving manner starting from a prior initial parameter set. The space is explored in the direction in which it is possible to improve the value of the criterion function. It stops when it is no longer possible to produce a significant improvement.

Local optimisation methods differ in particular according to the type of information used to explore the parameter space and the strategy adopted for this exploration (choice of direction and displacement length). Direct methods such as the Rosenbrock (1960), Pattern Search (Hooke and Jeeves, 1961) and Simplex (Nelder and Mead, 1965) methods are among the most widely used in hydrology. Gradient descent methods (e.g., Newton and conjugated gradient methods) proceed in the direction of the steepest slope from any point and converge faster than the direct methods. However, the objective function must be differentiable and the derivative must be continuous over the domain. Generally speaking, local methods are often limited for different reasons: 1) the parameter space generally contains several attraction zones where an algorithm can converge, 2) for each attraction zone, several local optima frequently exist and 3) the objective function and its derivative can exhibit major discontinuities over the explored domain.

Global approaches

Global optimisation methods have been developed to remedy these limitations. They can be used to explore a much larger part of the parameter space and therefore in principle the optimal parameter set without stopping at any local optimum.

In deterministic global methods, the space is explored in a systematic manner on the basis of a fi ne discretisation. The value of the objective function is calculated for each point of the discretisation. Stochastic global methods explore the parameter space in a random or pseudorandom

manner. This exploration can also involve a random choice of one or more starting points for the search algorithm. Some of these methods consist in modifying a population of parameter sets so as to identify the optimal set. Genetic algorithms are an example of this type of method. They use the notion of natural selection developed by Darwin to adapt the population (Holland, 1975). For an application in hydrology, see Franchini *et al*. (1998). Many other global optimisation methods can be found in Duan *et al*. (2003).

Combined approaches

Optimisation methods can also result from the combination of global and local methods. A good example of this is the relatively complex method proposed by Duan *et al*. (1992). This method combines the global method of the genetic algorithm type to randomly generate the starting points with a local method of the simplex type to fi nd the associated optima. The main limitation of global and combined methods is related to their complexity. For example, they generally include different parameters themselves that must be calibrated before the algorithm can find the optimum.

Whether manual or automatic, local or global, all these optimisation methods are based on the assumption that there is a single real optimum on the response surface. However, in reality, many parameter sets offer equivalent performance. The major uncertainty related to the values assigned to model parameters has led certain hydrologists to alter their way of dealing with the estimation of model parameters. The methods proposed characterise this uncertainty and subsequently characterise the resulting uncertainty of the simulation results. This point will be discussed in more detail in Section 3.4.2.

3.4 Multi-criteria and Multi-objective Calibration

Advantages and principles

A model can be calibrated for a single objective and on the basis of a single criterion or on multiple objectives using multiple criteria. This choice influences the ability of the model to reproduce any or all of the hydrological variables to be simulated. For example, a Nash criterion calculated on the basis of all the discharge values obtained over a given time period leads to better results for the simulation of flood discharges to the detriment of low flows. Moreover, the Nash criterion provides no information on the ability of the model to reproduce the observed hydrological balance. On the other hand, a bias criterion can optimise the reproduction of the hydrological balance but provides no information on the ability of the model to reproduce the temporal evolution of the observed discharges. Use of a single performance criterion therefore provides only a very partial view of model performance, even if the model is evaluated only for a single simulation objective (e.g., good reproduction of recessions). Because it does not make full use of the information contained in the data, single-criterion and single-objective calibration necessarily produce a sub-optimal parameter set with respect to all the variables simulated by the model. For this reason, the calibration procedures should be multi-criteria and multi-objective. One or more criteria are then assigned to each of the targeted objectives (e.g., good reproduction of peak flood discharges AND good reproduction of recessions).

There are many possible objectives and criteria. However, the method only works well if they are independent. Ideally, they should concern:

- different characteristics of a given model output variable (e.g., flood discharges and low flows),
- different output variables (e.g., discharge, water table height, snow cover extent and snowpack depth),
- different temporal windows and time scales (e.g., specific seasons or events, hourly, weekly and monthly time steps),
- different spatial resolutions if the spatial variability of the variable is simulated by the model (e.g., runoff volumes for the complete drainage basin and its subbasins).

The temporal windows to be considered can for example concern periods of the year during which the dominant processes have been identified to occur. For certain drainage basins in mountainous areas, for example, spring discharges are mainly related to snow melt, summer discharges to glacier melt and autumn discharges to groundwater releases, while rainfall events throughout the year affect numerous specific temporal windows.

The time scales considered can correspond to the time scales of certain processes. For example, mean annual discharge series can be evaluated to test the ability of a model to correctly describe, on the average, meteorological forcing (e.g., basin precipitation) or reasonably simulate the processes that influence the annual hydrological balance (e.g., interception, evapotranspiration). On the other hand, mean monthly discharge series can be evaluated to focus on highly seasonal processes (e.g., snow melt, groundwater recharge or discharge). Finally, the evaluation of hourly discharge series looks mainly at infra-daily processes (e.g., melting, evapotranspiration) or those related to the generation of flood discharges (e.g., infiltration and spatial redistribution of sub-surface water).

The use of multiple objectives is of interest in so far as model parameters often strongly influence a limited number of model processes or modelled hydrological phenomena. In conceptual models, for example, the parameters of the rainfall excess module determine a major part of the hydrological balance. This is for instance generally the case for the storage capacity of reservoirs used to manage sub-surface water reserves. On the other hand, the hydrological balance is not influenced at all by the parameters of the rainfall-runoff model. These parameters however determine the shape of the flood hydrograph.

Implementation strategies and limits

Even if the advantages of multi-criteria calibration are now widely accepted its implementation leads to an optimisation problem that is far more complex than for single criterion calibration. First of all, the different criteria have objectives of varying importance with respect to the fi nal objective of the hydrological model. Next, the different criteria do not generally lead to the same optima or to the same acceptable regions within the parameter space (Figure 26a). Finally, the criteria are often conflicting. Improvement of one criterion is nearly always achieved at the detriment of another (Figure 26b). It is for this reason that no unique optimal parameter set exists but rather different optimal parameter sets in the sense of Pareto optimality.[3] Note however that for practical reasons, it is generally impossible to identify all Pareto optima.

The main difficulty of multi-criteria approaches is therefore to find a method that makes it possible to combine conflicting information produced by different criteria and to consider the relative importance of the objectives targeted by each. Once again, many strategies can be followed to achieve this (Fenicia *et al.*, 2007).

One strategy consists in transforming the multi-criteria optimisation problem into a single-criterion optimisation problem with the sole criterion to be optimised based on a combination of individual criterion. A wide variety of rules can be devised to implement the combinations (e.g., arithmetic weighting of the different criteria, rules based on fuzzy logic). The search for the optimum value of the selected global criterion can then be carried out using classical optimisation methods. This approach is relatively convenient from a practical viewpoint. It is however of limited interested in that it globalises the optimisation and does not allow differentiated improvement of the different components of the model. Another strategy consists in carrying out the calibration in several steps. An example is illustrated schematically in Figure 27.

Each new step is aimed at estimating the values of the different supplementary parameters so that these values are optimal with respect to one or more selected criteria. The optimum values of the parameters calibrated in a previous step are eventually adjusted so as to optimise the performance of the model for the *n* first steps. In this sequential approach, calibration can be manual, automatic or semi-automatic depending on the step.

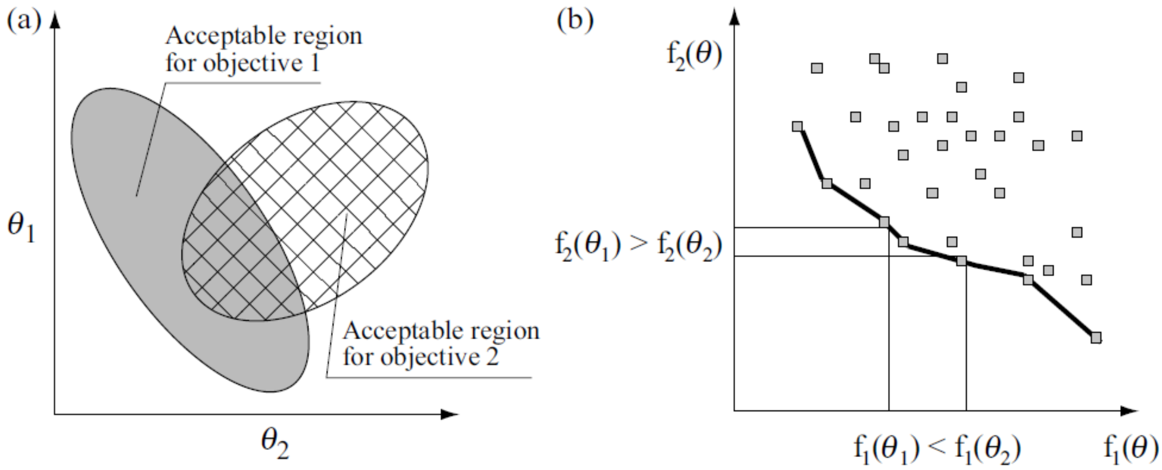


Figure 26. Typical problem related to a multi-criteria optimisation. a) For a two-parameter space (parameters θ_1 and θ_2), this graph shows two regions that are acceptable for a certain level of performance with respect to two distinct criteria. b) For a two-criteria space (criteria

[3] A Pareto optimum is a set of parameters for which model performance cannot be improved with respect to a given criterion without reducing performance with respect to another criterion. According to this notion, the parameter space can be divided into two sub-spaces. The fi rst concerns parameter sets for which it is possible to improve certain criteria by modifying the value of one or more parameters without deteriorating other criteria. The second sub-space concerns parameter sets for which improvement of one or more criteria leads to deterioration of at least one other criteria. This sub-space defi nes the Pareto set for the problem. For concrete problems, it is not possible to obtain an approximation of the Pareto set. The diffi culty lies in the fact that it is impossible to distinguish between "good" pairs and "better" pairs. One method proposed to solve this problem consists in transforming the multi-objective calibration into a series of single-objective calibrations by aggregating the various objective functions considered (Yapo *et al.*, 1998).

f_1 *and* f_2*), this graph shows the set of optimal parameter sets in the sense of Pareto optimality (solid line).*

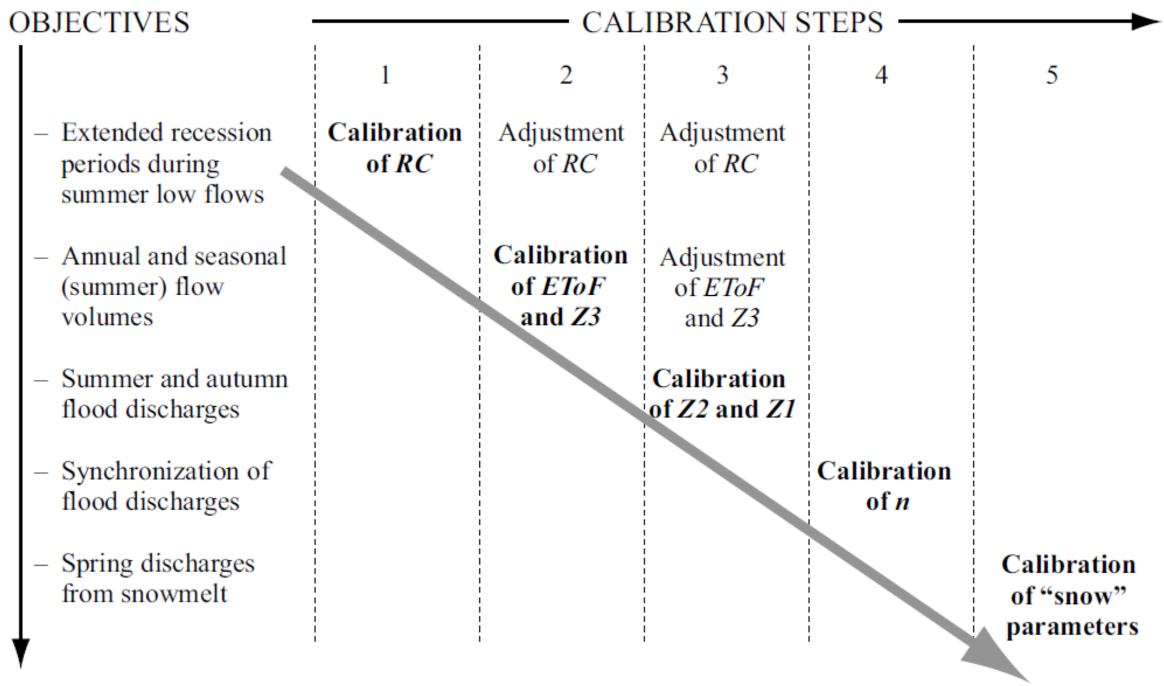


Figure 27. Schematic representation of a carefully reasoned sequential and multi-objective calibration process (adapted from Turcotte et al.*, 2003). Bold text: Main iterations used to estimate the indicated parameters on the basis of specific criteria. Normal text: Secondary iterations used to readjust the indicated parameters and satisfy the preceding criteria.* CR *is a recession coefficient. The evapotranspiration multiplication factor* ET_0F *and the total depth* Z_3 *of the soil layer control the flow volumes.* Z_2 *and* Z_1*, respectively the depths of the second and first soil layers, determine the division of surface runoff as well as the redistribution of water in the ground. The Manning coefficient* n *governs the flow velocity of surface runoff.*

To be optimal, the calibration must be carefully reasoned (Turcotte *et al*., 2003; Peugeot *et al*., 2003; Hingray *et al*., 2010). For example, it is often of interest to start by calibrating the parameters that affect the processes with the longest time scales (e.g., the rainfall excess parameters affecting the hydrological balance of the basin). Furthermore, to estimate the parameter(s) targeted in a given step of the calibration process, it is first necessary to identify the hydrological quantities that are most informative. This approach therefore requires a high level of modelling expertise. It generally makes best use of some or all of the information available in the available time series. **(end of 'not examinable')**

3.5 Choosing a Calibration Method and other Considerations

Influence of initial conditions on calibration

The initial conditions can influence the output of the model for a certain time during simulation (Figure 28). The duration depends on the response time of the associated processes. For example, the influence of initial conditions related to the amount of water stored in the hydrographic network can last from several hours to several days depending on the drainage basin. The influence of initial conditions related to soil moisture, snow cover or groundwater storage can last several weeks to several months.

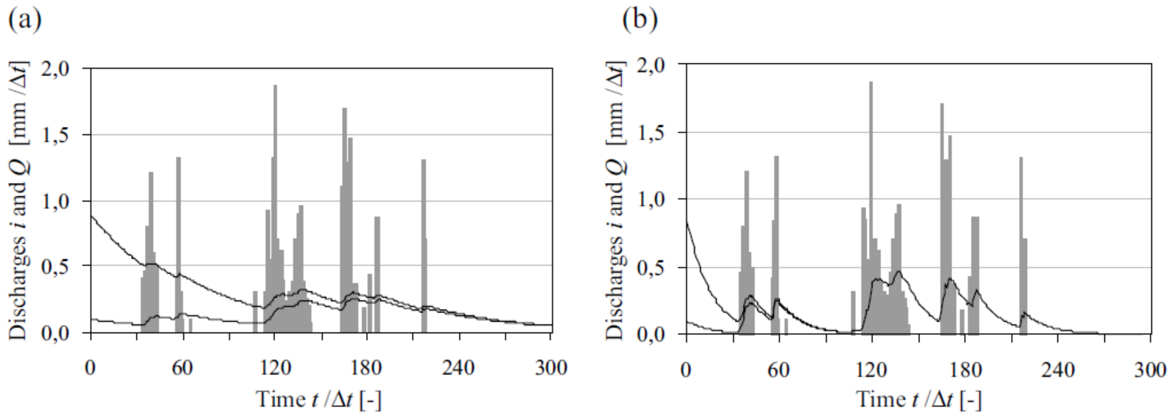


Figure 28. Influence of initial conditions on the outputs of a conceptual linear reservoir for different reservoir time constants K *[h]: a)* K *= 60Δ*t *and b)* K *= 15Δ*t *(graph b therefore represents a faster system response than graph a). The inflow* i(t) *into the reservoir is described by the grey histograms (in mm/Δ*t*) and the outflow* q(t) *from the reservoir by the black curves (in mm/Δ*t*). The initial conditions are defined by the value of the reservoir outflow discharge* q(t) *at time* $t = t_0$*. For each value of* K*, the simulated system response is given for two initial conditions:* q_0 *= 0.1 mm/Δ*t *and* q_0 *= 0.9 mm/Δ*t*.*

For the calibration of event-based models, the initial conditions are commonly considered as additional parameters to be calibrated. On the other hand, for continuous simulation models, if the simulation period is long enough, the influence of the initial conditions on the simulation results becomes negligible after a certain time. The results of a continuous simulation are therefore always evaluated without taking into account the results obtained for the first time steps of the simulation. The excluded period, which may last from a few days to a few months, depending on the context, is referred to as the initialization or warm-up period.

Whatever the simulation context, it is always important to carefully estimate the initial conditions of a model. Field observations can sometimes be used for this. If the hydrological behaviour of the drainage basin is very seasonal, another possibility is to use the mean values of the state variables estimated by simulation over a number of years (e.g., 10 years) or for the first month of the simulation. Note that system initialization is easier if the start of the simulation period is chosen within the period of the year in which storage volumes in the different compartments of the basin are known or assumed to be known (e.g., snow cover is for instance inexistent at the end of the snowmelt period).

Exercise 4

Use the HMS model to assess the impact of the initial storage on the hydrograph dynamics.

Predicting calibration parameters

Most conceptual models have only a few parameters (on the order of 5-10). Physically-based models may have many more parameters. A pertinent manual or automatic calibration is therefore generally not possible. In this case, a good approach is to predict as many

parameters as possible on the basis of observations, measurements, thematic maps, regional models or the expertise of the modeler. The remaining parameters can then be obtained by calibration (Fortin *et al*., 1995; Szilagyi and Parlange, 1999). The values of predicted parameters can if necessary be adjusted during this calibration phase.
In all cases, it is important to start calibration with reasonable prior estimates of the different model parameters (Fortin *et al*., 1995; Szilagyi and Parlange, 1999). This often makes it possible to take full advantage of the expertise acquired during past hydrological studies and to obtain a final parameter set that is consistent with the optimal configurations of parameters used for these studies. The need for such consistency is in particular essential for model parameter regionalization work that may be undertaken in a later phase.

Calibrating a distributed or semi-distributed model **(beginning of 'not examinable')**
Hydrological model calibration complexity increases with the number of elements selected for discretisation of the drainage basin. Even if the structure of the model is the same for all the elements of this discretisation, it is generally necessary to estimate the values of each of the parameters differently for each of these elements. If this estimation involves calibration, it rapidly becomes a tedious operation. To simplify the procedure, the default method often used consists in assigning a unique value to one or another of the parameters for all the sub-basins. A better strategy consists in using an assumed relationship between the parameter and various known characteristics of the sub-basins. For example, as a first approximation, it is reasonable to consider that the rainfall-runoff model parameter expressing the basin response time with respect to surface runoff increases with the size of the basin and decreases with the slope. This relationship can for example be formalised by a regional shape model:

$$\theta(X_1, X_2) = \theta_0 \cdot X_1^a \cdot X_2^b \tag{8}$$

where ϑ is the rainfall-runoff parameter to be estimated and X_1 and X_2 are respectively the size and mean slope of the basin. Estimating parameter ϑ for the different sub-basins of the model then consists simply in calibrating the parameters of the regional model, i.e., calibrating the scaling parameter ϑ_0 and the two exponents a and b of the scaling relationship in the present case.
This approach is particularly advantageous if the number of sub-basins is high and if a pertinent relationship of this type is expected to exist. First of all, it reduces the number of parameters to be estimated (here the three parameters ϑ_0, a and b instead of the n parameters ϑ_i corresponding to the n sub-basins). It subsequently ensures regional consistency in the spatial variation of the parameter. **(end of 'not examinable')**

Numerical or visual evaluation of performance
Objective functions measure modelling performance in a global, objective and reproducible manner. They are generally appreciated for these reasons. However, they have a number of drawbacks. For example, given their global character, it is impossible to identify the reason for the good or poor performance of the model. A model that simulates flood discharges well but low flows poorly can produce the same performance value as a model offering average performance for both flood discharges and low flows. Moreover, numerical criteria are often used to estimate the ability of a model to reproduce discharges over long time periods (typically daily discharges observed over one or two decades). This practice is far from

optimal. It only assesses the ability of the model to reproduce the average behaviour of the basin. It provides no information on its ability to produce exceptional events. Note that the reason for developing a model is often related precisely to such events. Furthermore, it is often these events that, by revealing flaws in the modelling concept, offer the possibility of improving it.
Evaluation of model performance on the basis of objective functions must therefore always be accompanied by a graphical evaluation. In particular, it is essential to use graphs to evaluate the ability of a model to reproduce the considered variables over time, in particular for selected events. This technique can detect certain anomalies that are difficult or even impossible to detect with numerical criteria. It can however be extremely tedious for long simulations. For this reason, graphical evaluation is often limited to certain specific or representative events.

Choosing the right level of calibration automation
Manual calibration is intrinsically subjective. The result of this type of approach therefore depends entirely on the expertise of the modeler. The modeler must therefore understand the hydrological behaviour of the basin to be modelled and the modelling concept on which the model is based. This approach takes full advantage of the results of calibrations that the modeler may have already carried out for example on other basins. Fundamentally, manual calibration is a multi-criteria method and gives very good results when the modeler has sufficient expertise. It also makes it possible to improve the modelling concept by allowing identification of certain shortcomings of the initial model (Fenicia *et al*., 2008). Manual calibration is in practice the method that is most widely used by hydrologists. It can however rapidly become difficult or tedious to implement for models with 3 to 5 parameters.
The great increase in the processing power of recent computers has led to the widespread use of automatic calibration methods. These methods allow a certain level of objectivity in the adjustment of the calibration parameters. They can easily be developed for simple cases on the basis of any of a number of existing optimisation methods. They generally allow fast calibration and account for uncertainties in the values of the parameters.
The development of multi-criteria and multi-objective automatic calibration methods is however difficult. Ultimately, these methods always require the expertise of the hydrologist to assess the pertinence of the calibration. The importance of this expertise increases with the number of parameters to be specified. For cases with many parameters, automatic calibration can lead to a set of parameters that gives satisfactory results for the wrong reasons (Klemes, 1986; Grayson *et al.*, 1992).
Ideally, hydrological models should be calibrated using a semi-automatic approach combining automatic and manual optimisation methods.

Exercise 5
Calibrate the HMS model using observed data.

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4 EVALUATING A MODEL
A model must be evaluated to ensure that it meets the objectives for which it was developed. This evaluation consists first in validating the model by testing the plausibility of its structure

and the parameter sets obtained during its calibration. The next step is to analyse and, if possible, to characterise the uncertainties of the model. These different aspects of the evaluation process are discussed below.

4.1 Validating a Hydrological Model

Validation versus verification

Model validation is not equivalent to model verification. The objective of verification is to check that the model is "realistic", i.e., that it provides a representation that is as close as possible to reality. Validation, on the other hand, will be considered to consist only in checking that the model provides a reasonable representation of this reality, adapted to the targeted objective.

In fact, whatever the ability of the model to reproduce the basin response for a given set of observations, it is never possible to conclude that the model is exact (or fully "realistic"). For example, it is always possible that future observations will be obtained for major hydrological events that will reveal a hydrological behaviour that could not be identified on the basis of past observations and therefore could not be accounted for in the development of the model. It is therefore impossible to verify that a model is fully realistic. It is however possible to conclude that it is not if one or more observations cannot be explained by the model.

On the other hand, it is in principle possible to validate a model, i.e., to check that it is reasonable for the targeted objective. However, no universal method exists for such a validation. Validation schemes and criteria used to evaluate the validity of models vary greatly.

Validation schemes

To systematise model evaluation, hydrologists have proposed various validation schemes that have different restrictions. The schemes are principally based on one or another of the five tests presented below. The type of test or tests that are best suited depends on the modelling context. It also depends on data availability and on the stationarity of the processes over the period covered by these data. This stationarity depends in particular on the stationarity of the basin characteristics and the hydrometeorological context. The five tests are:

- the split sample test,
- the proxy basin test,
- the differential split sample test,
- the proxy basin differential split sample test,
- the blind validation test.

The ***split sample test*** (Klemes, 1986) consists in testing the model over a time period that was not used for calibration. In practice, available data are split into two datasets. For continuous simulation models, the two datasets correspond to two sub-series of the initial series. For event-based models, the two datasets come from catalogues of distinct observed events. The catalogues are constructed either by randomly sampling pertinent events or in such a way as to provide a complete range of events of varying magnitudes in each catalogue. In both cases, one of the datasets is used for model calibration and the other for model validation or vice versa. The results must be similar for both datasets. The way the two datasets are split can be modified as desired. The split sample test can be applied to a drainage basin for which the duration of the available series is sufficiently long. The hydrometeorological context and characteristics of the basin must be stationary over the considered period.

The ***proxy basin test*** (Klemes, 1986) consists in testing the model on a basin providing data that has not been used for calibration. Calibration is carried out on one basin and then the parameters are transferred to the other basin for validation. The two basins must have comparable hydrological behaviours and sufficient data for model implementation and evaluation. This test requires a stationary hydrometeorological context over the considered period. It is often used when the calibration basin does not have enough data to both calibrate and validate the model. This test is always used in hydrological regionalisation studies. In such a case, several basins are used to calibrate the hydrological model to be regionalised and several others to validate it.

The ***differential split sample test*** (Klemes, 1986) consists in testing the model on a time period that has a different hydrometeorological context than the corresponding calibration period. This test is carried out when the model must simulate the behaviour of a basin in a disturbed or non-stationary hydrometeorological context (e.g., climate change, land-use change). For example, if the objective is to estimate the change of the hydrological regime that would result from a wetter climate in the future, the test would include: 1) identification of wet and dry periods in historical data, 2) calibration of the model using the dry period and 3) validation on the wet period. The model must show that is capable of performing well for both contexts. The characteristics of the basin must be stationary over the considered period.

The ***proxy basin differential split sample test*** (Klemes, 1986) is similar to the previous test except that the basin used for validation is different than the basin used for calibration. This test can be used when there is not enough data for the basin of interest and when the objective is to simulate the behaviour of a basin under non-stationary conditions. This approach is for example used to study the impact of climate change on the hydrological regime of ungauged drainage basins.

In ***blind validation*** (Ewen and Parkin, 1996), model performance is evaluated by a third-party once the parameters have been estimated. The modeler receives the data required to describe the drainage basin and prepare the model input data, but does not receive the observation data that the model is supposed to simulate. Once the model parameters have been estimated, the model is used in simulation mode. The simulation results are then transmitted to the third-party that compares them with the observed data. The hydrometeorological context and the basin characteristics must be stationary over the considered period.

Internal and external validation

Whatever the test used, hydrologists distinguish between internal and external validation.

Internal validation is designed to test the ability of the model to reproduce the basic variables for which it was calibrated. This validation can use the same evaluation criteria as those used for calibration (Sections 3.2 and 3.4). The robustness of the model is generally quantified by the difference (usually a decrease) between the average performance obtained in the calibration and validation phases. Internal validation is the most widely used approach. The data used are traditionally measurements of the discharge at the basin outlet. There are two reasons for this: 1) discharge is the most frequently measured hydrological variable and 2) it integrates all the hydrological processes that determine the behaviour of the drainage basin.

External validation is designed to test the ability of the model to reproduce the variables or hydrological behaviours that were not considered in the calibration phase. Examples include

fluxes other than discharges (e.g., evapotranspiration, sediment transport), state variables of certain drainage basin compartments (e.g., water table height, soil moisture, snow depth), hydrological variables integrated over different time steps (e.g., daily and monthly discharges) and the spatial variability of certain hydrological variables (e.g., discharges, water table height).
External validation based on the spatial organisation of hydrological variables is for example possible when measurement stations exist for the same variable at different points in the drainage basin (e.g., discharges on nested sub-basins). Spatial data obtained by remote-sensing techniques can also be used in this way. They are however often limited by the fact that they do not provide a direct measurement of hydrological variables and that they often include high uncertainties. Confidence in the model is increased if external validation tests are passed.

Exercise 6
Validate the HMS model using observed data.

4.2 Uncertainties

Simulations carried out with a given hydrological model are obviously imperfect. For each model output variable that can be compared with observations, this imperfection produces errors that vary with time (Section 1.2). These errors are often considered as realisations of random variables. Ideally, they should be homoscedastic, independent and normally distributed with a mean of zero. Whatever the nature of the errors, they always imply that the simulated value of a given variable is uncertain.
Although not widely done, it is extremely important to characterise the uncertainties associated with hydrological model outputs. This makes it possible for example to assign a confidence interval to the simulated values or estimate the significance of performance tests (Pappenberger and Beven, 2006). The following sections deal with sources of uncertainty, possible methods to analyse the importance of uncertainties and to characterise the uncertainty on model parameters and on the associated simulations. For more information on this topic, see the work of Zin (2002).

Sources of uncertainty

There are many sources of error and uncertainty. They are mainly related to: 1) the model itself and 2) the data used to calibrate, validate and more generally use the model operationally.
Uncertainties related to the hydrological model concern its structure and its parameters. Uncertainties induced by the structure are the consequence of simplifications used in the model to represent the physical medium and processes. Whatever the approach used, a model is always a rough representation of a complex natural system. Hydrological processes can therefore be simulated only in a very imperfect manner. Uncertainties in the parameters result from the fact that the optimal parameter set depends on the calibration strategy used, the performance of the calibration algorithm and the calibration data sample. They are partly related to equifinality questions already discussed in Section 2.2. Note also that the ability of a model (structure + parameters) to represent the hydrological behaviour of the physical

medium and processes varies with time, particularly because the relative importance of the hydrological processes operating within the hydrosystem also varies with time. Consequently, the statistical properties of the errors can vary greatly with time. Uncertainties related to model input data and variables are many and varied. They are in part related to:

- measurement errors (e.g., precipitation) and the poor spatial representativeness of certain point measurements (e.g., physiographic characteristics of the medium, precipitation),
- data acquisition and processing errors (e.g., estimation of discharges on the basis of water levels),
- errors induced by the aggregated representation of certain variables at the spatiotemporal scales of the model (e.g., basin precipitation),
- the imperfection of models used to calculate unobservable input data (e.g., reference evapotranspiration),
- lack of knowledge of certain anthropogenic disturbances of system behaviour (e.g., diversions, irrigation off-takes).

Other sources of error include those related to the initial state of the considered hydrosystem and, in the case of a drainage basin for example, the state of basin saturation and water stored at the surface (e.g., soil moisture, extent of saturated zones, extent and density of snow cover).

In general, the majority of uncertainties are often related to the estimation of these state variables and meteorological forcing variables (e.g., precipitation and temperature).

Analysing uncertainties

For a given modelling context, the effects of all potential sources of uncertainty in the simulations should ideally be analysed. Then, the main sources should be characterised and a method found to take them into account and combine them, propagating them if necessary, to determine the overall uncertainty of each model output.

In practice, it is often impossible to take into account all sources of uncertainty and deal with them simultaneously. Most the time, only one or another of the sources is identified and considered. Moreover, characterization of the sources of uncertainty is not necessarily obvious, often due to a lack of knowledge. Finally, propagating the sources of uncertainties to deduce the uncertainties on output variables requires an appropriate sampling and simulation strategy, often difficult to implement. Then, even when only a single source of uncertainty is concerned, uncertainty analyses are limited to a sensitivity analysis designed to estimate the influence of the source of uncertainty on the model outputs - or possibly on the estimation of model parameters.

Analyses of uncertainties related to model input data and variables often depend on such a sensitivity analysis. For precipitation, for example, a sensitivity test can be used to assess the influence of different point data spatialisation schemes or the influence of the number of stations used as a basis for this spatialisation (Obled *et al*., 1994; Arnaud *et al*., 2002; Vischel and Lebel, 2007). For evapotranspiration, the influence of the method used to produce the reference evapotranspiration time series may be tested (Andréassian *et al*., 2004). When carried out, characterization of the uncertainty in the output variables resulting from the uncertainty in the input variables concerns mainly precipitation. This requires an error model capable of producing different precipitation scenarios that can each be used as input to the simulation model to provide different possible realizations of the hydrological variable of interest (Kelly and Krzysztofowicz, 2000; Zin, 2002). Such a set of realizations, that can also be obtained by taking into account other sources of uncertainty, represents what is commonly referred to as an ensemble prediction or forecast, depending on the simulation context.

Analyses of uncertainty concerning the model structure are relatively rare. They generally consist in: 1) calibrating different types of models or different versions of the given model over a given period and 2) comparing results of their simulations in another context (Butts *et al*., 2004; Schaefli, 2005). Uncertainties related to model structure are often taken into account using a multi-model approach. Different models are applied to the basin of interest and the corresponding simulation results are combined, possibly with different weights, to provide different possible realizations of the variable of interest (Butts *et al*., 2004; Schaefli, 2005). Analyses of uncertainties more often concern uncertainties associated with model parameters. These analyses can be designed to identify the most important parameters in the model, i.e., those that have the greatest influence on the simulation results. They can also be used to identify interdependencies between these parameters. In this case, they mainly involve analysis of the sensitivity of model performance to parameter variations. Performance is evaluated by a selected numerical evaluation criterion. The response surface associated with this criterion is described for a given domain within the parameter space. In a two-parameter space, this produces curves such as those presented in Figure 25. At each point of the considered domain, the variations of the values of this function with respect to the relative variations of the values of one of the parameters determine the sensitivity of model performance to this parameter. In other words, the sensitivity *Si*, at point *M*, of performance criterion *C* with respect to parameter ϑ_i is defined by:

$$S_i = \frac{dC / d\theta_i}{\theta_i} \tag{9}$$

where the values of all the other parameters $\vartheta_{j,j\neq i}$ are constant. For more information on the methods associated with this type of analysis, see the work of Singh (1995) or the more general work of Saltelli *et al*. (2000).

Analyses of uncertainties related to parameters is also and in particular aimed at estimating the uncertainty of the resulting simulation. This point will be discussed below.

Characterising simulation uncertainties related to parameters **(beginning of 'not examinable')**

In practice, to characterise simulation uncertainties induced by uncertainties in parameters, different parameter sets are sampled in the parameter space according to a method and selected criteria. For each, the hydrological model is then applied to simulate an estimate of the hydrological variable(s) of interest - and possibly any changes with time. The *n* estimates provided by the *n* considered parameter sets are then combined to estimate the probability distribution of each of these hydrological variables. In this way, these distributions can be used to characterise the associated uncertainties.

The difficulty in deriving the required probability distribution lies in the choice of parameter sets to be considered and/or the weights to be assigned to the results produced by each. All parameter sets do not have the same probability. Those that produce good model performance are intuitively more likely than those that produce poor performance. Two types of approaches are used in this context. From a very schematic point of view, the first consists in uniformly sampling the "entire" possible parameter space and then assigning a weight to each of the considered parameter sets depending on its likelihood.

This is the principle of the methods proposed by Van Straten and Keesman (1991) and by Beven and Binley (1992). The second approach is referred to as Generalised Likelihood Uncertainty Estimation (GLUE). Both these methods have three important drawbacks. The first is that their results depend strongly on the choice of the likelihood function used to weight the simulations (Gelman *et al.*, 1995). The second is the excessive number of parameter sets that must be considered to obtain a sufficiently dense sample of the parameter space. The importance of this problem increases with the number of parameters. The third drawback is related to the necessity of choosing a threshold for selection of the parameter sets that lead to simulations considered satisfactory with respect to observations. This choice can only be subjective.
The other type of approach consists in identifying the probability distribution of the parameter sets using a Bayesian inference process. In this case, prior knowledge of the model and the probable values of the parameters are used to define a so-called prior probability distribution of these parameters. This prior distribution can for example be produced on the basis of values obtained for these parameters on drainage basins other than the basin of interest. The likelihood between the observations and the outputs of the hydrological model is then used to revise or update this prior distribution and produce the posterior distribution of the parameters. Bayesian updating can be formally expressed as follows:

$$f(\boldsymbol{\theta}|\mathbf{Z}) = \frac{f(\mathbf{Z}|\boldsymbol{\theta})\cdot f(\boldsymbol{\theta})}{\int f(\mathbf{Z}|\boldsymbol{\theta})\cdot f(\boldsymbol{\theta})\cdot d\boldsymbol{\theta}} \tag{10}$$

where $f(\boldsymbol{\theta}|\mathbf{Z})$ is the posterior distribution of $\boldsymbol{\theta}$ if the observations $\mathbf{Z}$ are known, i.e., it is the probability distribution that is to be determined for the parameters. $f(\mathbf{Z}|\boldsymbol{\theta})$ is the likelihood of observations $\mathbf{Z}$ if $\boldsymbol{\theta}$ is known, i.e., it is the information concerning $\boldsymbol{\theta}$ that is provided by the data. $f(\boldsymbol{\theta})$ is the priori probability distribution of $\boldsymbol{\theta}$. The denominator, representing the likelihood of the data sample, is a constant. The likelihood $f(\mathbf{Z}|\boldsymbol{\theta})$ of observations $\mathbf{Z}$ knowing $\boldsymbol{\theta}$ is equivalent to the likelihood of the error model $\varepsilon_Z(\boldsymbol{\theta}) = \mathbf{Z}\text{-}G(\mathbf{X}, \boldsymbol{\theta})$ knowing $\boldsymbol{\theta}$. The latter can be determined by considering that the errors are realizations of a random variable that can only be described by a stochastic model. Given the structure of equation 10, the effect of updating the prior distribution clearly depends on the pertinence of the information available in the observations. If there are very few observations, for example, the posterior distribution will be greatly influenced by the prior distribution and little influenced by the data itself and vice versa if many observations are available.
This posterior distribution of $\boldsymbol{\theta}$ knowing observations $\mathbf{Z}$ does not have an analytical expression for the general case. The value of $f(\boldsymbol{\theta}|\mathbf{Z})$ for a given parameter set $\boldsymbol{\theta}$ is a function of the error values $\varepsilon Z(\boldsymbol{\theta})$ and the stochastic model chosen to describe them. Different algorithm schemes can however be used to reconstruct this distribution $f(\boldsymbol{\theta}|\mathbf{Z})$, including Metropolis (Metropolis *et al.*, 1953) and Metropolis-Hastings (Hastings, 1970) type algorithms. These algorithms can be used to sample the parameter space in such a way that the final sampling density corresponds at every point in space to the posterior probability density of the parameters

(Addendum 7). These Bayesian inference methods are increasingly used in the hydrology community for a wide range of applications (Overney, 1997; Kuczera and Parent, 1998; Talamba, 2004; Schaefli *et al*., 2007). Among other advantages, they correctly represent distributions that are bimodal or strongly conditioned by the interdependence of certain parameters. They are also relatively easy to implement. They still however require careful reasoning on the part of the hydrologist concerning the parameter estimation procedure. The identification of informative and pertinent prior distributions is in particular a crucial step for this estimation. **(end of 'not examinable')**

Figure 29 shows two options, ensemble modelling and Monte Carlo modelling, to consider uncertainty compared to a single simulation.

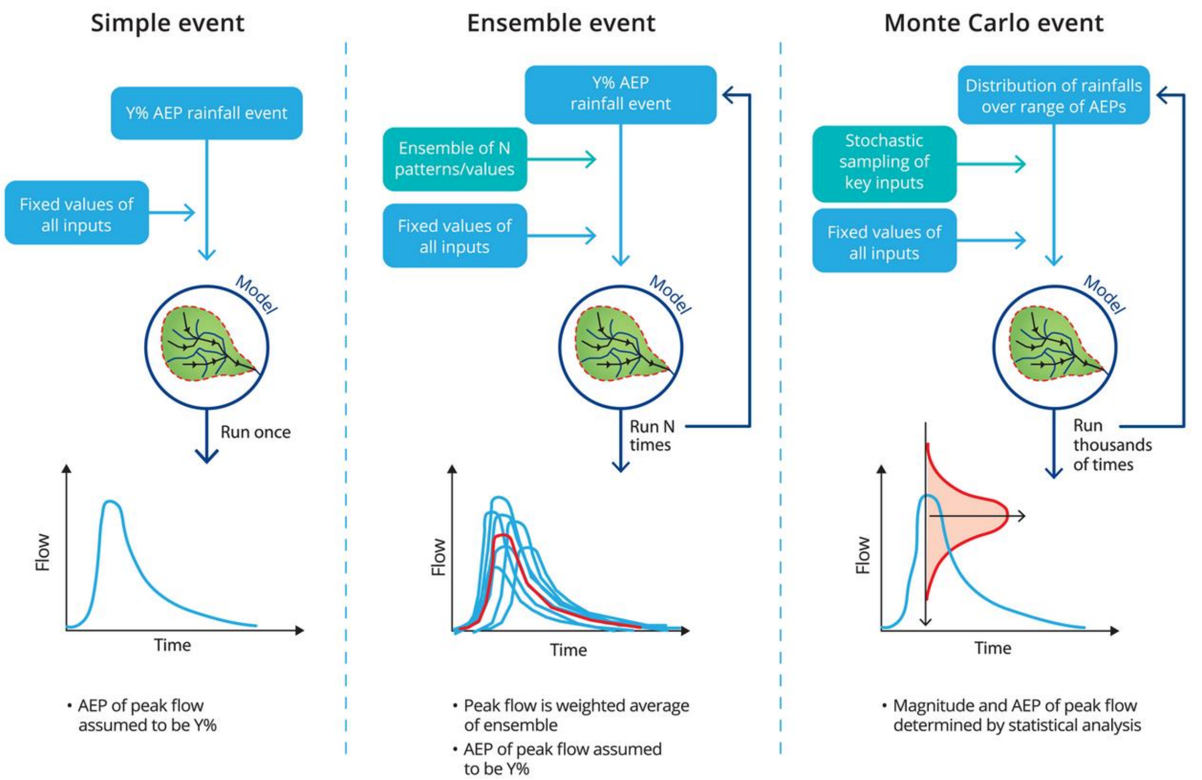

Figure 29. Single simulation, ensemble simulations and Monte Carlo simulations (Ball et al., 2019 – this is the Australian Rainfall Runoff Guideline, available at https://arr.ga.gov.au/arr-guideline).

These three approaches to dealing with probability neutrality can be described as:

- Simple Event, where all hydrologic inputs are represented as single probability neutral estimates from the central range of their distribution;
- Ensemble Event, where the dominant factor influencing the transformation is selected from a range of values representing the expected range of behaviour, and all other inputs are treated as fixed; and
- Monte Carlo Event, where all key factors influencing the transformation are stochastically sampled from probability distributions or ensembles, preserving any

significant correlations between the factors, and probability neutrality is assured (for the given set of inputs) by undertaking statistical analysis of the outputs.

Addendum 7 Principles of Metropolis type algorithms (beginning of 'not examinable')

Metropolis algorithm schemes combine Monte-Carlo and Markov chain simulations to sample the parameter space from a posterior distribution of **θ**, knowing observations **Z**. At the nth step of the Markov chain, *n* parameter sets $\boldsymbol{\theta}_0,\boldsymbol{\theta}_1,\boldsymbol{\theta}_2,\ldots\boldsymbol{\theta}_n$ are available. To select parameter set $\boldsymbol{\theta}_{n+1}$, the algorithm proceeds via the following steps:

1. A parameter set candidate $\boldsymbol{\theta}^c$ is drawn randomly from a multivariate and symmetrical probability distribution $J(\boldsymbol{\theta}^c|\boldsymbol{\theta}_n)$ depending on the values of the two parameter sets **θ**n and $\boldsymbol{\theta}^c$.

2. Parameter set candidate **θ**c is selected as the parameter set for step *n*+1 on the basis of its acceptance probability defi ned by the probability of transition from $\boldsymbol{\theta}_n$ to $\boldsymbol{\theta}^c$:

$$p(\boldsymbol{\theta}^c|\mathbf{Z}) = \min\left(\frac{f(\boldsymbol{\theta}^c|\mathbf{Z})}{f(\boldsymbol{\theta}_n|\mathbf{Z})},1\right) \qquad (11)$$

where $f(\boldsymbol{\theta}^c|\mathbf{Z})$ and $f(\boldsymbol{\theta}^n|\mathbf{Z})$ are the posterior distributions of respectively $\boldsymbol{\theta}^c$ and $\boldsymbol{\theta}_n$, knowing observations **Z**.

3. To determine if $\boldsymbol{\theta}^c$ is accepted or refused, a random number *u* is drawn from a uniform distribution over the interval [0,1] and compared to the above transition probability as follows:

if $p(\boldsymbol{\theta}^c|\mathbf{Z}) > u$ then $\boldsymbol{\theta}_{n+1} = \boldsymbol{\theta}^c$ (12)

if not, remain at current position, i.e., $\boldsymbol{\theta}_{n+1} = \boldsymbol{\theta}_n$ (3.13)

4. *n* is incremented and the procedure is repeated from step 1 as long as $n<N$.

The result of the algorithm is a sequence of parameter sets $\boldsymbol{\theta}_0,\boldsymbol{\theta}_1,\boldsymbol{\theta}_2,\ldots\boldsymbol{\theta}_N$ for which the probability distribution corresponds, if the chain is sufficiently long, to the posterior distribution of **θ**, knowing observations **Z**. Given that the result is relatively sensitive to the parameter set $\boldsymbol{\theta}_0$ used to initialize the chain, the algorithm should generally be applied several times using different initial parameter sets $\boldsymbol{\theta}_0$. Moreover, the first part of the chain generated by the Metropolis algorithm should not be taken into account for the inference. The distribution $J(\boldsymbol{\theta}^c|\boldsymbol{\theta}_n)$, referred to as the transition function, is defined as being symmetrical when $J(\boldsymbol{\theta}^c|\boldsymbol{\theta}_n) = J(\boldsymbol{\theta}_n|\boldsymbol{\theta}^c)$. A multivariate normal distribution $N(\boldsymbol{\theta}_n, \Sigma)$ is generally used as the transition function, where Σ is the covariance matrix of **θ**, updated during the process if necessary to take into account intermediate results of the chain. An asymmetrical distribution is sometimes used for the transition function, in which case the Metropolis- Hastings criterion must be used to calculate the acceptance probability of the new parameter set (Hastings, 1970).

The acceptance/refusal phase is the key step in the Metropolis algorithm. The algorithm will always move towards a parameter set providing better performance than the parameter set $\boldsymbol{\theta}_n$. However, there is also a non-zero probability (probability $p(\boldsymbol{\theta}^c|\mathbf{Z})$) of moving towards a parameter set providing poorer performance. This makes it possible to explore in a dense manner the high-probability regions of the parameter space and also the regions of lower probability. This algorithm also avoids the problem of local optimum zones.

For further information on this algorithm and on precautions to be taken when implementing it, see for example Kuczera and Parent (1998) or Gaume (2007). **(end of 'not examinable')**

5 USING A MODEL

Once a hydrological model has been developed, calibrated and successfully validated, it is ready to be used. The way it is used obviously depends on the objectives of the study for which it was developed (Section 1.1). In all cases, using the model implies taking into account a number of constraints that will now be discussed.

5.1 Initial Conditions and Boundary Conditions

First of all, the initial conditions and boundary conditions of the simulation must be specified. The way to specify initial conditions depends on the type of application. For example, if the model is to be used to simulate a design flood resulting from a given design meteorological event, the initial conditions are generally defined in such a way that they correspond to the average initial state of the drainage basin during a typical hydrological year. If the model is to be used to forecast flood discharges over the coming hours, initial conditions are estimated in such a way that they correspond to the initial state of the drainage basin when the forecast is made. However, if the model is used to simulate the system response over a long time period, estimation of the initial conditions is less critical given that they influence only the initial time steps of the simulation (Section 3.5). For subsequent analysis of the simulation results, the corresponding initialization period, of specified duration, must however be removed from the total simulation period. Whatever the case, initial conditions must be determined with careful reasoning.

The way to specify boundary conditions and input variables also depends on the simulation context. For example, meteorological forcing input can come from observed data or various scenarios constructed elsewhere (e.g., design rainfall or meteorological forecasts. In all cases, the quality of the input data is a critical factor given that it determines the quality of the model output data.

5.2 The Importance of Recognising the Difficulties Inherent to Modelling

Over recent decades, progress in hydrology has been driven much more by computers and hydrological models than by scientific research aimed at improving the understanding of associated processes. Although hydrological modelling remains at the technological forefront, this must not lead us to forget certain weaknesses that are inherent to hydrological models. Final decisions related to the technical aspects of a given project cannot be based solely on the output of models, however sophisticated they may be. The interpretation and application of model output always requires the know-how and expertise of hydrologists and engineers.

It is important to recognise and expose the difficulties inherent to modelling. Although the performance of models presented in scientific publications and study reports is outstanding, it often concerns relatively straightforward drainage basins and events, i.e., those for which the model encounters no particular problem. The real world is often very different. Difficulties in correctly simulating observations are relatively frequent. Sooner or later, hydrologists encounter a hydrological event that the selected model cannot correctly reproduce or a drainage basin for which the behaviour cannot be simulated. Note that such failures should never be concealed, as is often the case, as this can make users overconfident with respect to simulation results. Hiding failures also prevents the modeler from having the opportunity to

subsequently deal with a problem, for example by modifying the modelling concept or refining the strategy used to observe the variables required for the description of meteorological forcing or the state of the system.
Moreover, care must be taken not to extract more information from the simulations than they really contain. In particular, model outputs for which the simulation cannot be validated due to a lack of appropriate observations should be treated with caution. This is for example the case for discharge series produced at various intermediate ungauged stations of the distributed or semi-distributed model.
Whatever the context and target application, it is recommended ultimately to inform users of modelling imperfections. It is also important to deliver not only the output variables required for the project, but also the associated uncertainties. If all the uncertainties cannot be characterised, at least the most important must be (Section 4.2). Ensemble simulations are relatively easy to implement and should be more widely used to obtain rough estimates of these uncertainties.

5.3 Extrapolation and Spatial and Temporal Transposition

In general, a model should not be used for a hydrometeorological context that is too different from the one for which it was developed. In particular, model result uncertainty increases if a model is used to extrapolate to exceptional hydrometeorological events not considered in the calibration/validation phases. A model should not be used for spatial transposition (e.g., to ungauged basins without parameter regionalisation) or temporal transposition (e.g., to a climate context different from that of the calibration period). If a model is nevertheless used in one of these contexts, different tests must be carried out to determine to what degree the results are pertinent (Section 4.1). In practice, such tests are unfortunately seldom carried out.
Note that all past and present hydrological studies (e.g., frequency analysis, modelling) are based on the assumption that the hydrological operation of drainage basins is stationary (Milly *et al*., 2008; Meylan *et al.,* 2012). This assumption is however far from reasonable for many applications. This is obviously the case for studies aimed at assessing the hydrological impact of climate change (Addendum 8). Is also the case for most studies concerning recent decades. The behaviour of drainage basins has often been modified continuously over the years by various development projects (e.g., urbanisation, hydraulic works) or by the management of natural resources and land use (e.g., agriculture, forestry). In the absence of anthropogenic influence, the behaviour of drainage basins can also be modified by changes in meteorological conditions or the natural evolution of the physical environment (Renard, 2006). Modification of the behaviour of the drainage basin can in fact be accompanied by a modification in the main processes that cannot be represented correctly by the hydrological model used. It is therefore always necessary to keep this supplementary source of difficulty in mind, in particular because modifications of the behaviour of the basin are not necessarily visible to hydrologists.

Addendum 8 Hydrological modelling and climate change

One of the key problems in hydrology today is the assessment of the hydrological impact of climate change. The current approach for this consists in producing the required hydrological scenarios by simulations based on meteorological scenarios generated for the considered

climate change scenario. The simulation is carried out using a hydrological model developed for the considered hydrosystem from observations and information available for the present climate context.
This approach supposes that a hydrological model that is valid for the present climate remains reasonably valid for a modified climate, which implies that the hydrological behaviour of the hydrosystem is stationary. This assumption is obviously unreasonable if the climate changes. If the regional climate and consequently the local meteorological conditions are modified, the surface of drainage basins will also be modified (e.g., vegetal cover, temporary and permanent drainage networks). The same is true for the water stored at the surface and sub-surface (e.g., snow cover, ice surfaces, soil moisture). The hydrological processes that are preponderant today and which serve as a basis for developing hydrological models may no longer be preponderant tomorrow. The question may therefore be posed as to the pertinence of using existing hydrological models to simulate a given future state.
In fact, there is no way of guaranteeing that today's models will be reasonable tomorrow. A number of analyses can however be carried out to obtain useful information in this concern. For example, it is essential to at least check that the model can reproduce the behaviour of the basin for different meteorological contexts observed in the past (e.g., validation of temporal transposition between wet and dry years, see Section 4.1). Similarly, it is important to check that the model can reproduce unusual situations sometimes observed in the present context but that could occur more frequently in the future (e.g., situations corresponding to basin saturation conditions and weather types that will be more probable in the future or extreme events that are expected to become more frequent).
As mentioned above, it is likely that a presently valid modelling concept will be inappropriate for a future changed climate context. When assessing the impact of future changes, it is therefore judicious to devise and apply a different modelling concept for the targeted basin, based on experience acquired on other basins for which the present behaviour could resemble the future behaviour of the target basin. In this approach, it is necessary to define pertinent similarity criteria to identify, among basins observed today, those that can reasonably be thought to resemble the target basin tomorrow. The search for analogue basins simply on the basis of geographic or physiographic similarity criteria is obviously insufficient here.
In all cases, producing hydrological scenarios for a given basin in the future climate context requires a description of what the basin could resemble tomorrow. Some expected modifications to the system can be estimated by simulations with an external model (e.g., glacier retreat model, Gerbaux *et al*., 2005; Schaefli, 2005; vegetal cover model, D'Almeida *et al*., 2006; Ishidaira *et al*., 2008) or on the basis of scenarios produced by an expert. These scenarios can also concern development projects carried out within the basin (e.g., urbanization, construction of hydroelectric projects, extension of irrigation parameters) or future resource management modes (e.g., adopted to a modified future demand). The production of these scenarios represents one of the most critical and uncertain aspects of such impact analyses.

6 CHOOSING A HYDROLOGICAL MODEL

The great diversity of existing hydrological models makes choosing a model a difficult task. As pointed out throughout this chapter, this choice first of all depends on the objective of the hydrological study. Obviously, it must also be based on the ability of the chosen model to

satisfy robustness, parsimony and validity criteria. Given two models offering equivalent robustness and validity, preference should in principle be given to the most parsimonious model (which does not necessarily mean a simplistic model). Other criteria can of course also be taken into consideration. For instance, a model can be required to explicitly take into account the spatial variability of meteorological conditions or certain hydrological processes. It can also be required to be applied to a different meteorological context than the one for which it was developed. In certain contexts, it may be necessary to easily transpose the model spatially, for instance to basins without data.
Other aspects to be considered include data availability, the type of drainage basin analysed, the adequacy of the model with respect to the main hydrological processes that explain its hydrological behaviour and the required nature and quality of results. Other less scientific criteria may also affect the choice of a model, for example the experience of the modeler, the fact that the model has been used successfully elsewhere and the available time and budget. The final choice often results from a compromise between these different criteria. In practice, the decision is often highly subjective, depending for instance on criteria imposed by the user. Whatever the criteria, it must be possible to justify the choice for each analysis carried out. Several aspects related to this requirement will now be discussed.

6.1 Data Availability, Model Complexity and Performance

As already mentioned in Section 2.4, the choice (or construction) of a model must be based on a balance between the degree of complexity required in the model structure (which depends on the objectives of the study) and the degree of complexity that is compatible with available data.
In other words, a model that is too simple to use all the information that can be derived from available data is sub-optimal. On the other hand, it is of no use to choose a model with an internal structure that is too complex to be verified on the basis of the available data. This principle is illustrated in Figure 30.

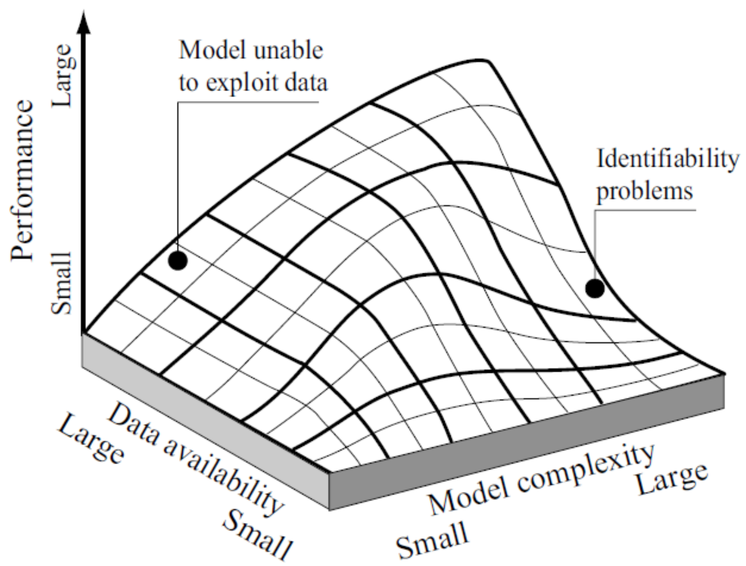

Figure 30. Schematic representation of the relationship between data availability, model complexity and performance (adapted from Grayson and Blöschl, 2000). For a fixed level of data, the performance of a hydrological model is optimal for a certain degree of complexity.

For a given model, an increase in data availability increases model performance up to a certain threshold above which performance is no longer improved.

Note that most of the time, the data required for the construction of a hydrological model do not exist. In such a context, it is best to choose parsimonious models for which the parameters can be predicted using regional models.

6.2 Model Output Requirements

The required model output variables can also greatly influence the choice of a certain type of model.

A distributed or lumped conceptual approach may be sufficient if only the discharges at the outlet of the given basin must be estimated. Such an approach is however inappropriate if the discharges must be estimated at different stations within the hydrographic network. Distributed or semi-distributed models are indispensable in such a case. Almost all hydrological models can simulate discharges. However, many cannot simulate other hydrological variables that may be indispensable to certain hydrological studies. Examples of such secondary variables include river stage, flow velocity and water table height. Physically-based models are often the only models that can be used to estimate such variables.

6.3 Choosing a Model Type

The different possible types of hydrological models are presented in Section 1.3. The choice of a certain type of model and then the choice of the model itself depends first of all on the objectives and available resources (data as well as computer and financial resources).

Conceptual or physically-based approaches?

The literature is rich in discussions comparing the respective virtues and shortcomings of conceptual and physically-based models (Beven, 1989, 1993; Grayson *et al*., 1992). Conceptual models, in particular the so-called reservoir models, are widely used in operational hydrology. They offer many advantages. First of all, they are generally easier to develop and implement. They require little data and often have few parameters. For a generally moderate investment in terms of personnel and cost, they usually provide very satisfactory results for common hydrological applications.

Their main limitation is related to their representation of the processes that is often far from reality. This representation is often based solely on the law of mass conservation, neglecting the conservation of momentum and energy. Although these models can reproduce observed behaviour after calibrating the parameters, nothing ensures they can reproduce behaviour that has not already been observed. It is even highly probable that they will be unable to do so. Such models are therefore *a priori* not transposable to different hydrometeorological contexts (application to a basin configuration modified by development projects, extrapolation to unobserved events, applications to another drainage basin). For the same reasons, they should not be used under non-stationary conditions, for example for impact studies concerning changes in land use or climate.

Physically-based models are in principle not subject to these limitations. As long as it is possible to accurately describe the physical medium, these models should be able to function "blindly" without calibrating their parameters. They are thus supposedly able to simulate, in

a realistic manner, the behaviour of the considered hydrosystem for a modified hydroclimatic context. However as already mentioned, sub-grid parameterisation of the elementary processes is inevitable and these models therefore generally require calibration of some of their parameters. In this respect, it would be misleading to think that it is sufficient to increase the spatial resolution of the discretisation and improve the physical description of the processes to achieve an accurate representation of the hydrological operation of the drainage basin. It is in fact impossible to describe the complete structure of the physical environment and the spatial variability of the physical characteristics that govern its hydrological behaviour. The difficulties encountered in defining the surface drainage network are a patent example. Over and above permanent drainage paths, how can temporary drainage channels or those that are only used, sometimes on a micro-scale, during flood periods be identified and described? Similarly, it is impossible to describe all the hydrological processes that coexist within the basin by means of physical equations. For example, the Darcy model, even generalised, is unable to represent the preferential flows that occur in various networks of conduits or macropores that can exist below the surface. It is important to remember that physically-based models are still only models and are therefore subject to the same limitations as conceptual models, even if to a lesser extent.
In practice, physically-based models are mainly used to study and understand processes taking part in the hydrological cycle. Useful for developing an understanding of these processes, they are widely used in the academic world and in scientific research (Higy, 2000). Their complexity however makes their development and use a tedious task. The considerable quantity of data required for the description of the physical medium and the estimation of their parameters represents the most severe obstacle. They also require a great deal of computing time. This stands in the way of any automatic calibration approach or any attempt to characterise the uncertainties in parameters and in the corresponding simulations.[4] Given the complexity and difficulties inherent to their calibration, these models can generally only be developed by specialists. Although seldom used in operational hydrology over past decades, this situation is rapidly changing today. Physically-based models are especially suited to studies aimed at assessing the impact of modifications of the physical environment on the behaviour of drainage basins.

Distributed modelling?

The choice of spatial discretisation rather than distributed modelling depends on the way the processes are represented. Spatial discretisation is for example required if the hydrological model is process-oriented. The choice of spatial discretisation is also closely related to the objectives for which the model was developed.
Distributed models are for example used to check observed data and reconstruct missing data or to extend historical discharge series by simulations (when longer time series of meteorological variables are available). They are also used to generate flood scenarios for development projects and design studies (e.g., hydraulic works, roadway culverts), to evaluate water resources by simulating hydrological balances or to carry out real time hydrological forecasting (for management of water resources and the associated hydrological risks).

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[4] With the computer resources of their time, Yang *et al.* (2000) indicated for example that the simulation of two years of data at an hourly time step required 72 hours of calculations for the MIKE SHE physically-based model and only 2 minutes on the same machine for the TOPMODEL distributed conceptual model.

Lumped models have a number of drawbacks that are specific to conceptual models. In addition, because they consider the drainage basin as a single and unique hydrological entity, they cannot account for the heterogeneity of the physical medium or the spatial variability of the inputs, meteorological forcing and hydrological processes. Neither can they account for the phenomena that spatially redistribute surface and sub-surface water nor the influence of hydraulic works located within the basin on discharges. They are therefore not suitable for very large basins or basins in which meteorological forcing exhibits high spatial variability. They are also unsuitable for basins subject to a high level of anthropogenic activities or presenting high spatial heterogeneity. For instance, a lumped conceptual model cannot represent the behaviour of a drainage basin made up of a rural part and an urbanised part. The hydrological behaviour of the two parts of the basin is too different.

An advantage of distributed models is that they can account for different types of hydrological behaviour within the drainage basin. This is a limitation of lumped models. The semi-distributed model type sits between the lumped and distributed model type. The main advantage of distributed models is precisely that they can consider the spatial variability of meteorological forcing, initial conditions and the characteristics of the physical medium that influence hydrological behaviour. Distributed models make it possible to simulate the hydrological variables of interest at any point within the spatial discretisation of the system and in particular at any point of the hydrographic network described by the model.

The main drawback of distributed models lies in the quantity of information required to describe the drainage basin. The spatial variability of the characteristics of the physical medium is for this reason often considered in a very simplified manner. In addition, meteorological forcing variables must be estimated at every point of the discretisation, requiring a suitable method to spatialise the observed local variables. The inevitable uncertainties related to this spatialisation can greatly reduce the accuracy and hence usefulness of the distributed description of the medium. Another difficulty lies in the discretisation of the basins which remains a delicate task in spite of the use of DEMs and GISs. The implementation difficulties associated with distributed models, in particular when physically-based, limited their use in operational applications.

Semi-distributed models generally represent a good compromise between lumped and distributed approaches, in particular when modelling large drainage basins. They are generally easier to develop and implement than distributed models. They do require a large quantity of data but can partially consider the spatial variability of meteorological forcing and state variables of the physical medium. The spatial variability of hydrological behaviour can also often be taken into account. This is for example the case when model parameters can be estimated on the basis of the key characteristics of the medium available elsewhere (e.g., average slope of the basin, percentage of impervious ground surface) and a scaling factor.

Whatever the approach considered, the spatial discretisation resolution can vary with the level of detail required. In all cases, the basin is often divided up in such a way that all the different spatial units have roughly the same area. This ensures a certain consistency in the representation in the same way as it is necessary to ensure a certain compatibility between the levels of complexity used to represent the physical medium and processes. The choice of the average size of the units depends on the modelling objective and, as already mentioned, the information available to characterise the variability of the medium and the processes. Ideally, the choice should be made in such a way as to limit, within each unit, the spatial variability of the key meteorological variables, the characteristics of the medium and the dominant processes. This aspect was already discussed in Section 2.4.

Event-based or continuous simulation models?

The choice between an event-based or a continuous simulation model depends largely on the modelling objectives.

An event-based model is generally sufficient to simulate discharges resulting from a given meteorological event. This is for example the case when determining a design flood on the basis of a given design rainfall. In such situations, the various other related hydrological processes (e.g., evapotranspiration) can generally be neglected. Continuous simulation models can be used to simulate the evolution of state variables (e.g., soil moisture) and output variables (e.g., discharge) over several hydrological years. It is therefore possible to simulate flood discharges, low flows and hydrological regimes as well as frequencies of failure related to the operation of existing or future hydraulic structures and projects. This is the main advantage of this type of model. The main difference with respect to event-based models is that certain processes that can be neglected during an event cannot be neglected in a continuous simulation because they become predominant between events. This is for example the case for evaporative fluxes that can be neglected in simulations of given rainfall-runoff events but which are preponderant during dry periods that separate rainfall events. Consequently, continuous simulation models require more data to describe the climatic forcing that determines the basin response (e.g., temperature, radiation and wind speed in addition to precipitation). Give their numerous advantages over event-based models, the use of continuous simulation models is growing rapidly (Boughton and Droop, 2003).

6.4 Comparing the Performance of Hydrological Models

The choice of a hydrological model is generally guided by its performance with respect to a given objective. Many studies have been carried out to evaluate the performance of individual models. However, relatively little work has focused on comparing objectively the performance of different types of models for a given application.

Intercomparison projects mostly concerning a limited number of models applied to a limited number of basins for a given time step, climate context and type of application (simulation, forecasting, transposition; see Refsgaard and Knudsen, 1996; Jiang *et al*., 2007). Larger scale intercomparison projects have also seen the day. One example concerns the intercomparison projects initiated by the World Meteorological Organisation (WMO) focusing on the ability of different reservoir-type lumped models to simulate observed discharges (WMO, 1986) or forecast discharges in real-time (WMO, 1992). More recently, Perrin (2000, 2002) compared the simulation performance of a large number of reservoir-type lumped models. The models were evaluated over more than a thousand drainage basins located around the world. In the USA, the Distributed Model Intercomparison Project (DMIP) compared in particular the performance of different distributed models for operational discharge forecasting and evaluated the interest of an ensemble forecast obtained by combining associated individual forecasts (Smith *et al*., 2004; Georgakakos *et al*., 2004).

These analyses show that no universal model structure exists. Furthermore, the models are often well-suited to the hydrometeorological contexts for which they were designed but show much poor performance when they are applied to other contexts.

6.5 Other Considerations

In practice, an important selection criterion is the cost of developing and implementing a model. Costs and time requirements related to acquiring and preparing data, constructing calibrating and validating the model and interpreting the simulation results increase greatly with the degree of sophistication of the model.

Model development for a given basin is much easier if it is possible to use an existing user-friendly modelling system. The cost of such a system, including the license and user training, can however be prohibitive and impose the development of an in-house model. The user-friendliness of the modelling system software interface can also be an important selection criterion. The user manual for certain modelling systems is sometimes insufficient for the user to understand and easily implement the corresponding model. Furthermore, the preparation of input data files and even the reading of output files can be extremely tedious for certain modelling systems.

The existence of a user-friendly interface does not eliminate the difficulties inherent to any hydrological modelling work. These difficulties are simply masked by the interface.

Care must be taken to never sacrifice the understanding of the hydrological phenomena and the various representations proposed by the selected model. In other words, it is important to use a model only if all its modules are fully understood.

7 APPENDICES

7.1 Example of a Lumped Conceptual Model—the GR4J Model

Developed at Cemagref (France), the GR4J model (French for *Génie Rural 4 paramètres Journaliers*) is a 4-parameter daily model designed for environmental and agricultural engineering applications. It is the most recent version of the GR3J model proposed by Edijatno and Michel (1989) and subsequently improved by Nascimento (1995), Edijatno *et al*. (1999) and Perrin *et al*. (2003). More recent versions of the model exist: with five parametres (GR5J; Le Moine, 2008) and six parametres (GR6J; Pushpalatha, 2013). They were developed to provide a better simulation of low flows.

Drainage basin description

GR4J is a lumped model. It can be used in semi-distributed mode (Baudez *et al*., 1999).

Process descriptions

The model is made up of rainfall excess, rainfall-runoff and routing functions. Four parameters must be estimated.

- Rainfall excess (one parameter): Reservoir with limited storage capacity, infiltrated rainfall and actual evapotranspiration as a function of present storage and respectively total rainfall and reference evapotranspiration.
- Rainfall-runoff (one parameter): Two unit hydrographs in parallel.
- Routing (two parameters): Non-linear reservoir with limited storage capacity.

Other characteristics

- Time step: Daily for the GR4J model and also hourly, monthly and annually for other models of the GR model family.
- Data requirements: Precipitation, reference evapotranspiration and discharge series.
- Parameter estimation: In principle by calibration but possible also using regionalisations.
- Resource requirements: Low computer, human resource and time requirements.

- Applications: Many drainage basins around the world.
- Web site https://webgr.inrae.fr/en/models/daily-hydrological-model-gr4j/ (last visited 23 April 2022).

7.2 Example of a Semi-Distributed Conceptual Model—the HBV Model

The HBV model (Swedish for *Hydrologiska Byråns Vattenbalansavdelning,* i.e., Water Resources section of the Hydrology Bureau) was initially developed by Bergström (1973, 1976, 1992). It is the standard hydrological forecasting model used in Nordic European countries.

Drainage basin description

HBV is a semi-distributed rainfall-runoff model in which the discretisation of the basin is based on elevation and the characteristics of the vegetation cover. Version HBV-96 is a semi-distributed version that allows division of the drainage basin into sub-basins. The description of certain processes (e.g., snow accumulation and snowmelt, evapotranspiration) has been modified.

Basic model description (HBV)

The HBV model is a 5-reservoir model with 7 to 9 parameters that must be estimated. It is made up of 4 modules.

- Snow module (3 parameters): Reservoir with accumulation and degree-day type snowmelt.
- Interception module (no parameters): Storage reservoir for forested surfaces.
- Rainfall excess module (6 parameters): 3 reservoirs in series (soil, intermediate and groundwater reservoirs; linear and non-linear reservoirs, with or without storage capacity and with or without overflow).
- Rainfall-runoff module (1 parameter): Unit hydrograph.

Other characteristics

- Time step: Daily with possibility of shorter time steps.
- Parameter estimation: 9 parameters to be optimised, possible also using regionalisations.
- Data requirements: Moderate; precipitation, reference evapotranspiration (Penman, 1948), discharge and air temperature (for the snow module) series. In forecasting mode, the model requires air temperature and precipitation forecasts.
- Applications: Many drainage basins around the world.
- Well-suited in particular to mountainous and Nordic drainage basins strongly influenced by snow.
- Resource requirements: Low computer, human resource and time requirements.
- Web site: The HBV model has been adopted by several federal agencies and research institutions. One version that has been made available to the public is 'HBV-light': https://www.geo.uzh.ch/en/units/h2k/Services/HBV-Model.html (last visited 23 April 2022)

7.3 Example of a Distributed Conceptual Model—the CEQUEAU Model

The CEQUEAU model, named after the "CEntre QUébécois des Sciences de l'EAU" was developed initially by Girard *et al*. (1972). The corresponding software is available from the "Eau, Terre, Environnement" center of the "Institut National de la Recherche Scientifi que" (INRS) (Quebec, Canada).

Drainage basin description

The drainage basin is divided up into elementary areas ("whole squares") which are in turn divided up into "partial squares" (depending on the water divides).

Model description

The conceptual model has 28 distributed parameters that must be calibrated. It implements two main functions.

- Rainfall excess (27 distributed parameters to be calibrated): Vertical balance conceptual model including 3 to 4 reservoirs. The water balance is carried out on each whole square and at each time step. The model deals with snow accumulation and snowmelt using a degree-day function for a ripe snowpack (monitoring of cold storage), evaporation and evapotranspiration, water in unsaturated zone, saturated zone and lakes.
- Rainfall-runoff (1 parameter to be calibrated): Conceptual model designed to forward the volume of water produced within each whole square from one partial square to the next until reaching the basin outlet.

Other characteristics

- Time step: 1, 2, 3, 4, 6, 8, 12 and 24 hours.
- Parameter estimation: From 11 to 28 parameters to be calibrated.
- Data requirements: Extensive; liquid/solid precipitation and air temperature (maximum and minimum) series along with discharges, daily reference evapotranspiration, physiographic (DEM) and thematic data for the drainage basin.
- GIS interface.
- Applications: Many drainage basins, in particular in Canada.
- Well-suited to drainage basins are very different sizes.
- Resource requirements: Relatively high computer, human resource and time requirements.

7.4 Example of a Distributed Physically-based Conceptual Model: TOPMODEL

TOPMODEL (*TOPography based hydrological MODEL*) was initially developed by Beven *et al*. (1984). A number of versions exist. For English-speaking readers, a detailed description of TOPMODEL, including its background and limitations, can be found in Beven (2012). Note that TOPMODEL formed the basis for the development of TopNet by NIWA.

Drainage basin description

The description is based on hydrological response units (HRUs) determined on the basis of a topo-hydrological similarity index calculated at each point of the basin from the local slope β_i and upstream drainage area *ai*. Parts of the basin that have the same values of this index are assumed to have a similar hydrological behaviour.

Process description

TOPMODEL is based on physical principles for certain processes but uses conceptual reservoirs to represent others. It includes 3 parameters to be calibrated for the simplest versions. The model has two components.

- Rainfall excess: Based on the concept of a variable contributing area that is determined at each time step on the basis of the value of the local saturation deficit. This deficit depends on the average saturation deficit on the basin scale and the local

value of a topographic index. The vertical balance is estimated for each index class from this local deficit using conceptual reservoirs connected in series.
- Rainfall-runoff: Different conceptual or empirical methods are used depending on the version (linear reservoirs, unit hydrographs).

Other characteristics
- Time step: Daily or shorter.
- One parameter is determined by field measurements (the topographic index and its distribution) and the others by calibration (e.g., conductivity at saturation at the top of the first layer, decay parameter for hydraulic conductivity at saturation, root storage depth, evaporation and interception rate, rainfall-runoff parameters).
- Data requirements: Moderate; precipitation series along with discharges, Penman evaporation and the distribution of the topographic index for the basin (currently obtained from DEMs).
- GIS interface.
- Applications: Many drainage basins around the world.
- Well-suited in principle to drainage basins with highly pervious soils.
- Resource requirements: Relatively high computer, human resource and time requirements.

7.5 An Example of a Distributed Physically-based Hydrological Model: The SHE Model

SHE (French for *Système Hydrologique Européen*) is a hydrological model that was developed back in the early 1980s (Abbott *et al.*, 1986a,b). More user-friendly and modular versions (e.g., MIKE-SHE) are commercially available from the *Danish Hydraulic Institute* (DHI, web page: https://www.dhigroup.com/).

Drainage basin description

The basin is divided into regular square grid cells. The spatial resolution is in principle chosen to match the particularities of the drainage basin however in practice it is limited by the resolution of the supporting DEM and available computer resources.

Modelled processes

SHE solves the water balance on the drainage basin entirely on the basis of physical equations and distributed parameters. The equations used and the other characteristics of SHE include:
- Snow: Accumulation and melt using energy balance or degree-day method.
- Interception: Rutter model.
- Evapotranspiration: Penman and Penman-Monteith model.
- Infiltration and percolation: One-dimensional Richards model for vertical fluxes in unsaturated soil and two-dimensional Darcy model for groundwater flow.
- Streamflow routing: One-dimensional Barré de Saint-Venant model.

Other characteristics
- Time step: Flexible (from a minute to a day).
- Calibration: In principle no parameters to calibrate but calibration is possible.
- Data requirements: Very extensive; meteorological data, discharge series (if calibrated), DEM, thematic data for the drainage basin, measurable soil characteristics, special features of the hydrographic network (hydraulic works, lakes).
- GIS interface.
- Applications: Many drainage basins and sub-basins around the world (in particular with derived models).

- Effective for modelling complex drainage basins.
- Resource requirements: Very high computer, human resource and time requirements.
- Requires purchase of license and updates. Web page: https://www.mikepoweredbydhi.com/products/mike-she .

7.6 The semi-distributed HEC-HMS (Hydrologic Engineering Center-Hydrologic Modelling System)

HEC-HMS is introduced here in more detail, as it is widely used in engineering applications. The Hydrologic Modelling System (HEC-HMS) is designed to simulate the complete hydrologic processes of dendritic catchment systems. The software includes many traditional hydrologic analysis procedures such as event infiltration, unit hydrographs, and hydrologic routing. HEC-HMS also includes procedures necessary for continuous simulation including evapo-transpiration, snowmelt, and soil moisture accounting. Advanced capabilities are also provided for gridded runoff simulation using the linear quasi-distributed runoff transform (ModClark). Supplemental analysis tools are provided for model optimization, forecasting streamflow, depth-area reduction, assessing model uncertainty, erosion and sediment transport, and water quality.

The software features a completely integrated work environment including a database, data entry utilities, computation engine, and results reporting tools. A graphical user interface allows the user seamless movement between the different parts of the software. Simulation results are stored in HEC-DSS (Data Storage System) and can be used in conjunction with other software for studies of water availability, urban drainage, flow forecasting, future urbanization impact, reservoir spillway design, flood damage reduction, floodplain regulation, and systems operation.

HEC-1, the predecessor of HEC-HMS, was originally designed in 1967, with the official release in 1973, to simulate surface runoff processes from precipitation over a catchment. The flood hydrograph package was developed over a number of years by the Hydrologic Engineering Centre. It must be noted that HEC-HMS can be used for both, event-based and continuous simulation. While the steps to set up the model are similar, the focus here is on event-based applications. The student should consult the latest HEC-HMS user's manual for more details (accessible at https://www.hec.usace.army.mil/software/hec-hms/).

The process of converting precipitation to direct runoff can be simulated for either small subbasins or large complex catchments. An original HEC model for any river basin has basic components for subbasin runoff, channel and reservoir routing, and hydrograph combining (see Fig. 1). Subarea boundaries are delineated so that lumped precipitation loss and catchment parameters can be used. The computations proceed from upstream to downstream in the catchment, and hydrograph data or plots can be provided at any convenient point. Historical rainfall events or design rainfall is transformed to runoff via UH methods. A discharge hydrograph is computed at the outlet of each subarea (see Fig. 1). For example, a storm hyrdrograph for a given rainfall would be computed for subareas 10, 20, 30, 40, and 50 in Figure 1.

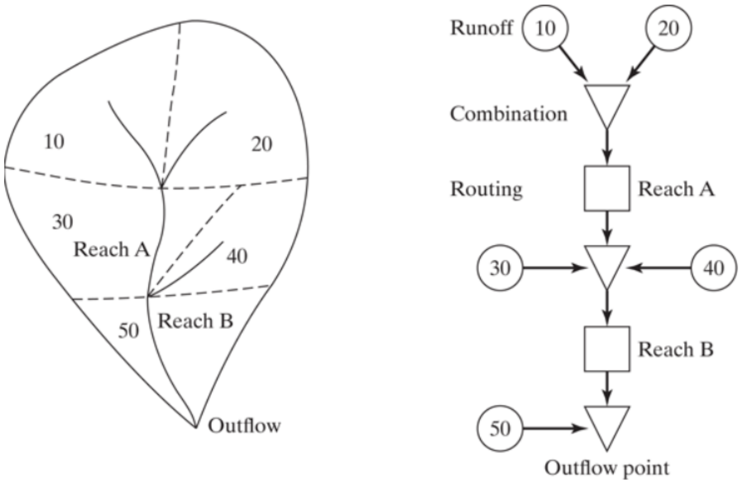


Figure 1 HEC-HMS model configuration.

The routing component in HEC-HMS requires input parameters to define the specific routing characteristics of a river reach or reservoir. Output consists of an outflow hydrograph at the downstream station. Hydrograph combining at key locations is essential for the overall system logic in HEC and allows for an optimal use of computer storage in the model. Combine steps occur at the triangle indicators in Figure 1, where two or more hydrographs are joined. Finally, simple river routing is accomplished for both reach A and reach B, as shown by the square indicators. The final output hydrograph is computed at the outlet or outflow point.

HEC-HMS is a general flood hydrograph package with the following capabilities:

- Simulation of catchment runoff and streamflow from historical or design rainfall
- Determination of flood flows at various locations for use in floodplain studies
- Evaluation of land use or topographic changes in a catchment area
- Simulation of flood control measures, such as reservoir storage and channel options
- Computation of damage frequency curves and expected annual damages for various locations and multiple flood control plans

Catchment Delineation and Organisation

HEC-HMS uses parameters averaged in space and time to simulate the runoff process. The size of subbasins, routing reaches, or computation interval is selected based on the basin physiography, available rainfall data, available streamflow data, and required accuracy. A catchment is subdivided into small and relatively homogeneous subbasins according to drainage divides based on mapped or digital topography, as shown in Figure 1. The size of a subbasin should generally be in the range of 1 – 15 km^2 because of limitations of UH theory, especially in urban areas. Routing reaches are identified, and the overall order of the runoff computation is defined (from upstream to downstream) for input to HEC. Routing reaches should be long enough so that a flood wave will not travel faster than the computation time step. Otherwise, numerical errors in flood calculations will occur. Various routing methods are contained in HEC-HMS.

Precipitation

Precipitation P is computed for each subbasin in Figure 1 using historical data or synthetic design storms. The model allows for (1) incremental P (depth of P for each time interval) for each subbasin, (2) total cumulative P and a time distribution, or (3) historical gage data together with areal weighting (Thiessen) coefficients for each subbasin. Standard design storms can be used in the form of (1) depth vs. duration data, (2) probable maximum precipitation, or (3) standard project precipitation. The precipitation data must be input at a constant time interval, but this interval may be different from the computational time interval in the model. Figure 2 shows a typical basin model for the HEC-HMS model indicating various subareas and river reaches that connect the subareas together.

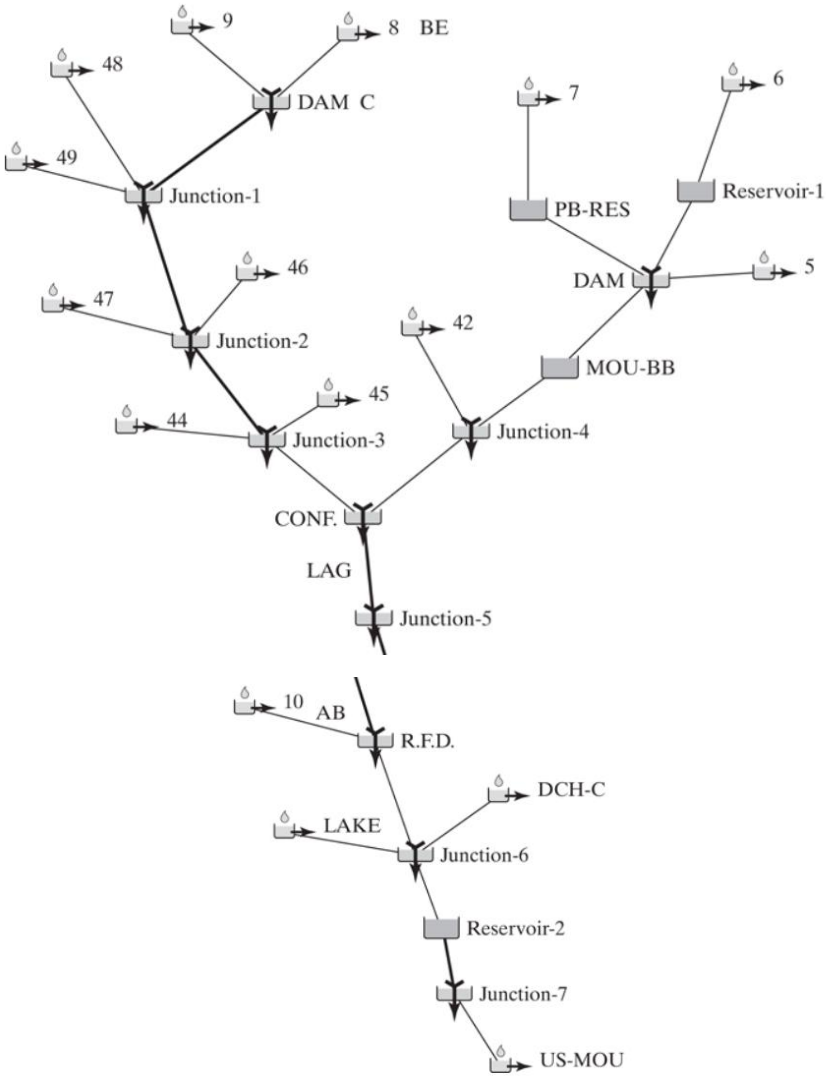


Figure 2 Example of a catchment in HEC-HMS.

HEC-HMS is capable of simulating snowfall and snowmelt. Up to 10 elevation zones of equal increments may be specified in each subbasin. Usually, an elevation increment of 300 m (~1000 ft) is used, but any increment may be specified as long as the air-temperature lapse rate corresponds to the change in elevation within the zone. Snowmelt occurs when the temperature is equal to or greater than the melt temperature and is calculated by either the degree-day method or an energy budget method. For more detail on snowfall and snowmelt calculations, see the HEC user's manual (available at https://www.hec.usace.army.mil/software/hec-hms/) and Runoff from Snowmelt (U.S. ACOE, 1960). Dingman (2002) covers snowmelt in more detail with examples.

Loss Rate Analysis

HEC contains several methods for computing the loss of precipitation to interception and infiltration, as shown in Table 1. The simplest is the initial and constant loss rate function, in which the initial loss volume is satisfied before the constant loss rate begins. The remaining constant loss is identical to the phi index method for infiltration and is often used in design storm analysis or where data are insufficient to allow calculation with more sophisticated methods.

Table 1 HEC-HMS Loss Methods

Method	**Description**
Initial and constant	Initial loss volume is satisfied, and then constant loss rate begins.
HEC exponential	Loss function is related to antecedent soil moisture condition and is a continuous function of soil wetness.
SCS curve number	Initial loss is satisfied before calculating cumulative runoff as a function of cumulative rainfall using SCS methods
Holtan method	Infiltration rate is computed as an exponential function of available soil moisture storage from Holtan's equation.
Green and Ampt	Infiltration rate is computed from the Green and Ampt equation as a function of soil moisture and hydraulic conductivity
Deficit/ Constant	First an initial deficit storage is filled, and then infiltration rates can be specified on a monthly basis.
SMA	Soil moisture accounting assigns a value of initial storage to all layers of the ground using a gridded method.

The SCS method employs the SCS curve number, CN, which is directly related to land use and hydrologic soil properties of a catchment. The SCS approach relates accumulated rainfall excess or runoff to accumulated rainfall with CN. It is popular because of its application to ungaugged areas and its large empirical database. The method requires input of CN, initial abstraction, and percent imperviousness (McCuen, 2005).

The Green and Ampt method was added to HEC in 1990 and probably is one of the best estimates for infiltration, since the parameters are all measurable and directly related to soil types in a catchment. With HMS version 4.2 and later four different parameters to apply the method are used; it is necessary to measure or estimate hydraulic conductivity, capillary suction at the wetting front, initial moisture content, and saturated moisture content. These parameters can be estimated effectively using physical soil characteristics, such as soil texture class and major soil horizon classification. Calculation of these parameters assumes that the soil is a homogeneous medium with uniform initial moisture content. The Green and Ampt method has received significant attention in recent years with the release of HEC-HMS.

Subbasin Runoff Calculation

Several methods for surface runoff or transform computations in HEC are available in the model and can be selected by the user. These are presented in Table 2 and include the UH methods of Clark TC & R (1945), Snyder (1938), and the Soil Conservation Service (SCS, 1984, 1986). Known UHs can also be directly input. The kinematic wave overland flow computation was added to HEC in 1981 and allows more accurate representation of urbanised areas for UH calculations (HEC, 1981).

Table 2 Surface Runoff Methods in HEC-HMS

Method
Direct input
Clark hydrograph method (TC & R method)
Snyder unit hydrograph method
SCS method (CN method + SCS UH)
Kinematic wave for overland hydrograph
ModClark
User-specified S-graph

Clark's method is based on the time–area curve method. The time–area histogram, determined from isochrones of the catchment, or contours from points at which water arrives at the same time, is convoluted with a unit design storm hyetograph to yield the hydrograph, which is then routed through linear reservoir storage to allow for peak attenuation. The Clark-method parameters are the time of concentration and the storage coefficient R. Hoggan (1997) provides more details on the Clark method.

The HEC model uses a synthetic time–area curve derived from a generalised basin shape, and the equations are applicable to most basins:

$$AI = 1.414T^{1.5}, \qquad 0 \leq T < 0.5,$$

$$1 - AI = 1.414(1 - T)^{1.5}, \qquad 0.5 < T < 1.$$

where AI is the cumulative area as a fraction of subbasin area and T is fraction of time of concentration, TC. Specific time–area curves can also be input to HEC model if they can be computed from some other GIS-type method.

The computed hydrograph based on a time–area method is routed through a linear reservoir to produce the final UH. A family of 2-hr UHs generated by HEC is shown in Figure 3 for a 4.3 mi^2 basin with TC = 0.45 hr. Note the wide range of hydrograph shapes that can be produced by the two-parameter UH. An SCS UH is shown for comparison. The larger the value of R/(TC+R), the flatter the UH response, more typical of natural catchments with flat slopes. Thus, the smaller ratios correspond to more urbanized catchments. The steepest hydrograph could also represent a basin with steep slopes and a small amount of storage (Hoggan, 1997).

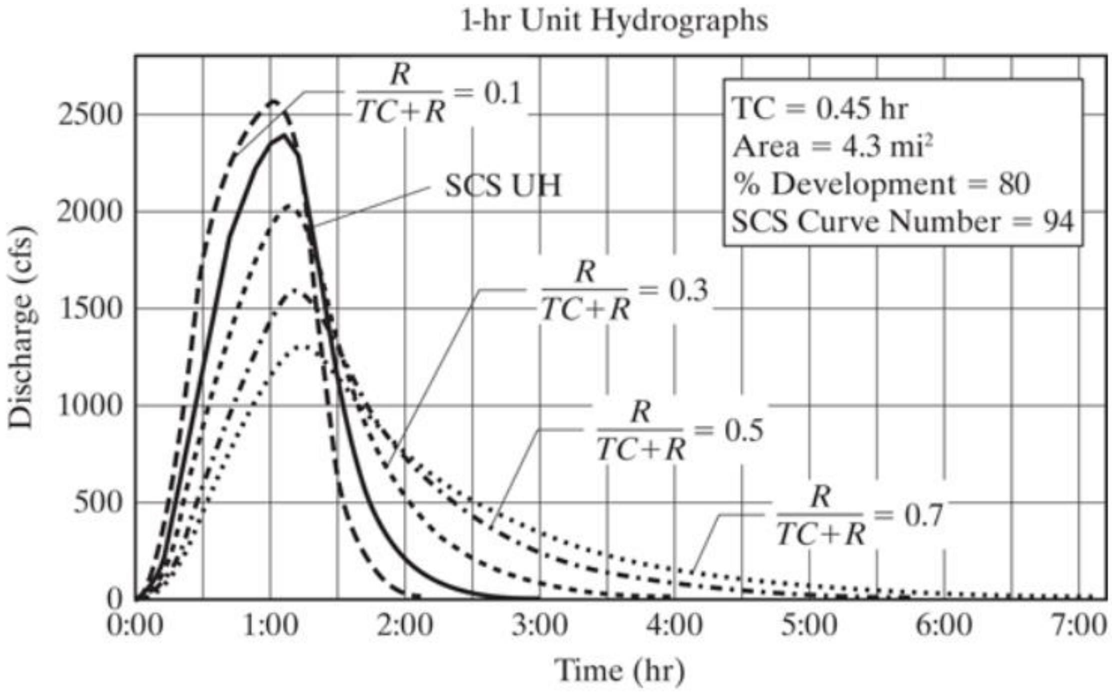


Figure 3 Family of 1-hr UHs for a 4.3-square-mile basin with 80% development and an SCS Curve Number of 94.

Snyder's method provides time to peak t_p and peaking coefficient C_p, which are insufficient to produce the curved shape of the hydrograph. Clark's method is therefore used to help smooth and define the shape of the hydrograph corresponding to Snyder's coefficients within the HEC model. The direct runoff ordinates are then calculated by convolution of the UH with the net precipitation for the subbasin.

The kinematic wave process, represents a nonlinear runoff response compared with linear UH methods. The kinematic wave method relies on parameters that are generally measurable from a basin, such as slope, land use, lengths, channel shape, roughness, and area. Overland flow, collector, and main channel elements are used to represent the characteristics of a drainage basin (Fig. 4).

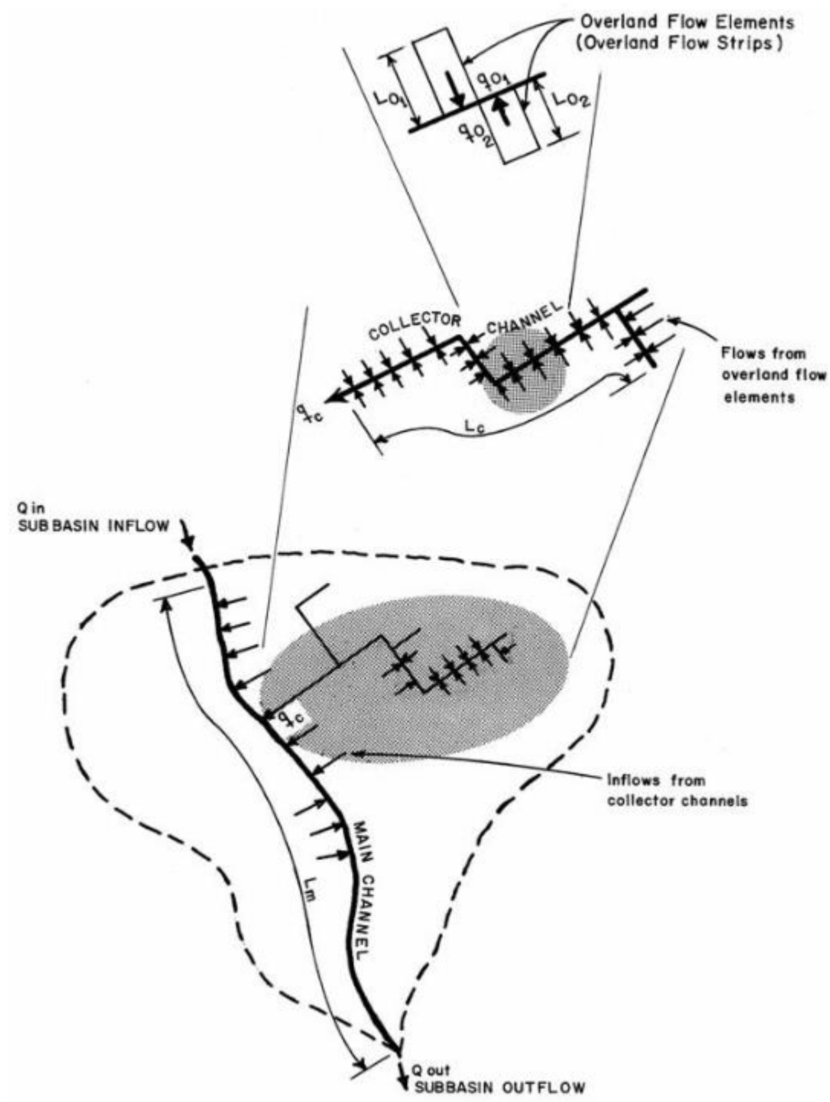


Figure 4 Relationship between flow elements in the kinematic wave model as implemented in HEC-HMS.

In the method, the main equations are Manning's equation of flow and continuity. These are solved numerically to produce overland flow as a function of time and space. The same equations are solved for channel elements (Fig. 4). The kinematic wave (KW) method is more applicable to the analysis of urban basins, since KW theory does not provide for the attenuation of flood waves. The area of the subbasin is required along with L, S, n, shape, and size, and any upstream hydrograph to be routed in the reach. KW routing is also available in the main channel.

Base flow Calculation

One of the methods implemented in HEC-HMS is discussed here. It simulates base flow of a flood hydrograph with three parameters: (1) Q_0, flow in channel prior to start of rising limb; (2) ratio of recession flows at t = 1 (RTIOR), exponential decay rate; and (3) point on falling limb called the recession threshold (Fig. 5). The above parameters can be obtained through a semilog plot of observed flow Q vs. time. The decay rate RTIOR is equal to the slope of the line, and Q_R/Q_{peak} is usually in the range of 0.05 to 0.15.

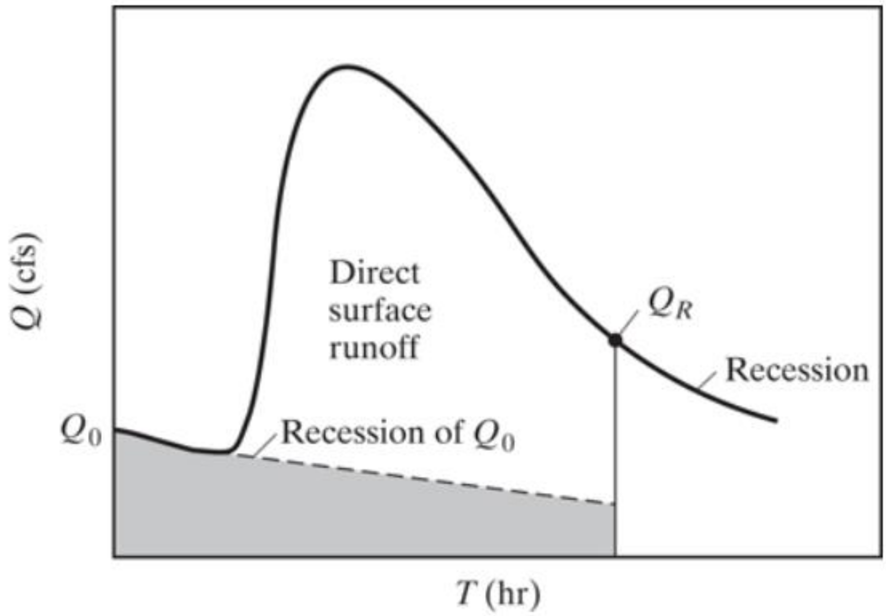


Figure 5 Components of streamflow hydrograph.

Equation (1) defines the parameters that are used in the model, in which the recession flow threshold and the decay constant RTIOR must be specified by the user. Figure 5 defines the relation between the streamflow hydrograph and user-defined parameters. The following equation computes recession flow Q as

$$Q = Q_0(\mathrm{RTIOR})^{-n\Delta t},$$

where

$$\begin{aligned} Q_0 &= \text{starting base flow prior to the rising limb,} \\ Q &= \text{base flow rate at end of } n\Delta t, \\ \mathrm{RTIOR} &= \text{ratio of recession flows at } t = 1 \text{ hr increment apart.} \end{aligned}$$

A range of applications of HEC are for urban flood events, where base flow is a relatively small percentage of the total hydrograph flow. As another option base flow can be entered as a constant value or, if not considered important for the given application, as zero. Several possibilities exist for computing base flow in HEC-HMS.

Flood Routing

Flood routing involves analysing the movement of a flood wave as it travels from reach to reach through a stream or river. Table 3 lists the major methods of flood routing that are available in HEC-HMS, where both storage-indication reservoir and river routing methods were covered. The Muskingum method requires a K and x, where K can be obtained from analyses or using the reach length divided by the wave velocity determined from

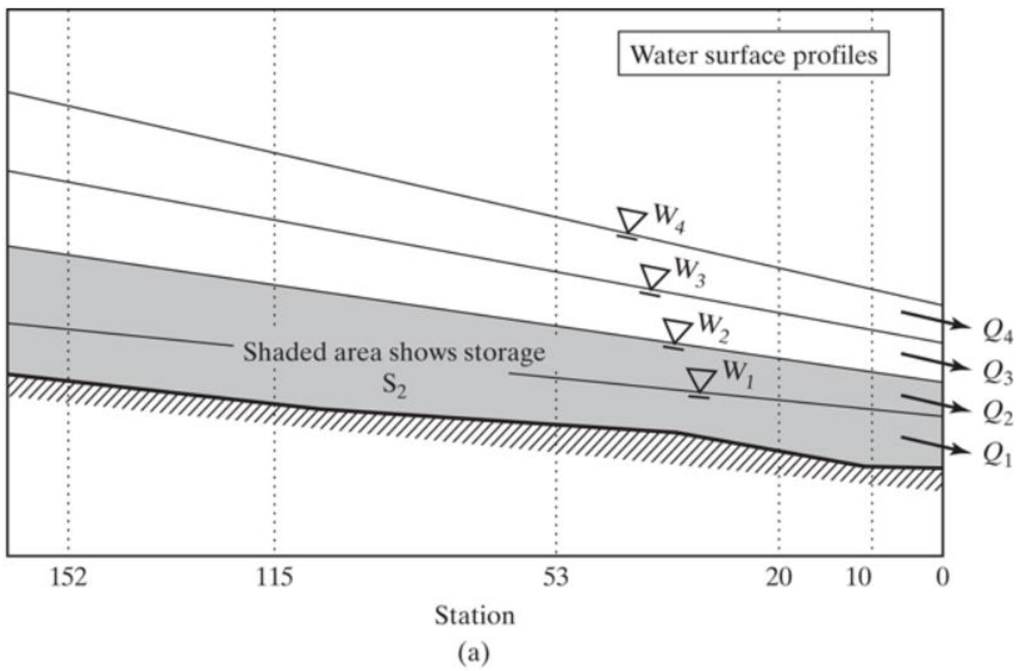
$$\frac{dx}{dt} = c = \frac{1}{B}\frac{dQ}{dy}$$
.

dQ/dy can be obtained from the slope of the rating curve at a typical discharge value to be routed. To avoid numerical instability, note that for the Muskingum method, $2Kx < \Delta t <= K$ and number of subreaches = $K/\Delta t$, where K = travel time and Δt = time interval.

Table 3 Flood Routing in HEC-HMS

Method	Description
Muskingum	Storage coefficient (x) plus travel time (K) through each reach
Modified Puls	Table of storage vs. outflow for each reach based on HEC-RAS water surface profile information—various storages are plotted vs. peak flows in the reach
Kinematic wave	Channel shape, length, slope, and n; outflow from each reach based on depth of flow in continuity equation and Manning’s equation
Muskingum-Cunge 8 point	Same equations and technique as the Muskingum–Cunge but channel is described with 8 station-elevation values
Lag	Direct lag of flow through channel with translation accounted for but not attenuation

The Modified Puls method can be used for a reservoir but can also be applied for rivers by developing a storage–discharge S–Q relation. One of several methods can be used: (1) steady-flow profile computation, (2) observed profiles, (3) normal depth profiles, (4) storage from inflow and outflow hydrographs, and (5) optimisation techniques. These are described in more detail in e.g. Hoggan (1997) and will not be repeated here. For method 1, several water surface profiles are generated: W_1 to W_5, corresponding to discharges, Q_1 and Q_5, using for example the HEC-RAS program. The storage volumes are computed from cross-sectional areas of the channel and their lengths for each W_i. An S–Q curve (rating curve) is determined (Fig. 6). The number of routing steps to be used for a river reach is NSTEPs, equal to $K/\Delta t$. This is a numerical parameter related to flood wave speed vs. the computational time step so that numerical stability is maintained. The greater the storage in a reach per unit of flow, the greater the attenuation of the flood wave as it is routed through.



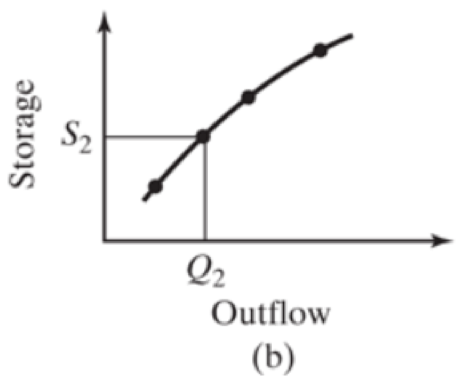


Figure 6 (a) Steady-flow water surface profiles. (b) Storage-outflow curve. (Source: Hoggan, 1997)

The kinematic wave channel routing method does not allow flood peak attenuation, which is a problem where the floodplain is extensive. Thus, KW can overestimate flows compared to other methods. But it has the advantage of allowing for laterally distributed inflow along the main channel (HEC, 1990). Manning's equation forms the basis for the kinematic wave method. All of the above flood routing methods exist for HEC-HMS and any one of them can be selected in the flood routing setup options. The model Vflo (Vieux, 2004) is a good example where KW overland flow combined with storage routing in channels can provide accurate results for most catchments.

Application of HEC-HMS to Catchments

A note upfront: In some parts of the world, and in particular in the USA, the term 'catchment' is used instead of the term 'catchment'. Yet a catchment can be comprised of several sub-catchments, which might be termed sub-catchments. Similarly, the term 'sub-basin' is ta times used instead of the term sub-catchment. It is usually clear what the different authors mean. One just needs to be aware that there are different terms. In the HEC-HMS model documentation there will often be a reference to ‘catchments’ and ‘basins’.

The HEC Hydrologic Modelling System is a Windows-based hydrologic model that continues to be improved and upgraded over the years. This section contains a brief description of the background, capabilities, and usage of the program; specific information is given on the selection and application of the various methods offered. A detailed applications manual is available from HEC (Ford et al., 2008). The beginning student is advised to refer to the most recent user manuals for the actual model setup and applications.

In HEC-HMS, a project consists of three separate parts: the basin model, the meteorologic model, and the control specifications. A background map containing subcatchment boundaries and streams can be entered from a GIS mapfile as a visual reference, but it is not used for any calculations. Various map file types can be used as a background by selecting “background maps” under the View file menu and entering the name of the existing file. The three model sections are accessed by the main screen, within the Catchment Explorer (WE) screen, which is the top window on the left side panel shown in Figure 7. This screen links to all the data and tools through the menus for the various main components. The user can select the models in the Catchment Explorer, and edit their components in the bottom window on the left side panel.

The HEC-HMS window. The catchment components are listed on the left and the catchment is displayed graphically on the right.

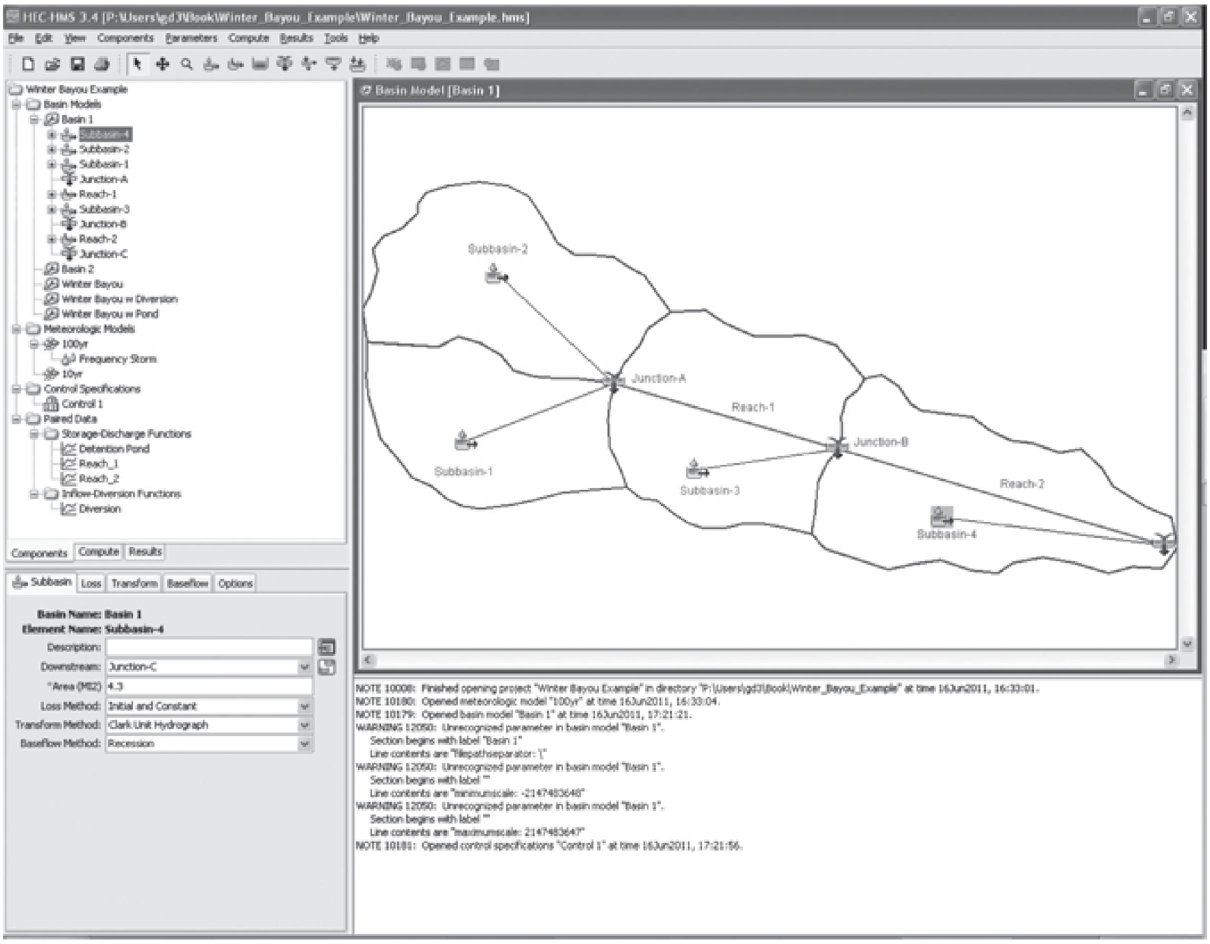


Figure 7 Screenshot of a basin model in HEC-HMS.

The Basin Model

The basin model contains the basin and routing parameters of the model as well as connectivity data for the basin. The GUI uses a simple click-and-drag method to place subbasins, reaches, reservoirs, junctions, diversions, sources, and sinks together to form the catchment. Each element can be given a name and description, and the user may select the method of calculation to be used by the model. The basin model is merely a representation of the actual catchment, and the visual location and sizes of each element do not matter as long as the numerical data and connectivity are correct. HEC-HMS then connects some objects placed at either end of routing reaches. Multiple elements can connect downstream to one element, but one element cannot have multiple downstream connections. The user must be careful to connect objects in the correct direction of flow, which can be checked by selecting the option to show flow directions from the toolbar. A series of icons are used to represent the various elements within the model such as subbasin, routing reach, junction, reservoir, or outlet and are arranged in a horizontal line beneath the main menu. The icon representations for subbasins and junctions can be identified in basin model in Figure 7.

Subbasins represent the physical areas within the basin and produce a discharge hydrograph at the outlet of their respective areas. The hydrograph produced is calculated from precipitation data minus the losses. The resulting precipitation excess is transformed using a UH methodology to compute runoff at the outlet, which is then added to base flow. Each component can be calculated using several methods. The area, loss rate, runoff, and base flow inputs for each subbasin are entered into the basin model for each one using the Catchment Explorer.

Loss rates can be simulated by one of several methods. For event modelling, techniques include initial and constant, SCS curve number, gridded SCS curve number, and Green and Ampt methods. The one-layer deficit and constant model can be used for simple continuous modelling. For modelling of complex infiltration and evapotranspiration environments, the five-layer soil moisture accounting model can be used. The method is changed in a simple menu-driven window, and the input data are entered on the “Loss” tab in the editor [Fig. 7].

Transform methods, which convert rainfall excess into surface runoff, can also be simulated using a variety of tools. HEC-HMS includes the popular Clark TC & R and SCS UH techniques, as well as the Snyder Method or user specified UH. Spatially distributed runoff can be computed with the quasi-distributed linear transform of cell-based precipitation and infiltration. The Modified Clark method (ModClark) is a linear quasi-distributed UH method that can be used with gridded precipitation data. If the ModClark transform with gridded rainfall is used, a file that contains characteristics of subbasin grid cells is required. HEC-HMS can handle grid cell depiction of the catchment for distributed runoff computations. The kinematic wave method with multiple planes and channels is also included. Input data are entered in a simple menu-driven window.

Base flow takes into account normal flow through a channel or the effects of ground water. HEC-HMS offers two methods for base flow calculation: recession and constant monthly. The recession method is an exponential decay function of a defined starting base flow. For the constant monthly method, the user simply enters a constant base flow value for each month. No base flow is also an option, and in simple hydrologic models over short time periods or highly urbanised basins with channels, base flow might have less relevance and might hence be neglected if the error is estimated to be negligible.

Flood routing in HEC-HMS offers a number of options for the reaches and routing of flood hydrographs. The Muskingum method can be used for general routing; routing with no attenuation can be modelled with the lag method. The most popular and accurate is the Modified Puls method, which can be used to model a reach with a user-specified storage–outflow relationship. The S–Q relationship is derived from a hydraulics model such as HEC-RAS.

Channels with trapezoidal, rectangular, triangular, or circular cross sections can be modelled with Modified Puls, the kinematic wave or Muskingum–Cunge method (see Chapter 4). Channels with overbank areas can be modelled with the Muskingum–Cunge eight-point method, which takes coordinates at eight points in a cross section of the channel. The addition of options for subbasins and reaches allows different users to run HMS using the methods most appropriate to their catchments. Input data are entered in a simple menu-driven window within Catchment Explorer.

Reservoirs store the inflow from upstream elements and produces an outflow hydrograph based on a monotonically increasing storage–outflow relationship. A reservoir can be entered with one of three possible types of relationships: storage vs. outflow, elevation vs. storage vs. outflow, or elevation vs. area vs. outflow. The outflow structure must be well understood in order to develop an accurate storage-discharge relation. The inflow entering the reservoir must be contained with the minimum and maximum values of the data entered. The user must also select an initial condition of storage, elevation, outflow, or select inflow equal to outflow. The model assumes that elements have a level pool, such as ponds, lakes, or reservoirs. The effect of adding a detention pond to a basin can be modelled by using a reservoir. The input window for a reservoir is used to relate storage to outflow. A reservoir icon is used to represent storage at any point in the catchment and is then connected downstream to a junction.

Sources are elements that represent a discharge into the basin as an observed hydrograph or a hydrograph generated by a previous simulation. They often are used to represent inflow from reservoirs, unmodeled headwater regions, or a catchment outside the region. This may be entered as gage data or a constant discharge. Sinks are elements that have an inflow and no outflow. The only inputs are the name and description of the sink. It may represent the lowest point of the drainage area or the outlet.

Diversions for hydrologic models use a simple table relating inflow to diverted flow and routed flow. These relationships can be determined using geometric calculations and hydraulic models. Diversions will have two "downstream" connections, one being the routed path and the other the diverted path. The user specifies the diverted flow through the use of a table (inflow-diversion), and whatever flow is not diverted will travel the main path. If the "connected" option is selected, then the diverted flow will return to the catchment at the downstream location.

Figure 8 shows a typical basin model with a mapfile in the background. In very complicated catchments where icons are often too crowded to display all labels, the user can identify objects by clicking on the buttons in the toolbar of the basin model. The selected item will be highlighted grey in the basin model. The toolbar also contains a magnifying button to zoom into a selected area, and the user can zoom out of an area by going to “Zoom Out” under View in menu the main menu. Objects can only be selected using the arrow tool.

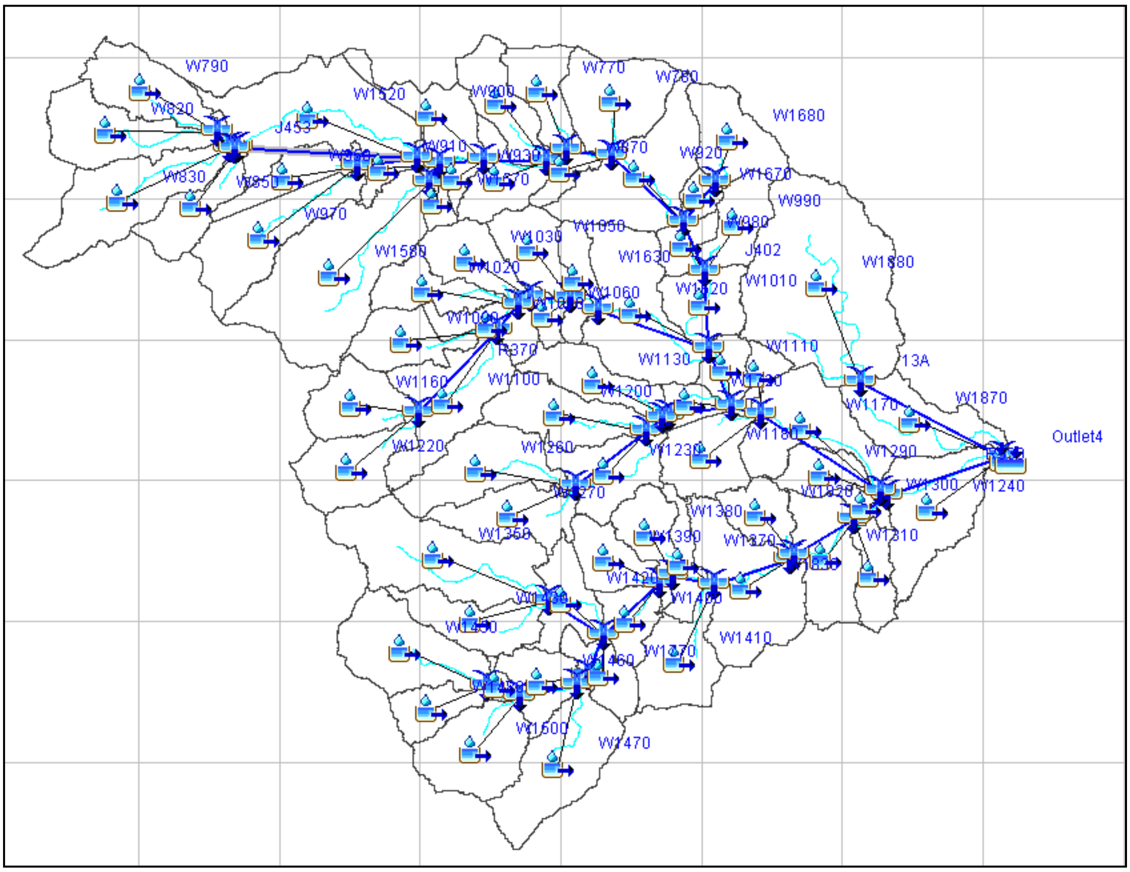


Figure 8. Basin model in HEC-HMS with sub-basins as mapfile in the background.

The Meteorologic Model

The meteorologic model contains the precipitation data, either historical or hypothetical, for the HEC-HMS model. The model contains a number of options for modelling precipitation and can also account for evapotranspiration. Examples of historical precipitation inputs include hyetographs, gage weighting, and inverse-distance gage weighting [Fig. 9]. The program can handle an unlimited number of recording and nonrecording gauges, and gage data can be entered manually, imported from an existing DSS file, or based on an Excel file, as described later. In addition, HEC-HMS has the capability to model gridded rainfall, such as NEXRAD-estimated radar rainfall. Hypothetical precipitation data can be used for design storms based on frequency storm analysis and standard project storm (SPS) models. Figure 10 presents an example of how the 1% AEP (annual exceedance probability) 100-year frequency (ARI annual recurrence interval) storm for Houston, Texas, would be inputted into the model. Note that by clicking on the meteorological model the user can select the type of storm [Fig. 9], and by clicking on the type of storm directly beneath the meteorologic model the user can edit the specifications for that storm [Fig. 10].

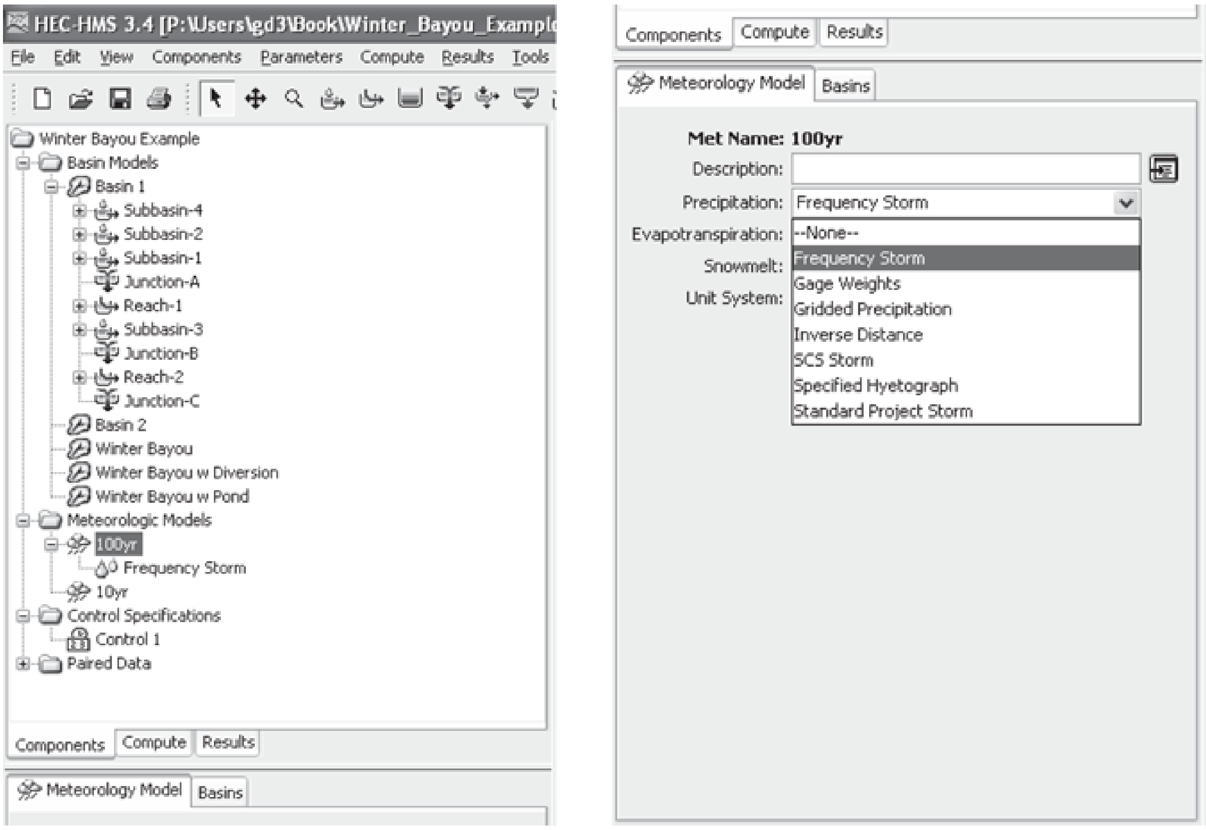


Figure 9 Meteorologic model options.

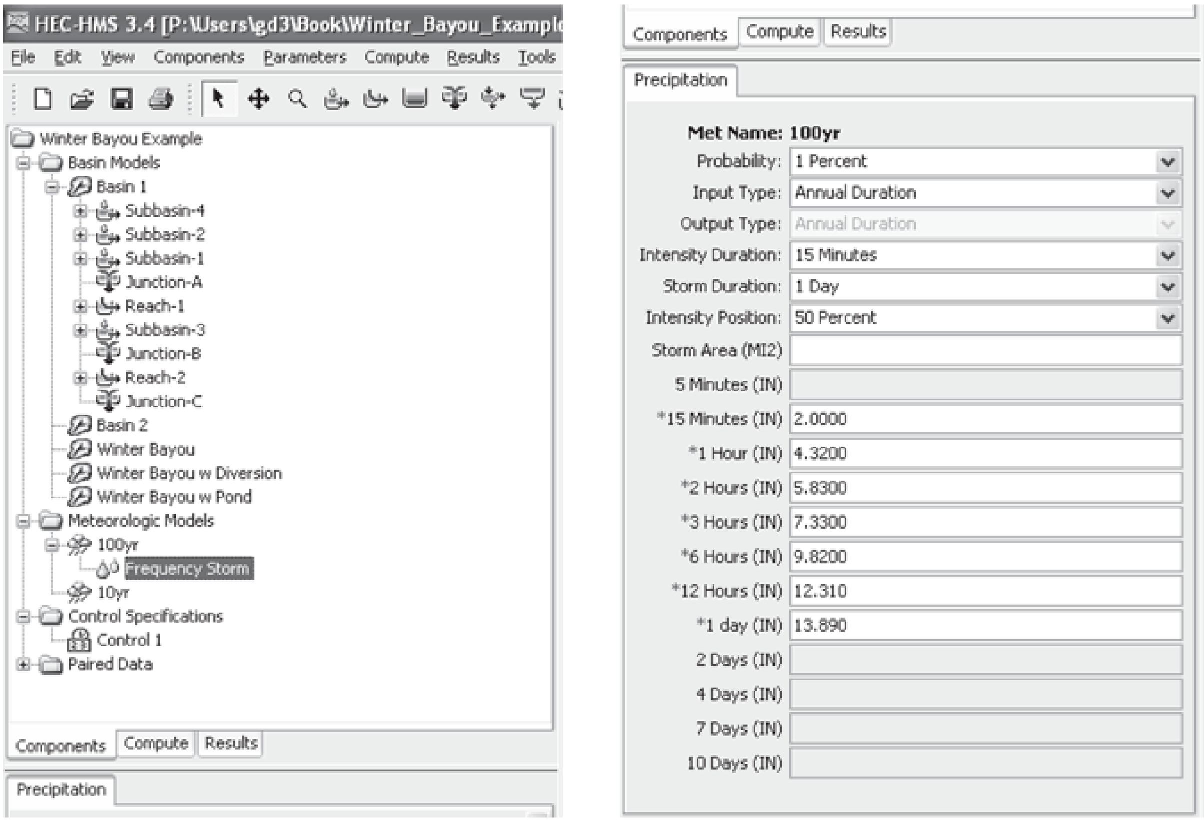


Figure 10 Inputting frequency storm information into the meteorological model.

Control Specifications

The control specifications contains all the timing information for the model, including the start time and date, stop time and date, and computational time step of the simulation [Fig. 11]. This allows for easy organisation of modelling data, which requires a separate dataset describing all aspects of each independent modelling run. A series of runs can be easily organised using this option with several different scenarios. HEC-HMS can also handle continuous hydrograph simulation over long periods. It is standard to run the model several days past the end of the storm to ensure that the falling limb of the computed hydrograph is completely captured.

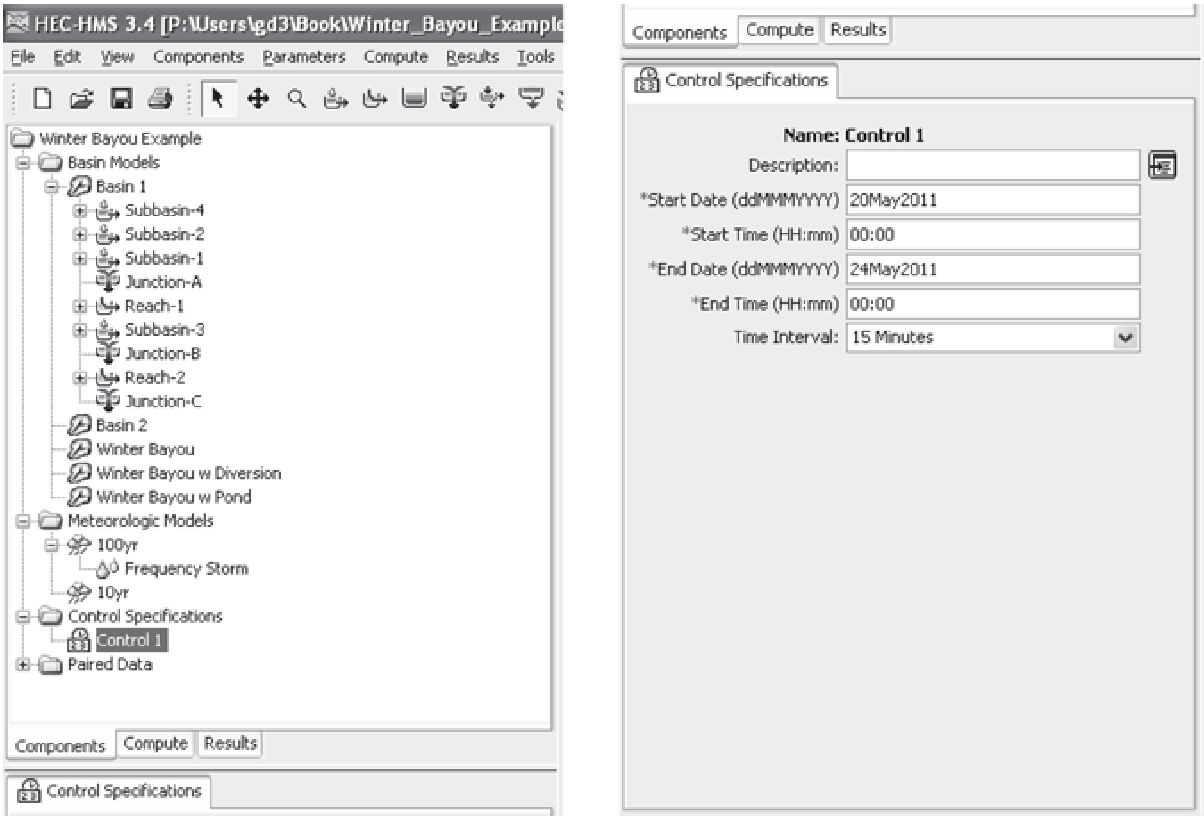


Figure 11 HEC-HMS Control Specifications.

Data Handling

A major improvement for HEC-HMS is the use of the Data Storage System, or HEC-DSS, used to manage time-series and tabular data. The system was the result of a need in hydrologic engineering to store a variety of types of data in a standardized format. Previously, data formatted for one program would need to be entered into another format for a different program by hand by each user. Each program would then use separate functions to analyse and graph the data. The HEC-DSS software is the result of an effort to make hydrologic data management more efficient and allow for the HEC family of programs to use the same database.

HEC-HMS stores all its output in a DSS file and can store much of its tabular or time-series input data in one or more DSS files. For example, reservoir storage–outflow relationships as well as precipitation and observed flow time-series data are stored in a DSS file. The user either can input the data manually into HEC-HMS using the appropriate dialog boxes or can tell HMS to access the data from a DSS file where they are stored. HEC-HMS can use more than one DSS file for a project. There is always one DSS file associated with the project, which stores output. A second DSS file (or more) can hold precipitation and observed flow data as well as other user-entered data (see user's manual for more detail).

While the HEC-DSS software package is an improved means of managing hydrologic data, there is no easy way to transfer the data from a spreadsheet to a HEC-DSS file. HEC-DSS utility software available for download from the USACE website will allow import and export of data from text files for input to HEC-HMS.

Running HEC-HMS and Viewing Results

The user may specify different data sets for each component within a project and then run the hydrologic simulation using different combinations of models. For example, one can run a 10-year or a 1% AEP (100-year frequency) storm model using the same basin model and control specifications to compare the resulting flows. Or, one can model the effects of adding diversions and reservoirs to a basin by saving it under a different name, altering the new basin model, and running the simulation under the same precipitation and control specifications. With several basin models saved in the same project, running simulations with various models is a simple task. In order to create the scenario for a particular run, go to "Simulation run" under Compute in the main file menu. Select one basin model, one meteorological model, and one control specification. Then run the scenario by selecting "Select run," and subsequently "Compute run," both of which are under Compute in the main file menu. If changes are made to one of the models after a run with the same configuration has been calculated, select the correct run to recalculate. Note that the "Compute" tab beneath the Catchment Explorer allows the user to easily select different runs.

Results can be viewed by going to the "Results" tab beneath the Catchment Explorer, clicking on any object (e.g., subbasin or junction) in the basin model, and selecting from the menu. The program gives the resulting times and flows for each basin element and its immediate upstream elements in the form of a graph, summary table, or time-series table. The graph shows time vs. flow, the summary table shows peak flows and corresponding times, and the time-series table gives the inflow and outflow for each time step. For subbasins, HMS also shows the precipitation, loss, excess, and base flow in each form of viewing results. Junction plots show tributary hydrographs and their combined result. Subbasin output hydrographs plotted with rainfall can be obtained by clicking on the appropriate icon. Summary table information is also available for each junction or subarea.

Other Features

HEC-HMS includes advanced features such as parameter estimation with optimization, soil moisture accounting, GIS and grid-cell hydrology, snowmelt simulation, and improved hydraulics. HEC-GeoHMS is a companion program that allows for the creation of HEC-HMS projects from GIS sources, including digital elevation models, land use data, and other electronic sources. In more recent versions of HEC-HMS GIS-capabilities are built-in.

The parameter estimation and optimisation function is used to compare resulting hydrographs to observed hydrographs, so at least one element in the basin must have observed data. The program automatically estimates the parameters in order to find the best fit of the generated hydrograph to the observed one for one element. It allows the user to set initial values for the parameters of each element along with maximum and minimum values so that the parameter values estimated by the program must fall in a reasonable range. The optimisation function measures the degree of variation between the two hydrographs based on four possible choices: peak-weighted RMS error, sum of squared residuals, sum of absolute values, or percent error in peak flow. The program is run iteratively until the user is satisfied with the objective function value, which is zero if the hydrographs match identically.

HEC-HMS is available to anyone for free. The development of HEC-HMS is a project that continues to evolve with time; users should periodically check the HEC U.S. Army Corp of Engineer's (ACOE) website for updates. For further information and help with HEC-HMS, visit the HEC-HMS website: https://www.hec.usace.army.mil/software/hec-hms/

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USACE HEC-HMS software and documentation available at: https://www.hec.usace.army.mil/software/hec-hms/